

Post Mining Land Use | Rezoning | Macquarie Coal Complex Pilot Project

Geotechnical Report

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Definitions and abbreviations

Abbreviation	Definition
ASS	Acid Sulfate Soils
CFA	Continuous Flight Auger
CSIRO	Commonwealth Scientific and Industrial Research Organisation
EbD	Enquiry by Design
GIS	Geographic Information System
m AHD	metres Australian Height Datum
m BGL	metres Below Ground Level
MCC	Macquarie Coal Complex
PLMU	Post Mining Land Use
RL	Reduced Level
TSF	Tailings Storage Facility
SA	Subsidence advisory
WBH	West Borehole
YW	Young Wallsend

1 Executive summary

This Geotechnical Engineering Report has been prepared to support the State-led rezoning of the Macquarie Coal Complex (MCC) (the Precinct), and Teralba Southgate for post-mining land use. The report provides a precinct-scale assessment of geotechnical conditions, constraints and risks, and establishes a risk-informed framework to guide land use planning, infrastructure provision, and staging.

The Precinct and Teralba Southgate comprises a highly modified post-mining landscape characterised by variable topography, anthropogenic ground conditions, and legacy mining impacts, including mine subsidence risk, disturbed terrains, a Tailings Storage Facility (TSF) and areas of potentially deep alluvium. These conditions result in spatially variable development constraints across the Precinct and Teralba Southgate.

A multi-criteria geotechnical assessment has been undertaken, considering slope, geology, regolith depth, soil landscapes, groundwater, salinity and acid sulfate soils. The assessment identifies a range of constraints, from low to high, across the Precinct and Teralba Southgate and confirms that development feasibility will depend on location, land use intensity and staging.

1.1 Key geotechnical constraints

The principal geotechnical constraints affecting the Precinct and Teralba Southgate include:

- **Mine subsidence:** all the sub-precincts are underlain by historic underground coal workings, resulting in potential subsidence risk that must be considered in land use planning and foundation design. Potential subsidence risks are not currently quantified as further studies are required. Development in these areas must include geotechnical assessments to characterise the risk and potentially solutions such as grouting of the former workings, if settlement has not already occurred.
- **Tailings Storage Facility:** the TSF comprises hydraulically placed tailings and fill materials with high variability, uncertain composition, and long-term settlement potential. The area is currently undergoing rehabilitation and remains a declared dam. It represents a high-risk area for development, with settlement potentially occurring over decades.
- **Alluvial and low-lying areas:** land adjacent to Cockle Creek and its tributaries is characterised by deep, compressible soils, shallow groundwater, flooding and acid sulfate soil risk, presenting significant constraints to structural development.
- **Disturbed and anthropogenic terrains:** areas of fill and disturbed ground exhibit variable and uncertain geotechnical properties, reducing bearing capacity and increasing settlement risk.
- **Salinity and reactive soils:** much of the Precinct and Teralba Southgate are affected by moderate to very high salinity hazard and reactive clay soils, with implications for durability of infrastructure and long-term maintenance.

1.2 Structure Plan sub-precincts and Teralba Southgate

- Lower constraint areas (e.g. Killingworth, West Lake, Teralba Southgate, Barnsley West and more broadly the eastern portion of Cockle Creek) are suitable for early-stage development.
- Moderate constraint areas (e.g. environmentally constrained areas) require targeted investigation and design responses prior to development, subject to subsidence risk investigation.
- Higher constrained areas, including the TSF within the Cockle Creek sub-precinct and floodplain environments, are subject to long-term considerations for development potential.

Given that all the sub-precincts are likely underlain by historic underground mine workings and the entirety of the Precinct and Teralba Southgate is mapped as a declared Mine Subsidence District, all future development will require approval from Subsidence Advisory NSW (SA) and may entail further site specific mine subsidence investigations and assessments.

1.3 Infrastructure and constructability

The Report identifies several implications for infrastructure design:

- Flood-prone areas require resilient design and scour protection
- Ground improvement (e.g. preloading, wick drains) may be required in soft or compressible soils
- Excavation in coal-bearing strata may require selective removal and treatment
- Utilities may require specialised trenching, dewatering, and corrosion protection

These factors will influence development cost, timing and feasibility across the Precinct and Teralba Southgate.

1.4 Data gaps and further investigations

The assessment identifies data limitations, including:

- Limited and dated geotechnical investigations
- Incomplete understanding of TSF material properties and depth
- Uncertainty in mine subsidence risks in details and condition of historic fill areas as well as groundwater behaviour.

To address these uncertainties, following rezoning, further site-specific investigations are required to:

- Confirm ground conditions and groundwater regimes
- Refine geotechnical risk mapping
- Support detailed design and future development approvals.

Development at later stages must be supported by intrusive investigations, modelling, and engineering validation.

In general, from a geotechnical perspective, the majority of the Precinct is considered to be suitable for the purposes of commercial and industrial use, with varying levels of localised mitigation measures based on geographically specific constraints.

Referral to Subsidence Advisory NSW (SA NSW) will be required to confirm site specific requirements and more detailed subsidence modelling may be required. It is expected that many of the site constraints can be managed through engineering intervention such as ground improvement and consideration of mine subsidence in structural design or ground stabilisation by means such as grouting of former workings.

2 Introduction

This Geotechnical Engineering Report (this Report) has been prepared by Aurecon on behalf of the Department Planning, Housing and Infrastructure (DPHI) to support the State-led rezoning of the Macquarie Coal Complex for Post Mining Land Use. The Macquarie Coal Complex Land Transformation Precinct (the Precinct) and Teralba Southgate form part of the NSW Government's investigation into future land use pathways for mining affected land.

This Report provides a desktop based geotechnical assessment to support the rezoning, planning and staged redevelopment of the Precinct and Teralba Southgate sub-precinct. It establishes an evidence base to identify, assess and manage geotechnical constraints, and to guide investigation, earthworks and foundation design requirements across in these areas.

The Report is intended to inform:

- State-led rezoning processes
- Structure planning and sub-precinct development strategies
- Infrastructure planning and delivery
- Future development applications and site-specific investigations.

The Precinct and Teralba Southgate are characterised by a highly variable ground environment, shaped by:

- Underground mining (West Wallsend Colliery)
- Open cut mining and backfilled voids
- Coal processing infrastructure and tailings deposition
- Progressive rehabilitation and landform recontouring

These legacy conditions result in a range of geotechnical challenges, including:

- Mine subsidence risk
- Variable and uncontrolled fill
- Soft and compressible alluvial soils
- Groundwater variability
- Stability and settlement considerations associated with the Tailings Storage Facility (TSF).

A strategic, risk-informed framework to manage constraints associated with the geotechnical conditions of the Precinct and Teralba Southgate should therefore be adopted.

2.1 Background

Open cut and underground mining operations have all ceased and current activities are focused on rehabilitation, closure planning, and long-term environmental management.

The Macquarie Coal Complex is a decommissioned mining operation previously operated by Glencore, which processed coal sourced from the nearby underground West Wallsend Colliery and the Westside open-cut mining operations. Mining activities at West Wallsend Colliery, Westside open-cut mining operation and Teralba Southgate respectively ceased in 2016, 2012 and 2001.

The Structure Plan has divided the Precinct and Teralba Southgate into three post mining land use land categories for conservation, development and recreation purposes, further divided into six sub-precinct areas as shown Figure 2-1 and listed in Table 2-1.

Table 2-1 Sub-precinct references

Precinct	Sub-Precinct	Reference
MCC	Cockle Creek	1
	Killingworth	2
	West Lake	3
	Barnsley West	5
	Environmentally constrained areas	6A, 6B, 6C, 6D, 6E and 6F
Not applicable	Teralba Southgate	-

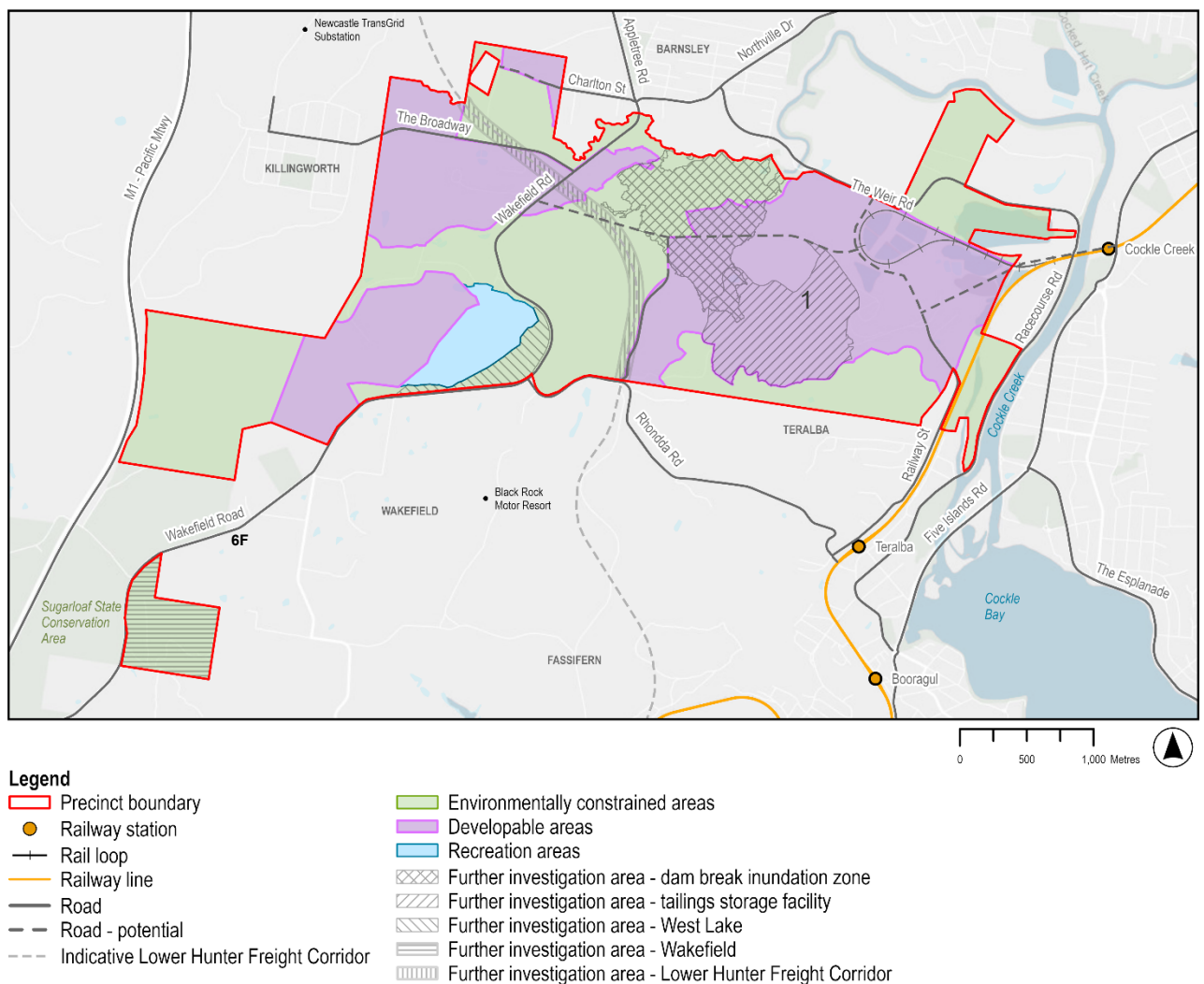


Figure 2-1 Macquarie Coal Complex Land Transformation Precinct Structure Plan (Master Plan, GHD 2026)

2.2 Purpose of this report

This Report provides an assessment of the geological and geotechnical setting across the Precinct and Teralba Southgate including site conditions, mining and rehabilitation history, engineered landforms, and subsidence constraints. It identifies key opportunities and constraints to inform the and Structure Plan and supports the development of Master Plan performance and staging triggers.

This Report has been prepared using publicly available information together with existing project data, including previous geotechnical investigations, borehole logs, laboratory testing records, mine rehabilitation documentation, and available topographic and subsidence information.

Specifically, the Report aims to:

- Provide a rezoning/master plan level geotechnical evidence base to verify developable land and inform land use compatibility, built form intensity and infrastructure feasibility
- Translate Enquiry by Design (EbD) identified constraints, including soils, steep terrain constraints and mine subsidence district considerations, into Master Plan mapping, performance criteria and staging triggers
- Provide a clear pathway for later-phase investigations (intrusive testing, stability modelling, settlement monitoring) without duplicating Development Application (DA) stage work.

2.3 Assumptions and limitations

This Report has been prepared to inform a Master Plan for post mining land use at the rezoning stage. The assessment is based on existing available information and does not constitute a DA level geotechnical design assessment.

The following limitations apply to this assessment:

- This assessment has been limited to publicly accessible data available to Aurecon and that provided by DPHI, and Glencore
- No geotechnical or hydrogeological modelling has been undertaken for the purpose of this assessment
- No detailed development analysis has been undertaken as part of geotechnical assessment
- No geotechnical investigations were carried out for this assessment.

The following items are specifically excluded from, or deferred beyond, the scope of this assessment:

- Comprehensive intrusive site investigation campaigns across the full Precinct and Teralba Southgate
- Detailed stability modelling, settlement predictions and foundation design
- DA stage geotechnical design certification and construction-phase verification
- Detailed data gaps identified for the geotechnical constraint assessment are further outlined in Section 5.2
- Mine subsidence is assumed to apply across the entire Precinct and Teralba Southgate, and is treated as a site-wide constraint rather than a differentiating factor between sub-precincts.

3 Desktop review

3.1 Site description and topography

The site is bounded by Cockle Creek to the north and east and the Pacific Motorway to the west. It is intersected by several Cockle Creek tributaries, including Diega Creek, and contains tailings dams and former open cut areas, as well as Black Swan Lake and West Lake. Historical mining activities have resulted in highly variable topography, with elevations ranging from less than RL 10 m along the Cockle Creek foreshore to approximately RL 75 m near the TSF in Cockle Creek sub-precinct. The site is predominantly bushland, with small water bodies and areas of rehabilitation. Numerous non-perennial creek lines are recorded across the Precinct and Teralba Southgate areas that drain the slopes. The Wakefield Road, local roads and private internal roads cross the Precinct and Teralba Southgate. It also includes sections of the Main Northern railway line between Cockle Creek and Teralba stations along the western boundary and Teralba Southgate sub-precinct.

Refer to Figure 3-1, Figure 3-2 and Figure 3-3 for structure plan site plan, topography and slope respectively. Higher-resolution versions of these figures are provided in Figures A1, A2 and A3 of **Appendix A – Figures**.

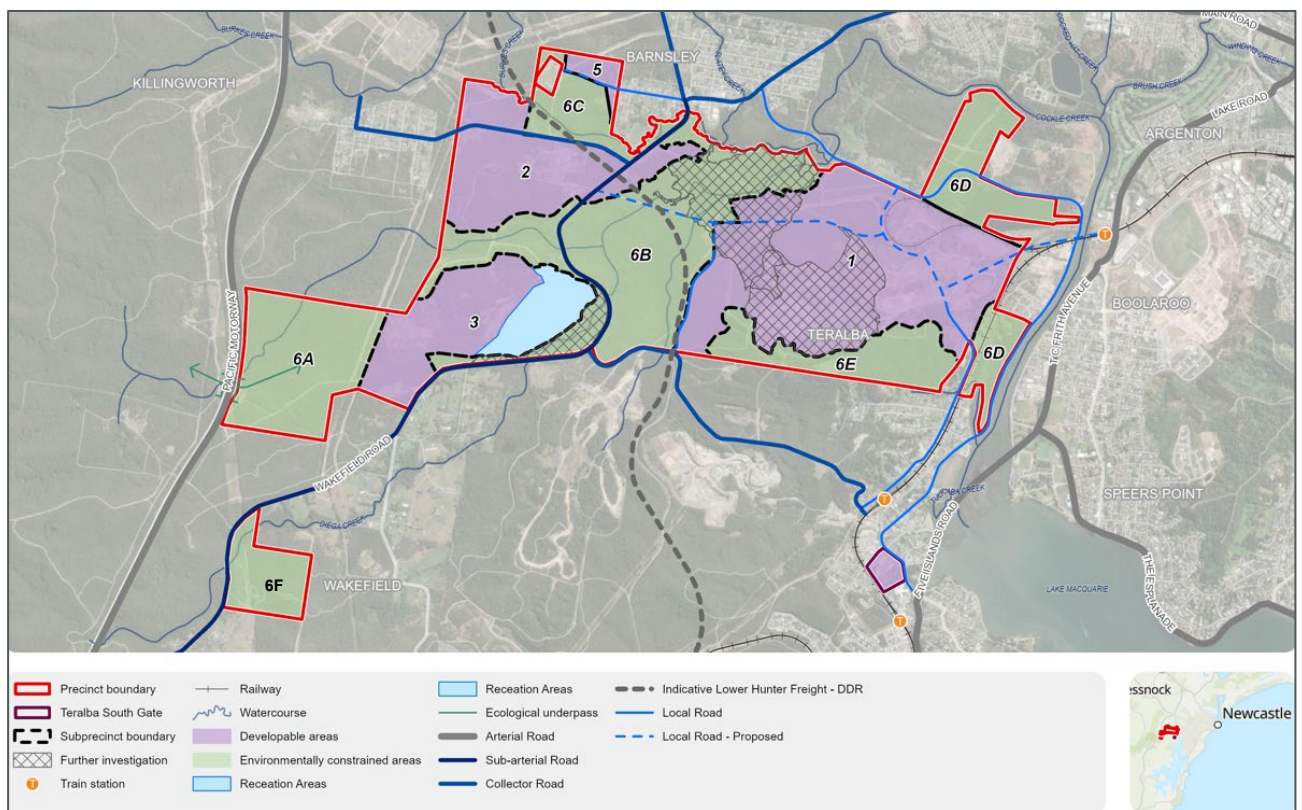


Figure 3-1 Structure Plan site plan and sub-precincts

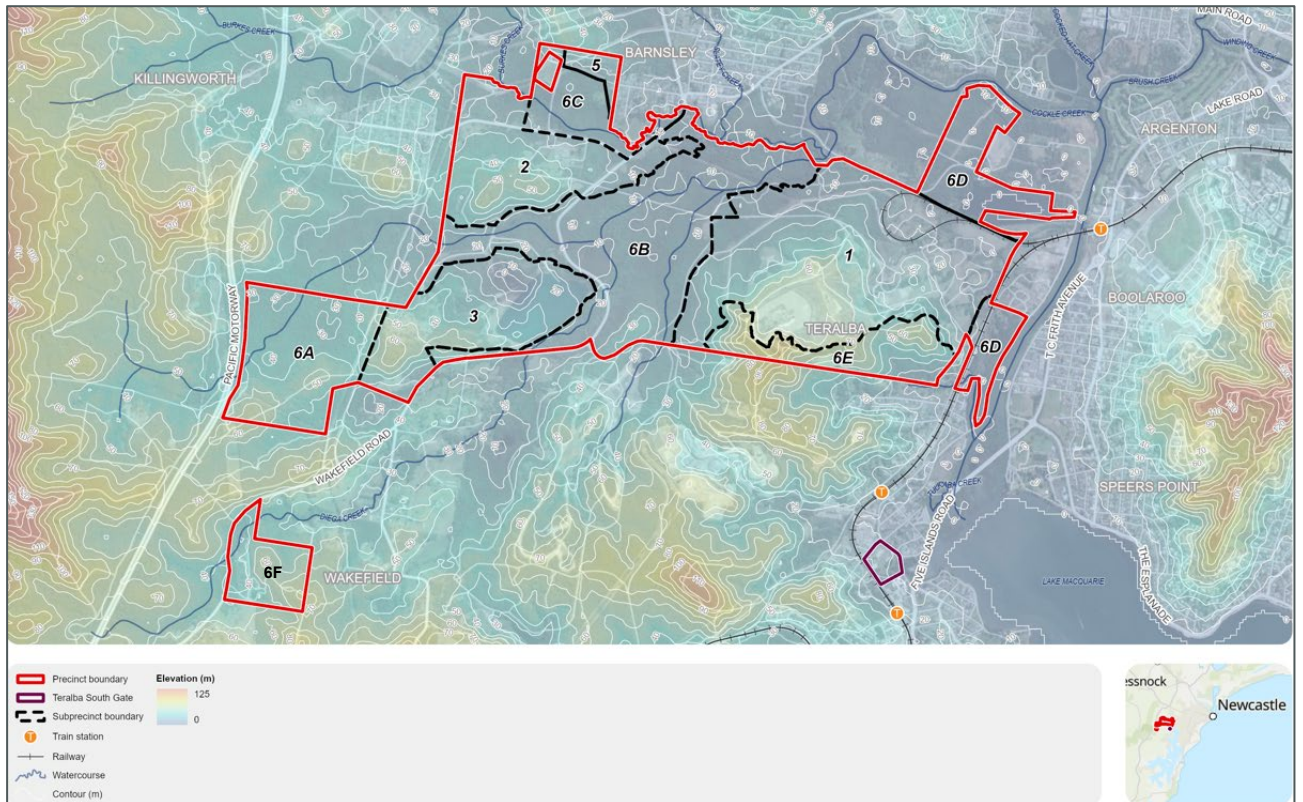


Figure 3-2 Topography across the Precinct and Teralba Southgate

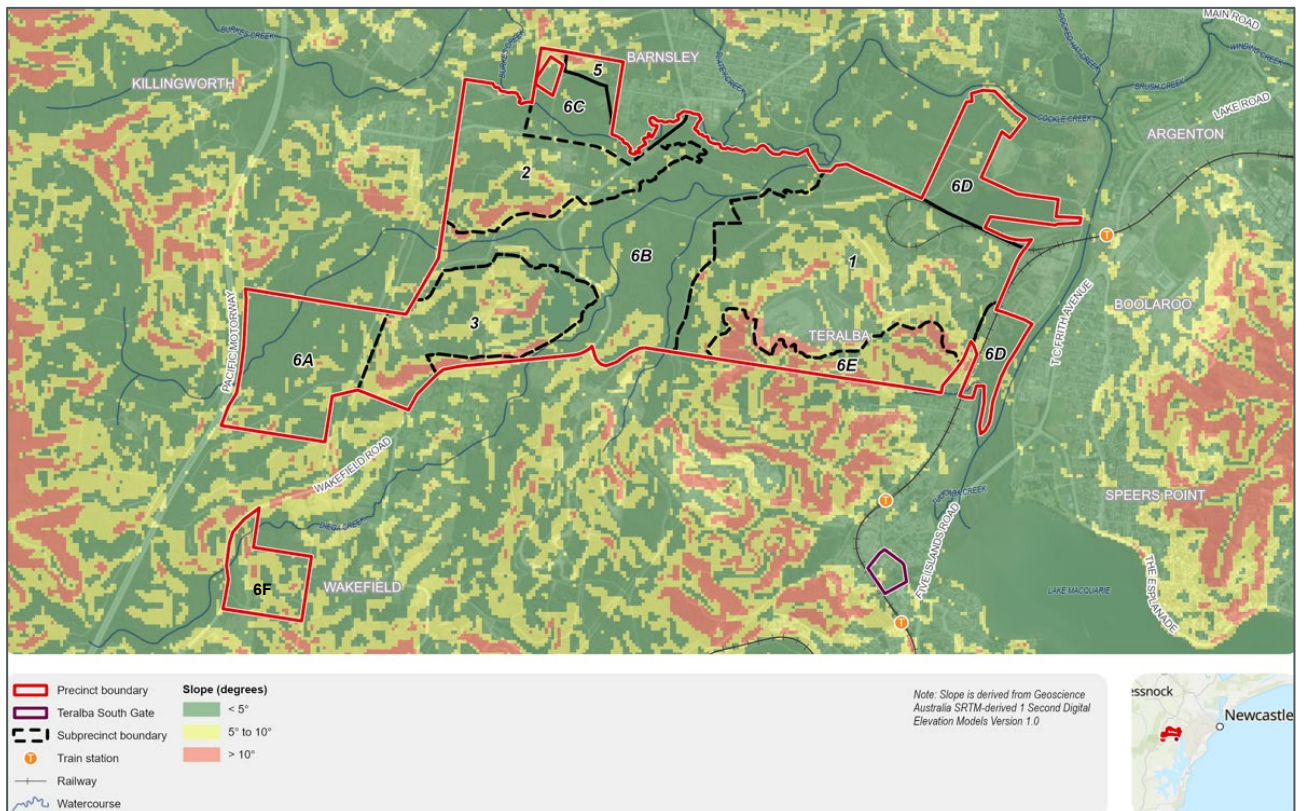


Figure 3-3 Slope across the Precinct and Teralba Southgate

Specific high-level descriptions and topography of each Structure Plan sub-precinct are further detailed in Table 3-1.

Table 3-1 Sub-precincts description and topography across the MCC site

Sub-precinct		Sub-precinct description	Topography	
No.	Name		Description	Typical slope range (degrees)
1	Cockle Creek	A mix of bushland and cleared areas, including a TSF landform currently subject to earthworks. Unnamed unsealed local roads surround the landform. Several watercourses from Cockle Creek cross the area.	Generally flat to undulating terrain, with slopes up to around 20 degrees, becoming steeper towards the TSF landform and to the south of the sub-precinct area. Elevations range from approximately RL 5 m AHD to RL 70 m AHD at the top of the landform.	5 - 10
2	Killingworth	Predominantly bushland with several unnamed dams and unsealed local roads. The West Wallsend Colliery compound is located in the centre of the area.	Generally flat to gently undulating terrain, with localised greater than 10 degrees to the south. Elevations range from approximately RL 10 m AHD to RL 60 m AHD, with higher and steeper ground in those areas.	< 5
3	West Lake	Predominantly bushland, with several discrete cleared areas, dams, and unnamed local roads throughout. The area is bounded by Wakefield Road to the south and Lake to the east, with the lake occupying approximately 40% of the site. Diega Creek traverses the western portion of the area, and several unnamed unsealed local roads are present near the lake.	Predominantly flat to undulating terrain, with slopes generally less than 10 degrees across most of the area, locally steepening to up to 20 degrees along Wakefield Road to the south, the southwestern portion of the site, and near West Lake to the east. Elevations range from approximately RL 5 m AHD at West Lake to RL 65 m AHD, with the highest and steepest ground occurring along the southern and southwestern margins of the area.	5 - 10
-	Teralba Southgate	Bounded by the railway to the west, Pitt Street to the north, and York Street to the south and east, and surrounded by a predominantly residential area. This area includes mining facilities and a dam, and is mostly cleared of vegetation.	Generally flat with slopes typically up to 5 degrees. Elevations range from approximately RL 5 m AHD to RL 15 m AHD.	< 5
5	Barnsley West	Bordered by an unnamed unsealed road extending south from Charlton Street. The area is partially cleared of bushland.	Flat to undulating with slopes up to 20 degrees in the western section. Elevations range from approximately RL 15 m AHD to RL 40 m AHD.	5 - 10
6A	Environmentally constrained areas	Predominantly bushland. Located west of the Pacific Highway, with several unnamed unsealed local roads visible throughout the area. Cockle Creek lies to the north, while a tributary of Diega Creek traverses the southern portion of the site.	Generally flat terrain with slopes typically up to 5 degrees, becoming locally steeper up to 10 degrees. Elevations range from approximately RL 30 m AHD to RL 50 m AHD.	< 5

Sub-precinct		Sub-precinct description	Topography	
No.	Name		Description	Typical slope range (degrees)
6B		Predominantly bushland, traversed by Wakefield Road and Private Mining Road, with several unnamed unsealed local roads to the east and north. Crossed by Cockle Creek, Diega Creek and tributaries, with associated floodplain areas.	Generally flat terrain with slopes typically up to 5 degrees, becoming locally steeper up to 10 degrees. Elevations range from approximately RL 30 m AHD to RL 50 m AHD.	< 5
6C		Predominantly bushland, bounded by The Broadway to the south and Wakefield Road to the east, and crossed by an unnamed unsealed road and Burkes Creek.	Generally flat terrain with slopes typically up to 5 degrees. Elevations range from approximately RL 10 m AHD to RL 15 m AHD.	< 5
6D		Predominantly bushland, crossed by The Weir Road to the south, with several unnamed unsealed local roads and multiple dams throughout the area, particularly in the north. Bordered by the railway to the west, Cockle Creek and Racecourse Road to the east, and earthworks activities to the north, with Black Swan Lake also located to the north. The area forms part of the Cockle Creek floodplain.	Generally flat terrain with slopes typically up to 5 degrees. Elevations range from approximately RL 0 m AHD to RL 15 m AHD, with most of the area lying below RL 5 m AHD.	< 5
6E		Predominantly bushland.	Generally undulating terrain with slopes typically up to 20 degrees. Elevations range from approximately RL 15 m AHD to RL 75 m AHD.	> 10
6F		Mostly bushland. Located west of Wakefield Road. Unnamed unsealed local roads are visible. Diega Creek runs to the north.	Generally flat to gently undulating terrain with slopes typically up to 10 degrees, becoming locally steeper. Elevations range from approximately RL 40 m AHD to RL 70 m AHD.	5 - 10

3.2 Historical land use

3.2.1 Disturbed terrain

Aerial imagery from 1954 onwards was reviewed, where available, to assess the progression of mining and surrounding land development and to identify potential areas of ground disturbance within the sub-precincts. The imagery indicates the early expansion of mining activities across the precinct, together with residential development in the surrounding localities of Killingworth to the west, Barnsley to the north, Wakefield and Teralba to the south, and Boolaroo to the southwest, as well as the development of the Pacific Motorway to the west.

The imagery was sourced from the NSW Government Historical Imagery Viewer for the years 1954, 1966, 1974, 1984, 1987, 1993, 1998 and 2011 (NSW Government Spatial Services, 2022), supplemented by Google Earth imagery (not shown in this report due to licensing constraints) between 2005 and 2026.

Key observations and visuals are summarised in Table 3-2 with a focus on areas that have been disturbed over time within the Precinct and Teralba Southgate. The most extensive disturbances are observed within the Killingworth, West Lake and Cockle Creek sub-precincts, with more localised disturbances identified within the Environmentally Constrained Area (6B) sub-precinct.

It is noted that the sub-precinct contours shown in the figures below are indicative only.

Table 3-2 Historical land use observations (1953 to 2026)

Observations and visuals	
Year 1954: ▪ The current alignment of Cockle Creek and tributaries, Wakefield Road and railway appear to have mainly remained unchanged since the aerial image was captured. ▪ The Precinct area is mostly a bushland, with tracks, with early residential developments in Teralba and Boolaroo areas visible. Between 1954 and 1966: ▪ Most of the existing unnamed unsealed tracks are visible in the aerial imagery. ▪ Localised areas of bushland have been cleared south of Wakefield Road and within Barnsley are associated with ongoing residential development. Between 1966 and 1974: ▪ Mining activities become clearly visible south of the Precinct, with more localised earthworks evident west of West Lake sub-precinct (Zoom-in No.1). ▪ Further bushland clearing has occurred south of Wakefield Road and within the Killingworth sub-precinct. Residential development has also continued to expand in Barnsley and the Boolaroo area.	

Observations and visuals

Between 1974 and 1984:

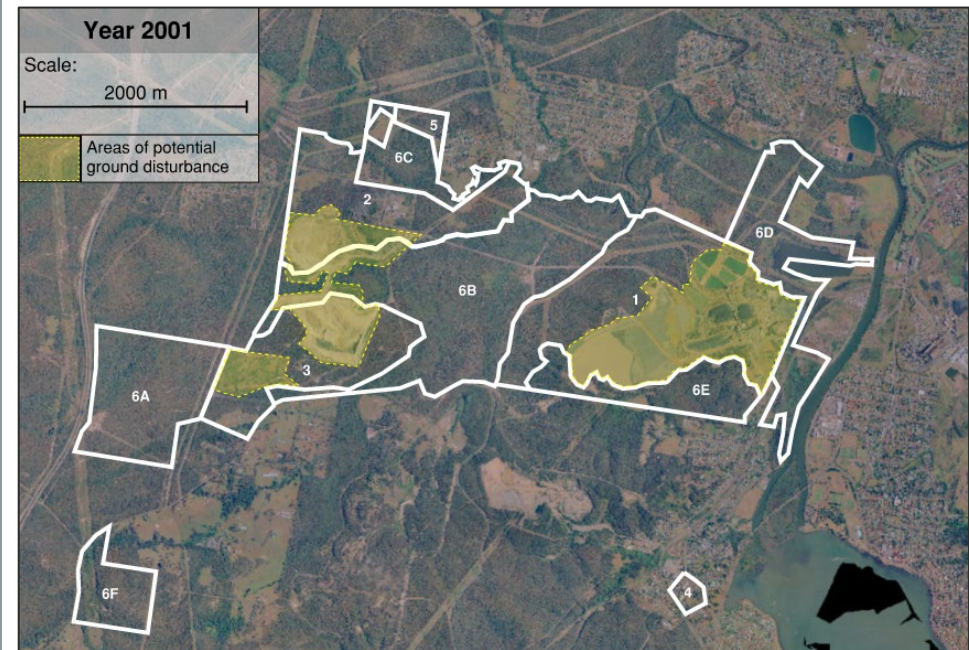
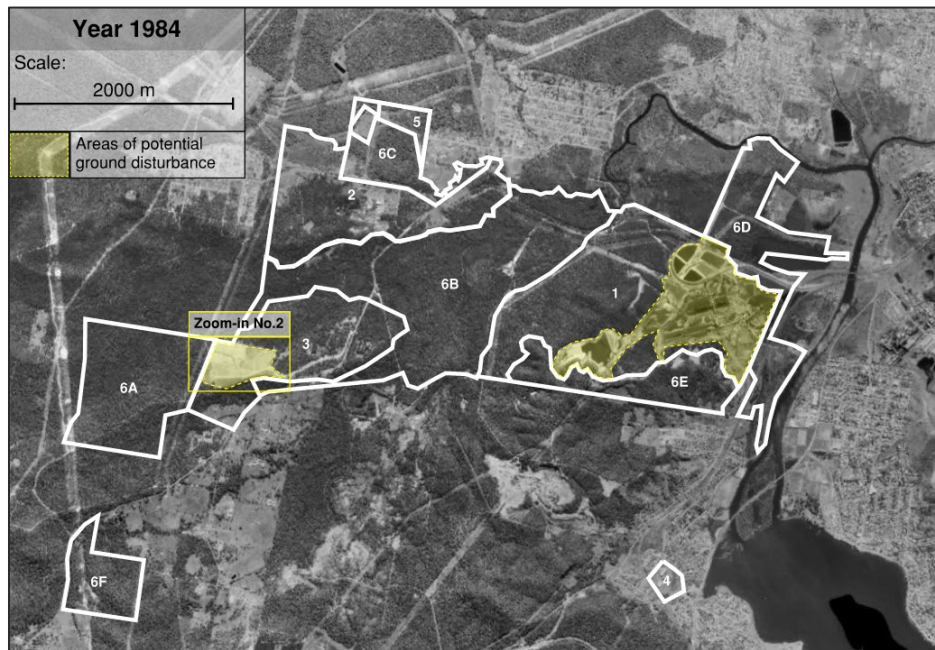
- The earthworks visible in the 1974 imagery within the West Lake sub-precinct appear to have been completed, with the affected area subsequently seemingly backfilled and levelled, suggesting the commencement of rehabilitation (Zoom-in No.2).
- Mining activities are now visible within the Cockle Creek sub-precinct, together with the associated rail loop and mining dams.

Between 1984 and 1993:

- Mining activities have expanded in Cockle Creek sub-precinct.
- Mining activities across the Killingworth and Environmentally Constrained Area (6B) sub-precincts are visible since 1984 and have progressively expanded across both sub-precincts between 1984 and 1993.
- Pacific Motorway is visible since 1993.

Between 1993 and 2001:

- Mining activities have expanded in Cockle Creek sub-precinct.
- Mining activities across the Killingworth and Environmentally Constrained Area (6B) sub-precincts have expanded across both sub-precincts, as well as in West Lake sub-precinct further south.



Observations and visuals

Between 2001 and 2005:

- Mining activities within the Killingworth and Environmentally Constrained Area (6B) appear to have ceased since 2001, with the previously exposed areas now largely replaced by grassed areas and trees.
- Mining activities within the West Lake sub-precinct have continued to progress southwards, extending to Wakefield Road, and across most of the area of the sub-precinct.
- Mining activities have locally expanded to the north in Cockle Creek sub-precinct.

Between 2005 and 2011:

- Mining activities within the northern section of the West Lake sub-precinct appear to have ceased around 2007, with previously disturbed areas progressively rehabilitated with grassed areas and trees.

Between 2011 and 2019:

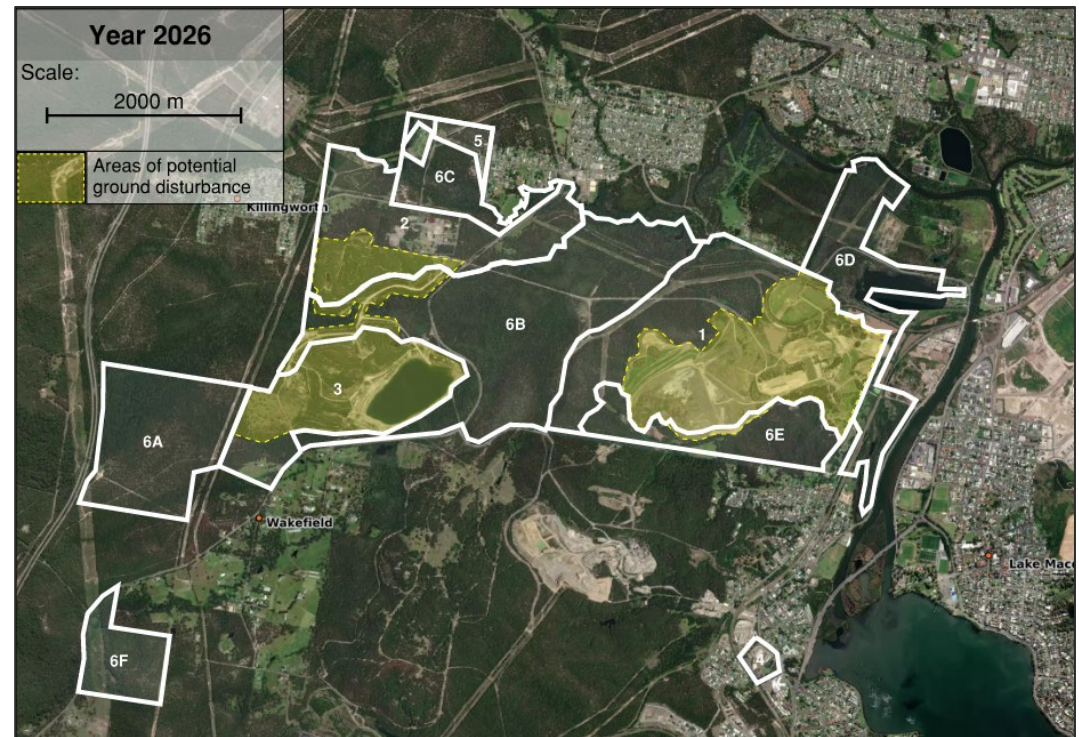
- The cessation of mining activities appears to have progressively extended westward from the active mining areas, with previously disturbed land increasingly replaced by grassed areas and trees within the West Lake sub-precinct, and the West Lake dam becomes visible from 2012 in the eastern section of the sub-precinct.
- Localised earthworks were undertaken near Wakefield Road within Environmentally Constrained Area (6B) in 2012, apparently replacing a former loop road associated with mining activities. The road is no longer visible in subsequent imagery, with the area appearing levelled and progressively revegetated with grassed areas by 2019.
- Similarly, localised disturbed areas surrounding the TSF appear to have been levelled and progressively replaced by grassed areas.

Between 2019 and 2026:

- Dams within the rail loop in the Cockle Creek sub-precinct appear to have been decommissioned and backfilled by end of 2019, with the area subsequently revegetated as grassed land.
- The areas around the TSF are also progressively replaced by grassed areas and trees, especially between 2019 and 2021. No major changes are observed between 2021 and 2026.

Notes:

- Historical aerial imagery from 1956 to 2001 was obtained from the NSW Government Historical Imagery Viewer (NSW Government Spatial Services, 2022).
 - Aerial imagery for 2026 was obtained from MinView.
- All contours are indicatives.



3.2.2 TSF landform

The central portion of the Cockle Creek sub-precinct comprises the TSF landform. The TSF was used for the disposal of fine coal processing residues, which were hydraulically placed and retained by earth fill embankments. Reference to the West Wallsend Tailings Dam Report indicates this TSF is currently undergoing rehabilitation works, including capping, regrading and revegetation. The landform consists of tailings with associated mine waste materials and should therefore be considered as an area of variable anthropogenic ground conditions and topography. The area is still under final construction with placement of fill and final capping in progress at the time of the preparation of this report. The TSF is a declared dam and the critical failure section is assessed as being the northern embankment, primarily due to the greater thickness of the eastern embankment. The eastern embankment is sufficiently large that this is the area where capping materials are being sourced for the final landform.

Reference to Figure 3-4 indicates that the final landform average slopes are designed to be 1V:8H for the area of borrow (or 7 degrees), 1V:100 H for the upper tailings capping area (or 0.5 degrees) and 1V:4H for the contour bank stripping area (or 14 degrees) following rehabilitation works (WSP, 2025). Surface drainage design has also been included in the final landform design.

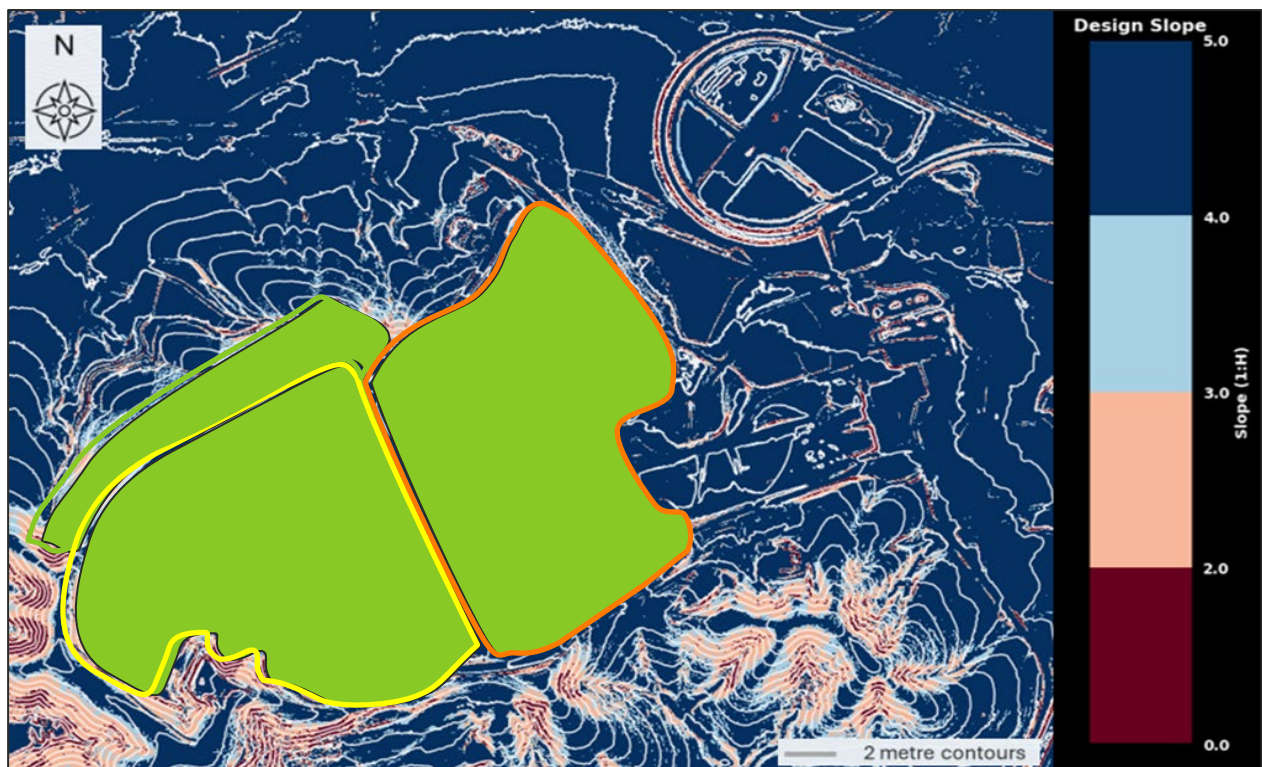


Figure 3-4 Visual representation of the final planned landform slopes, with the area of borrow in orange, upper tailings capping area in yellow, and contour bank stripping area in red (WSP, 2025)

3.3 Regional geology and geomorphology

Reference to the Seamless Geology layer via MinView (Colquhoun et al., 2019) indicates that the Precinct and Teralba Southgate are predominantly underlain by alluvial deposits (Q_{ab} , Q_{af} , Q_{ap} , Q_{at} and Q_{av}) along Cockle Creek and its tributaries in low-lying areas, potentially including soft to firm clays, silts and loose cohesionless materials.

The rock units of the Newcastle Coal Measures (P_{nb} , P_{nmc} and P_{nme}) are expected in relatively higher elevation areas. The Narrabeen Group (T_{nc}) and anthropogenic deposits are also present locally to the southwest and northeast of the site, respectively, with anthropogenic deposits occurring west of Cockle Creek around Black Swan Lake and its surrounds.

In structural geological terms, the area lies close to the northern margin of the Sydney Basin and at the edge of the relatively undisturbed central region and closer to the more disturbed and deformed northern marginal region of the basin. The Precinct and Teralba Southgate are well mapped due to the local mining history and a number of north-west to south-east trending structural features are recorded including:

- The Macquarie Syncline, which extends approximately 50km from Wyong to Buttai with the axial trace running less than 1 km to the west of the TSF site
- The Lochinvar Anticline, located to the northwest of the site
- Unnamed thrust fault mapped to the south of the rail loop
- Faulting, including thrust and normal faults, predominantly trending NW–SE
- Igneous intrusions, comprising a dominant NW–SE dyke trending series aligned with the Macquarie Syncline to the west, and an additional E–W trending dyke set to the north of the site
- Surface exposures of named coal seams (e.g. Fassifern coal seam and Great Northern coal seam).

These structural features only affect the Newcastle Coal Measures rock units in the Precinct and Teralba Southgate and will be more relevant once certain development types become known. Especially where new development will be above former mine workings, and the structural details will affect settlement predictions.

Refer to Figure 3-5 for the geological setting. A higher-resolution version of this figure is provided in Figure A4 of **Appendix A – Figures**.

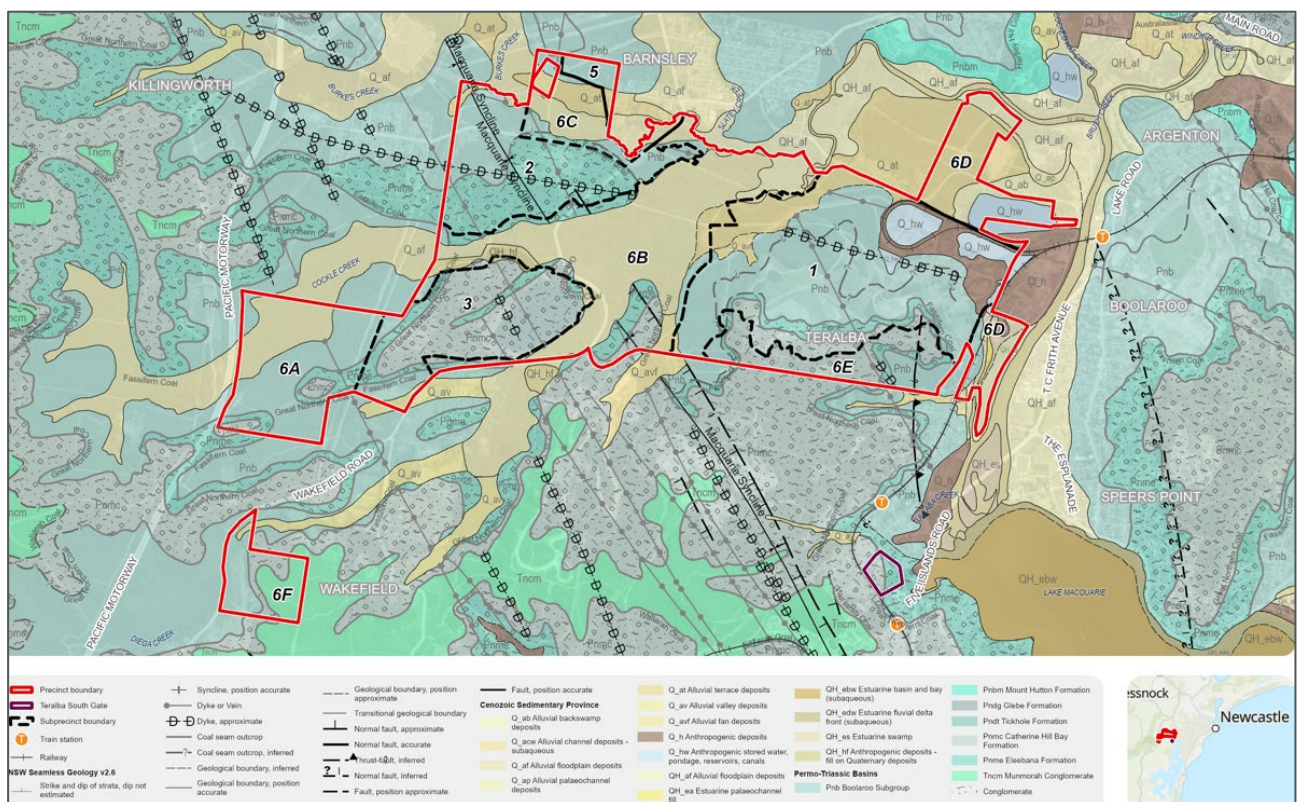


Figure 3-5 Geological setting

Table 3-3 provides a detailed description of the lithological units encountered across the MCC site and Teralba Southgate, together with their characteristics and typical limitations.

Table 3-3 Regional geology – stratigraphy, lithology and unit characteristics and typical limitations

Stratigraphy	Category	Name	Lithology	Unit	Characteristics and geotechnical considerations
Quaternary sediments	Anthropogenic deposits	Anthropogenic stored water, pondage, reservoirs, canals	Thinly laminated muds and silts with humic to biogenic debris (as bottom sediment to the overlying stored waters).	Q_hw	Anthropogenic water bodies may be associated with elevated groundwater levels, soft saturated soils, seepage and excavation/slope stability challenges.
		Anthropogenic deposits	Anthropocene deposits varying from large man-made clasts (concrete blocks to building demolition rubble) to quarried natural boulders, with interstitial sand-sized to clay matrix.	Q_h	Highly variable fill composition and compaction, resulting in increased ground condition uncertainty.
		Anthropogenic deposits - fill on Quaternary deposits	Land surface raised >1m above natural level by placement of fill on undifferentiated Quaternary deposits over an extensive area.	QH_hf	Shallow foundations may experience excessive or differential settlement and could require preloading. Deep foundations may need increased depths, with potential excavation challenges from obstructions, uncontrolled fill, and other unpredictable ground conditions.
	Alluvium	Alluvial backswamp deposits	Organic-rich mud, peat, silt, clay.	Q_ab	Highly compressible organic matter. These soils can cause significant total and differential settlements that can be challenging and costly to control.
		Alluvial terrace deposits	Silt, clay, (fluvially-deposited) fine- to medium-grained quartz-lithic sand, polymictic gravel.	Q_at	Highly reactive soils. Movements associated with moisture content require additional considerations in foundation design.
		Alluvial valley deposits	Silt, clay, (fluvially deposited) lithic to quartz-lithic sand, gravel.	Q_av	High water table. water management controls may be more expensive or impractical. This can also lead to potential impacts on adjacent areas.
		Alluvial fan deposits	Fluvially-deposited quartz-lithic sand, silt, gravel, clay.	Q_avf	Seasonal waterlogging: changes in strength of the soils as well as volume with moisture levels can lead to defects and failures if not adequately managed in design.
		Alluvial floodplain deposits	Silt, very fine- to medium-grained lithic to quartz-rich sand, clay.	Q_af	
		Alluvial palaeochannel deposits	Poorly sorted gravels, clayey sand and variably magnetic pisoliths.	Q_ap	
		Alluvial floodplain deposits	Silt, very fine- to medium-grained lithic to quartz-rich sand, clay.	QH_af	
		Estuarine swamp	Organic-rich mud, peat, clay, silt, very fine- to fine-grained sand (marine-deposited), fine- to medium-grained sand (fluvially deposited).	QH_es	
		Estuarine palaeochannel fill	Organic mud, peat, clay, silt, fine- to medium-grained lithic-carbonate-quartz sand (marine-deposited).	QH_ea	

Stratigraphy	Category	Name	Lithology	Unit	Characteristics and geotechnical considerations
Narrabeen Group	Rock	Munmorah Conglomerate	Medium- to coarse-grained lithic to quartz-lithic sandstone, granule to pebble polymictic conglomerate; minor siltstone and white claystone, thin lenticular coal seams including the Wallarah coal seam.	Tncm	Potentially highly reactive residual soils. Movements associated with moisture content require additional considerations in foundation design. Rock excavation may be required with possible constraints for earthworks, and potential limitations on material re-use due to coal.
Newcastle Coal Measures		Boolaroo Subgroup	Sandstone, conglomerate, siltstone, claystone, coal and tuff.	Pnb	Coal outcrops are subject to spontaneous combustion; therefore, where coal seams are encountered in cut areas, excavate and backfill or blend coal for reuse in foundation treatment.
		Catherine Hill Bay Formation	Quartz-lithic pebble polymictic conglomerate, quartz-lithic sandstone, shale, carbonaceous shale, coal and tuff.	Pnmc	
		Eleebana Formation	Conglomerate, sandstone, siltstone, claystone, coal and tuff.	Pnme	

The expected geological units and their indicative extents within each sub-precinct are presented in Table 3-4. These extents also incorporate areas of historical ground disturbance identified in Section 3.2.1 which may potentially contain anthropogenic deposits.

Table 3-4 Expected geology and areas of disturbed terrain across the sub-precincts

Sub-precinct		Expected geology and disturbed terrain (% of sub-precinct area)		
No.	Name	Anthropogenic deposits / disturbed terrain	Alluvium	Rock
1	Cockle Creek	65 % ^{[1], [2]}	5 %	30 %
2	Killingworth	30 % ^[1]	10 %	60 %
3	West Lake	85 % ^[1]	5 %	10 %
-	Teralba Southgate	0 %	0 %	100 %
5	Barnsley West	0 %	25 %	75 %
6A	Environmentally constrained areas	0 %	30 %	70 %
6B		10 %	70 %	15 %
6C		0 %	75 %	25 %
6D		25 %	70 %	5 %
6E		0 %	0 %	100 %
6F		0 %	5 %	95 %

Sub-precinct		Expected geology and disturbed terrain (% of sub-precinct area)		
No.	Name	Anthropogenic deposits / disturbed terrain	Alluvium	Rock

Notes:

[1] Reference to Section 4.2.2 of the Master Plan (Department of Planning, Housing and Infrastructure, 2026) and Section 3.2.1 indicate that portions of Cockle Creek, Environmentally Constrained Area No.6B and West Lake sub-precincts were impacted by former mining activities visible from 1974. These sub-precincts are expected to include disturbed terrain and fill materials that are not currently captured by the Seamless Geology dataset, but have been included as part of this assessment.

[2] The Seamless Geology dataset indicates Anthropogenic stored water, pondage, reservoirs and canals (unit Q_hw) within the rail loop area north of sub-precinct; however, this does not reflect current conditions, where fill material is present rather than surface water (refer to Sections 3.2.1 and 3.7).

3.4 Depth of regolith

Reference to the Soil and Landscape Grid National Soil Attribute Maps – Soil Depth, published by CSIRO on the MinView website indicates generally shallow surface soils (0 - 3 m) across higher elevations, including the TSF area, where they predominantly comprise residual soils derived from underlying rock units. Regolith thickness increases towards low-lying areas, particularly around Cockle Creek, Diega Creek and associated tributaries, where depths locally reach 10 - 20 m and are expected to comprise alluvium over residual soils. This trend is reflected across the Structure Plan sub-precincts, with thinner residual soils on higher ground and thicker alluvial-residual sequences in drainage lines. It is noted regolith CSIRO defines regolith as all soil, sediment and extremely weathered material overlying competent bedrock. Regolith depth therefore indicates depth to fresh rock only and does not reflect material consistency or relative density.

Refer to Figure 3-6 for the depth of regolith across the site. A higher-resolution version of this figure is provided in in Figure A10 of **Appendix A – Figures**.

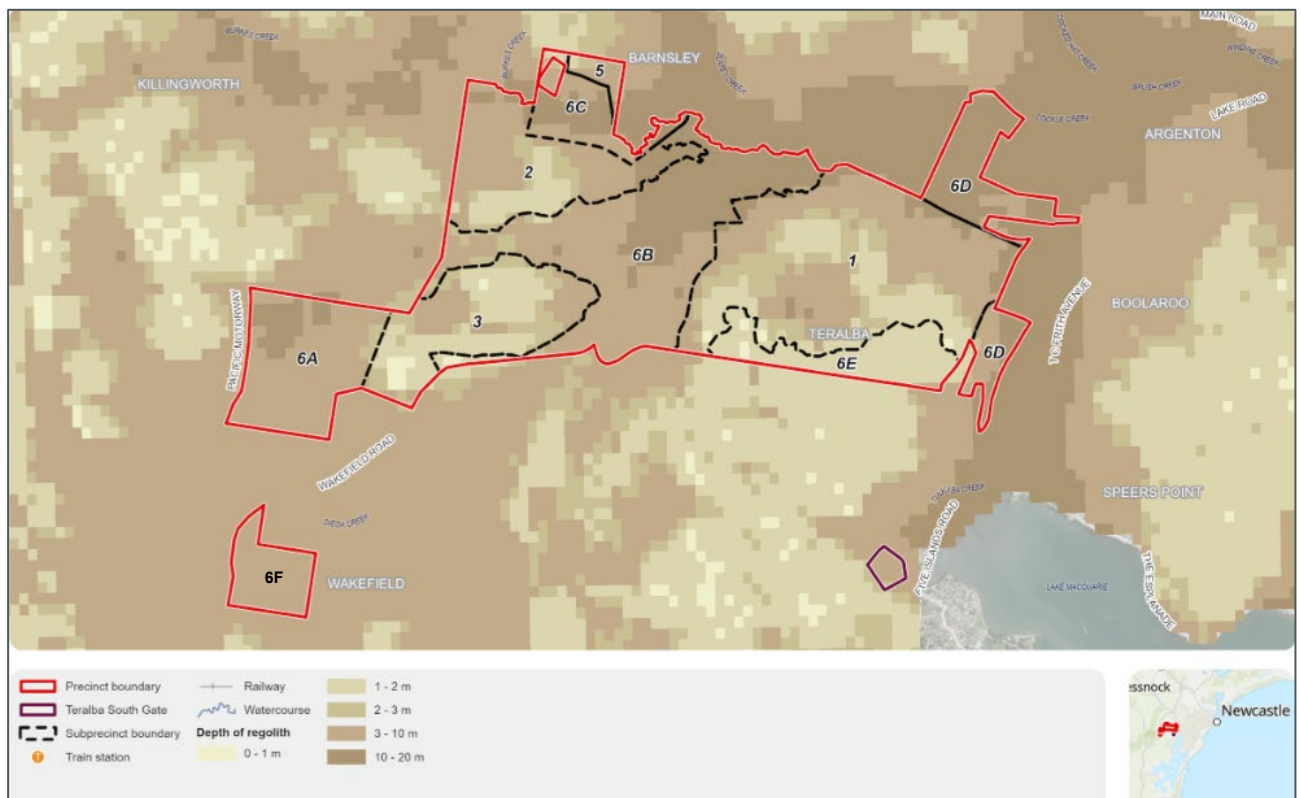


Figure 3-6 Depth of regolith

Depth of regolith and expected geological origin of surface soils across the sub-precincts are further summarised in Table 3-5.

Table 3-5 Depth of regolith and anticipated soil profiles across the sub-precincts

Sub-precinct		Typical depth of base of surface soils	Expected geological origin of soil	Comment
No.	Name			
1	Cockle Creek	1 - 2 m locally 3 - 10 m (north)	Residual soils, locally fill and alluvium	Depth to competent rock is typically 3 - 10 m within the TSF area and northern part of the sub-precinct, reducing to around 1 - 2 m elsewhere.
2	Killingworth	3 - 10 m locally 1 - 2 m (south) locally up to 20 m (east)	Residual soils, locally alluvium	Surface soil depths are typically 3 - 10 m, becoming shallower (1 - 2 m) in the southern higher-elevation areas and deeper towards Burkes Creek.

Sub-precinct		Typical depth of base of surface soils	Expected geological origin of soil	Comment
No.	Name			
3	West Lake	1 - 2 m locally 3 - 10 m	Residual soils, locally alluvium	Soil depths increase towards Diega Creek, where deeper alluvial deposits may be present in the eastern part of the sub-precinct.
-	Teralba Southgate	3 - 10 m	Residual soils	-
5	Barnsley West	3 - 10 m	Residual soils, alluvium	-
6A	Environmentally constrained areas	3 - 10 m (locally up to 20 m)	Residual soils, locally alluvium	Soil depths increase towards Cockle Creek, where deeper alluvial deposits may be present.
6B		3 - 10 m (locally up to 20 m)	Alluvium, locally residual soils and fill	Deep alluvial deposits are expected towards and around Cockle Creek.
6C		10 - 20 m (locally 3 - 10 m)	Alluvium, locally residual soils	Deep alluvial deposits are expected towards and around Burkes Creek.
6D		10 - 20 m (locally 3 - 10 m)	Fill, alluvium, locally residual soils	Deep alluvial deposits are expected towards and around Cockle Creek, with fill materials present locally in the north near the rail.
6E		1 - 2 m	Residual soils	-
6F		3 - 10 m	Residual soils	-

3.5 Seismicity

In accordance with the Australian Standard AS 1170.4-2007, the site has a hazard design factor (Z) of 0.11. The following sub-soil classes are likely to be encountered:

- Class Ce – Shallow soil site (likely across most of the sub-precincts)
- Class De – Deep or soft soil site (potentially towards Cockle Creek and Diega Creek within sub-precinct No.6).

Further investigation is required to confirm the indicative sub-soil classification. The hazard factor indicates that the MCC site is located within a low seismicity risk area.

3.6 Slope stability geo zones

Lake Macquarie City Council host a web portal “*Development in geotechnical areas*” that provides zoning against 6 geotechnical risk categories based on slope and ground conditions criteria, including the presence of coal, claystone and degree of induration of quaternary sediments, as outlined in Table 3-6. The geology appears to be derived from the NSW Seamless Geology dataset; however, the source and date of the slope/topographic data are not specified. As the mapping was developed in 2019, the slope stability geo zone classifications may not reflect areas where significant earthworks have since modified landforms, including parts of the West Lake and TSF areas and should be treated as indicative in these areas.

Refer to Figure 3-7 for Lake Macquarie slope stability geo zone zoning. A higher-resolution version of this figure is provided in Figure A11 of **Appendix A – Figures**.

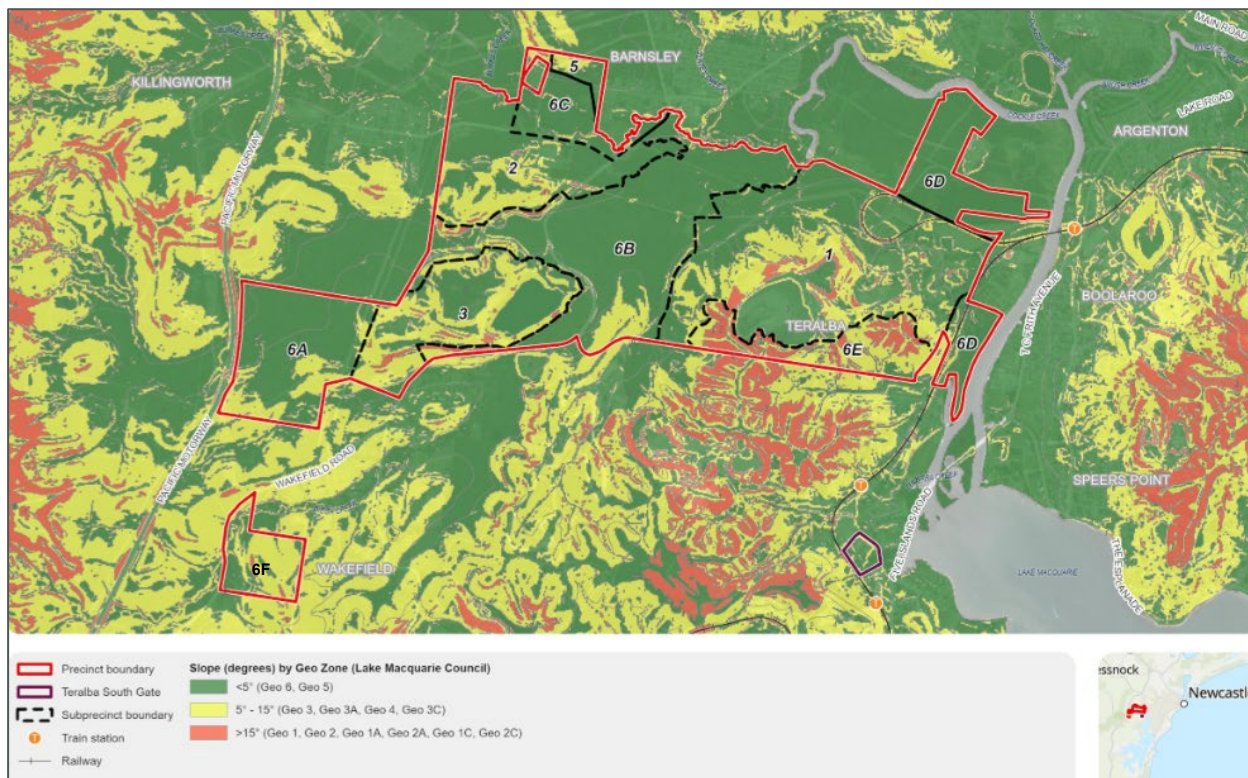


Figure 3-7 Lake Macquarie Council geo zones

Table 3-6 Slope stability geo zones

Slope	Stratigraphy					
	Newcastle Coal Measures		Narrabeen Group		Quaternary Sediments	
	With coal or tuff	Without coal or tuff	With Claystone-shale	Without Claystone-shale	Not indurated	Indurated
≥ 0° to <5°	Geo 5	Geo 6	Geo 6	Geo 6	Geo 6	Geo 6
≥ 5° to <15°	Geo 3	Geo 4	Geo 3A	Geo 6	Geo 3C	Geo 6
≥ 15°	Geo 1	Geo 2	Geo 1A	Geo 2A	Geo 1C	Geo 2C

Note: The severity of the geo zones takes the following order i.e. 1 being the most severe and 6 being the least severe:
 1. Geo 1, Geo 1A, Geo 1C
 2. Geo 2, Geo 2A, Geo 2C
 3. Geo 3, Geo 3A, Geo 3C
 4. Geo 4
 5. Geo 5
 6. Geo 6

Source: Lake Macquarie City website, Geotechnical Slope Stability Guidelines, <https://www.lakemac.com.au/Development/Planning-controls-and-development-guidelines/Development-in-geotechnical-areas>

Based on this zoning, the Precinct and Teralba Southgate are predominantly characterised by low to intermediate landslide risk, with Geo 5 and Geo 6 classifications generally dominant and associated with low-gradient terrain (<5°) and Quaternary sediments (indurated and non-indurated). Intermediate risk areas are associated with Geo 3, typically occurring on steeper slopes and/or in areas underlain by rock with coal or tuff.

Localised areas of higher relative slope stability risk occur where Geo 1 and Geo 2 are mapped, generally in steeper terrain underlain by rock with coal or tuff. These areas are mostly located in sub-precincts No.1 and 6E along the TSF landform slopes, and in the sub-precinct No.3 towards Wakefield Road.

The potential for landslide risk across the sub-precincts, based on Lake Macquarie zoning criteria, is further summarised in Table 3-7.

Table 3-7 Geo zones across the Structure Plan sub-precincts

Sub-precinct		Geo zone(s) ranked by area (largest to smallest)	Potential risk of landslide
No.	Name		
1	Cockle Creek	Geo 5, Geo 3, Geo 1, Geo 2	Generally low to intermediate. Locally high on the TSF slopes.
2	Killingworth	Geo 5, Geo 3	Generally low to intermediate.
3	West Lake	Geo 3, Geo 5, locally Geo 1 and Geo 6	Generally intermediate. Locally high in the south of the sub-precinct across Killingworth (9232kia) soil landscape.
-	Teralba Southgate	Geo 5	Generally low.
5	Barnsley West	Geo 5, Geo 3, Geo 6	Generally low to intermediate.
6A	Environmentally constrained areas	Geo 5, Geo 6, Geo 3	Generally low to intermediate. Intermediate in the south of the sub-precinct.
6B		Geo 6, Geo 5, locally Geo 3	Generally low
6C		Geo 6, Geo 5	Generally low.
6D		Geo 6	Generally low.
6E		Geo 3, Geo 2, Geo 1	Generally intermediate to high.
6F		Geo 3, Geo 5	Generally intermediate.

3.7 Site visit observations

A site inspection was undertaken on 6 May 2026 with representatives from the Department of Planning, Housing and Infrastructure (DPHI). Representatives from Glencore were also in attendance and provided an overview of the site, including current conditions and ongoing activities within the areas visited.

Given the scale of the project area, access across the site was primarily facilitated by vehicle, with targeted stops at key locations. Areas visited included the former rail loop, the TSF, and the open cut lake. Site access was limited in some locations due to ongoing operations and safety considerations associated with both site activities and the number of attendees.

Photography was not permitted during the site visit.

3.7.1 Rail loop

This area has had significant regrowth of vegetation and as such the landscape was not clearly visible. The rail and ballast were found and are now becoming overgrown and the rail ballast fouled as would be expected given the age.

3.7.2 Tailing Storage facility

The northern slope of the dam wall was driven via existing access roads and vegetation is established across the downstream face, including some larger individual plants.

The top of the facility was active with plant movements completing the cut and fill operation to cap the storage to final landform. Areas of cutting in to the original landscape were visible at the western edge of the storage. Glencore advised that coal ash for fill material was being provided from Eraring power station to fill up to the underside of the final landform capping layer.

3.7.3 Open cut lake site

This site is well revegetated and the lake apparently stocked with fish. The uppermost portion of the original open cut slopes are still visible above the lake edge on the western side. Glencore Staff mentioned ongoing water testing and that the bathymetry of the open cut was partly backfilled prior to allowing water levels to rise. Some areas of fill material could be seen around the lake edge and in the access road material including limited man-made material.

3.8 Soil landscapes

According to the eSPADE 2.2 tool from the NSW Department of Planning, Industry and Environment (Department of Planning and Environment, 2024), the Precinct and Teralba Southgate include various soil landscapes, predominantly Killingworth (9232ki) (~ 34 %) in higher areas and Cockle Creek (9232cc) (~ 35 %) along Cockle Creek and its tributaries in low-lying areas. This is followed by Doyalson (9232do) (~ 11 %), Disturbed Terrain (9232xx) (~ 10 %), and more localised occurrences of Warners Bay (9232wa), Gateshead (9232ga), and Stockington (9232sn) (each ≤ 5 %).

Upland areas (Killingworth soil landscape) are constrained by erosion susceptibility, steep slope, and mine subsidence risk. Alluvial areas (Cockle Creek) are constrained by flooding, waterlogging, elevated water tables, run-on effects, and dispersive soils. Disturbed Terrain presents the greatest uncertainty due to highly variable ground conditions, potential uncontrolled fill, and thus severe geotechnical constraints (e.g. low bearing capacity, compressible or loose materials, potential buried obstructions and voids, etc.). The remaining soil landscapes contribute to localised constraints including steeper slopes, shallow soils, and variable foundation conditions.

Refer to Figure 3-8 for the soil landscapes across the site. A higher-resolution version of this figure is provided in Figure A5 of **Appendix A – Figures**.

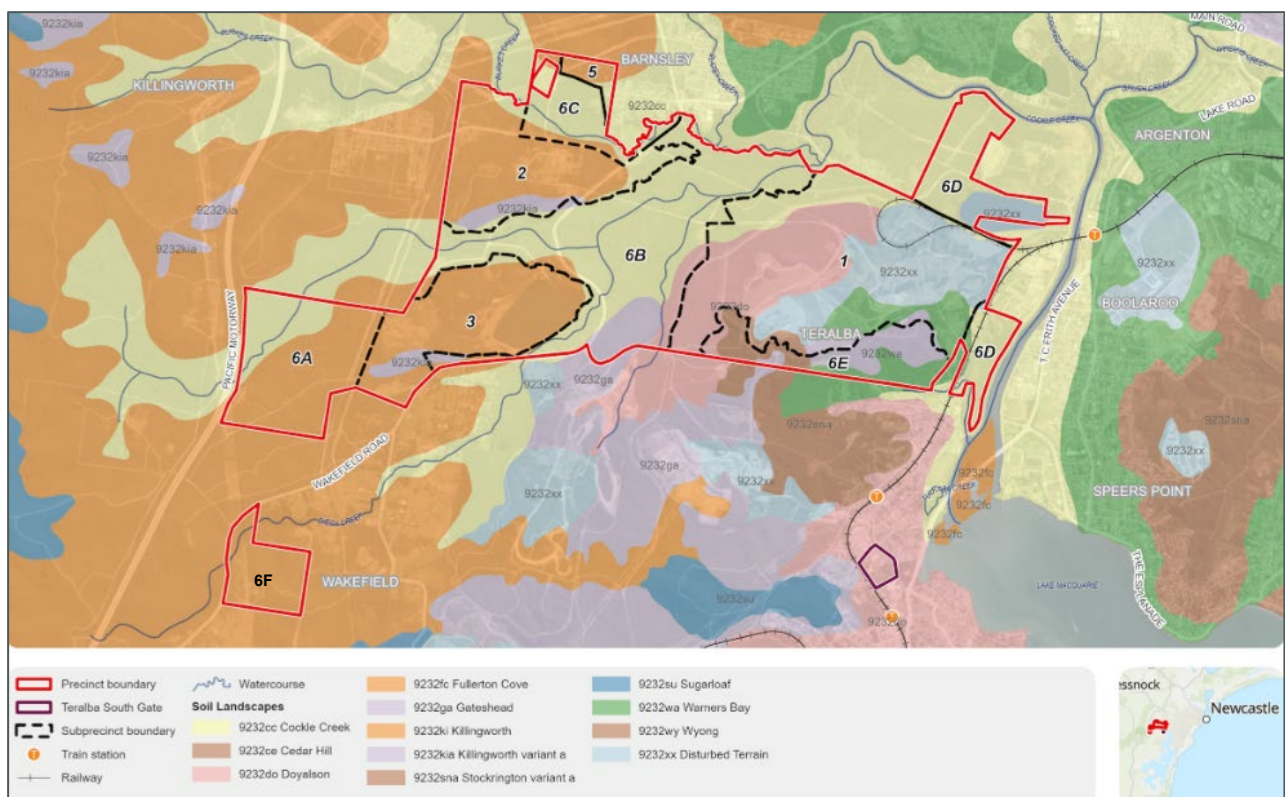


Figure 3-8 Soil landscapes

Detailed characteristics and typical limitations of the mapped soil landscapes are summarised in Table 3-8.

Table 3-8 Soil landscapes - topography, soil characteristics and typical limitations

Soil landscape	Topography	Soil characteristics	Limitations
Killingworth (9232ki)	The Killingworth soil landscape is an erosional landscape comprising undulating to rolling and low hills developed on the Newcastle Coal Measures of the Awaba Hills region. Elevation ranges from 50 to 160 m, with local relief of 30 to 100 m and slopes of 3 -20%. The area is predominantly covered by uncleared tall open forest. The landscape variant, Killingworth (9232kia), comprises steeper slopes (20-25%).	Soils are generally shallow (<0.6 m) to moderately deep (<1.5 m), typically well to imperfectly drained. Profiles commonly comprise loamy to sandy loam topsoils overlying clay loam to clay subsoils. Soil types include Yellow Podzolic soils and Soloths on crests and hillslopes, with Gleyed variants in lower or poorly drained areas. Shallow structured and bleached loams, as well as lithosols, occur locally on crests.	Soil erodibility is generally high. Other limitations include shallow soils (localised), seasonal waterlogging, and the presence of sodic/dispersible soils with low wet strength. Soils are typically strongly acidic with low fertility, with localised foundation hazard (high shrink swell subsoils) and subsidence hazard (Mine Subsidence District). Overall limitations for urban development are moderate. The landscape variant, Killingworth (9232kia), comprises steeper slopes (20–25%), with associated high limitations for urban development.
Cockle Creek (9232cc)	Alluvial flats, alluvial fans, and drainage plains, with slopes of 0–2% on floodplains and <5% on fans and plains. Elevation ranges from <1 to 50 m with minimal relief (<1 m). Landforms include moderately broad (<500 m) alluvial flats and broad (up to 1,000 m) deltaic deposits along lower Cockle Creek, with localised terraces, levees, and point bars (relief up to 3 m), and alluvial fans (<500 m) in upper reaches and tributaries.	Soils are generally deep (>2 m). Imperfectly to poorly drained Yellow Soloths and Yellow Podzolic Soils occur on floodplains. Moderately well to poorly drained Yellow Earths and Grey Earths occur on delta and fan deposits, with imperfectly to well-drained Yellow Podzolic Soils also present.	Flood hazard and high water erosion hazard, with localised areas of permanently high watertables and periodic to permanent waterlogging. High run-on conditions are present, with acid, infertile, sodic/dispersible soils of low wet strength.
Doyalson (9232do)	The Doyalson soil landscape is characterised by generally undulating rises with local relief of up to 30 m and gentle slope gradients.	Variable profiles reflecting sandstone/conglomerate and local finer-grained lenses. Typically, up to 0.1 m of loamy sand overlies 0.1–0.3 m of hardsetting sandy clay, underlain by 0.3–0.6 m of sandy clay loam grading to pale grey or strongly pedal clay. Total soil depth generally ranges from 0.5–1.5 m, with mainly gradational boundaries. I In finer-grained areas, shallow sandy soils overlie >1 m of clay (total depth ~0.75–2 m) with sharper boundaries. In drainage lines, loamy sand overlies sandy clay and >1 m of grey clay.	Erosion hazard with localised seasonal waterlogging and high run-on in footslope areas. The site is within a Mine Subsidence District, with generally low but localised high foundation hazard linked to clay-rich subsoils, high shrink–swell, low wet bearing strength, and high erodibility. Soils are strongly acidic, very low fertility, and may be stony, with localised dispersion, sodicity, and aluminium toxicity. Overall, moderate constraints for urban development with localised high to severe limitations.
Disturbed Terrain (9232xx)	The Disturbed Terrain landscape comprises plains to hummocky ground extensively modified by human activity, including complete removal, disturbance, or burial of original soils. Local relief and slope are highly variable. Areas include landfill consisting of soil, rock, building, and waste materials. Original vegetation has been fully cleared and replaced with turf or grassland.	Highly variable, with original soils removed, buried, or heavily disturbed. Bedrock may be exposed in quarries and pits, while landfill areas contain transported soils, sediments, and industrial or household waste. Surfaces may be artificially topsoiled or covered with concrete or bitumen.	Highly variable, including erosion, mass movement/rockfall, steep slopes, seasonal waterlogging, shallow or unconsolidated materials, rock outcrop, and Mine Subsidence District constraints. Soils may be low fertility, acidic, dispersive, stony, and highly variable in permeability, water-holding capacity, erodibility, and shrink–swell behaviour. Urban development is often severely constrained, with high to severe limitations for agricultural use.

Soil landscape	Topography	Soil characteristics	Limitations
Warners Bay (9232wa)	The Warners Bay soil landscape comprises undulating to rolling low hills and rises developed on fine-grained sediments of the Newcastle Coal Measures in the Awaba Hills. Elevation ranges from 40–80 m, with local relief of 30–80 m and slopes of 3–20%. Crests are broad (200–300 m), with long, gently convex slopes (500–1,200 m) and broad drainage lines. Rock outcrop is rare. The area is predominantly cleared tall open forest.	Soils are moderately deep to deep, typically 1–>1.5 m, and imperfectly to poorly drained. Moderately well-drained podzolic and soloth soils occur across slopes, with poorly drained structured loams in drainage lines.	Limitations include high water erosion hazard and Mine Subsidence District constraints, with localised steep slopes and mass movement hazard. Seasonal waterlogging and high run-on occur on lower slopes. Soils are strongly acidic, of low fertility, with moderate to high shrink–swell and plastic subsoils.
Gateshead (9232ga)	The Gateshead soil landscape comprises undulating to rolling rises on Permian conglomerate, shale, and sandstone in the Awaba Hills. Elevation reaches up to 130 m, with local relief of 60–100 m and slopes of 5–15%. The landscape is predominantly cleared woodland and open forest, with broad crests (250–500 m) and long slopes (500–1,000 m). Drainage lines are narrow and deeply incised at their heads, grading into broader flats (up to 5 m wide) downstream. Rock outcrop is rare (<2%).	Soils are moderately deep to deep, ranging from 1 to 4 m, and are moderately well to imperfectly drained. Yellow Podzolic Soils and yellow Soloths occur on conglomerate crests and sideslopes, with localised shallow, rapidly drained Lithosols. On shale parent material, Red Podzolic Soils, red Soloths, and some yellow Soloths occur, typically with moderate to imperfect drainage.	Water erosion hazard and Mine Subsidence District, with localised steep slopes and shallow soils. High run-on and seasonal waterlogging occur on lower slopes. Soils are typically acidic and of low fertility.
Stockrington (9232sn)	The Stockrington soil landscape comprises steep rises developed on conglomerates of the Newcastle Coal Measures (Adamstown Subgroup). Elevation ranges from 80 - 160 m, with local relief up to 180 m. Upper slopes are steep (25–>40%) with slope lengths of 200 - 500 m, while colluvial benches and lower slopes are gentler (15–20%). The area is uncleared tall open forest, with terraced and hummocky surfaces common. Rock outcrop is absent, although conglomerate boulders may occur on lower slopes and benches. A landscape variant, Stockrington (9232sna), includes sideslopes with lower (<20%) gradients.	Soils are moderately deep to deep (from 0.7 m to > 3 m). Rapidly drained earthy and friable loams occur on upper slopes, while deep (>2 m), well-drained podzolic and soloth soils occur on midslopes and benches.	Limitations include steep slopes, mass movement and water erosion hazards, and foundation hazard.

The indicative extent of each soil landscape within the sub-precincts, together with the associated key limitations, is summarised in Table 3-9.

Table 3-9 Expected soil landscapes and key limitations across the sub-precincts

Sub-precinct		Indicative soil landscape extents (% of sub-precinct area)							Key landscape limitations
No.	Name	Killingworth (9232ki, 9232kia)	Cockle Creek (9232cc)	Doyalson (9232do)	Disturbed Terrain (9232xx)	Warners Bay (9232wa)	Gateshead (9232ga)	Stockrington (9232sn)	
1	Cockle Creek	0 %	10 %	30 %	35 %	15 %	5 %	5 %	Flooding, waterlogging, elevated water tables, run-on effects, dispersive soils. Highly variable ground conditions in disturbed terrains.
2	Killingworth	80 %	20 %	0 %	0 %	0 %	0 %	0 %	Erosion, steep slopes, reactive soils, and mine subsidence hazard.
3	West Lake	95 %	5 %	0 %	0 %	0 %	0 %	0 %	Erosion, steep slopes, reactive soils, and mine subsidence hazard.
-	Teralba Southgate	0 %	0 %	100 %	0 %	0 %	0 %	0 %	Erosion, reactive soils, mine subsidence hazard.
5	Barnsley West	70 %	30 %	0 %	0 %	0 %	0 %	0 %	Erosion, steep slopes, reactive soils, and mine subsidence hazard. Flooding, waterlogging, elevated water tables, run-on effects, dispersive soils.
6A	Environmentally constrained areas	70 %	30 %	0 %	0 %	0 %	0 %	0 %	Erosion, steep slopes, reactive soils, and mine subsidence hazard. Flooding, waterlogging, elevated water tables, run-on effects, dispersive soils.
6B		5 %	80 %	5 %	0 %	0 %	10 %	0 %	Flooding, waterlogging, elevated water tables, run-on effects, dispersive soils. Localised steep slopes and mine subsidence.
6C		15 %	85 %	0 %	0 %	0 %	0 %	0 %	Flooding, waterlogging, elevated water tables, run-on effects, dispersive soils and mine subsidence hazard.
6D		0 %	65 %	0 %	30 %	5 %	0 %	0 %	Flooding, waterlogging, elevated water tables, run-on effects, dispersive soils. Highly variable ground conditions in disturbed terrains.
6E		0 %	0 %	5 %	0 %	35 %	40 %	20 %	Waterlogging, run-on effects, dispersive soils, erosion, steep slopes, mine subsidence.
6F		100 %	0 %	0 %	0 %	0 %	0 %	0 %	Erosion, steep slopes, reactive soils, and mine subsidence hazard.

3.9 Acid Sulfate Soils

Acid Sulfate Soils (ASS) risk maps indicate the likely distribution of soils containing iron sulphides, primarily pyrite (FeS_2) based on geomorphic setting. These soils are generally benign when undisturbed but, when exposed to air through excavation or drainage, oxidise to produce sulphuric acid ($\text{pH} < 4$), potentially reducing the durability of concrete and buried foundations and infrastructure. In addition, the costs associated with safe management of excavations and removal of construction spoil is significantly increased if ASS are present.

ASS potential across the Precinct and Teralba Southgate has been assessed using the eSPADE portal (Department of Planning and Environment, 2024), which indicates ASS risk broadly consistent with alluvial deposits in low-lying areas along Cockle Creek and its tributaries. Mapped risk ranges from low to high, notably predominantly low across sub-precinct No.6B, with higher risk closer to Cockle Creek in the north. Upland areas generally don't present known occurrence of ASS, and disturbed terrain is also present to the northeast of the MCC site. Sub-precinct No. 6D exhibits the highest likelihood of ASS across most of its area.

Refer to Figure 3-9 for the soil ASS probability across the site. A higher-resolution version of this figure is provided in Figure A5 of **Appendix A – Figures**.

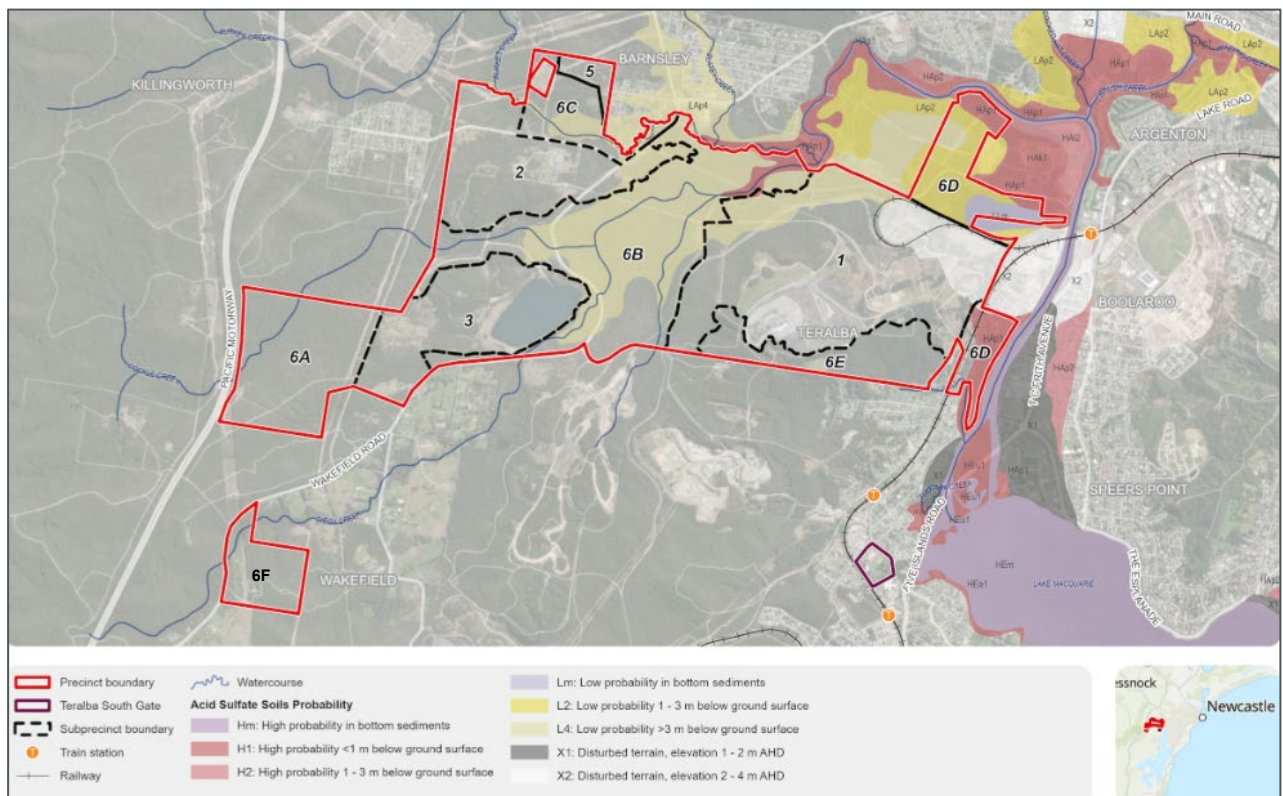


Figure 3-9 ASS probability

ASS probability for each sub-precinct is summarised in Table 3-10.

Table 3-10 ASS probability across the sub-precincts

Sub-precinct		ASS probability
No.	Name	
1	Cockle Creek	Generally no known occurrence; locally low probability below 3 m BGL near Cockle Creek and within disturbed terrain in the north. The disturbed terrain areas identified in Section 3.2.1 are not shown and are expected to include potential for ASS.

Sub-precinct		ASS probability
No.	Name	
2	Killingworth	Generally no known occurrence; locally low probability below 3 m BGL near Cockle Creek in the east. The disturbed terrain areas identified in Section 3.2.1 are not shown and are expected to include potential for ASS.
3	West Lake	No known occurrence. The disturbed terrain areas identified in Section 3.2.1 are not shown and are expected to include potential for ASS.
-	Teralba Southgate	No known occurrence.
5	Barnsley West	No known occurrence.
6A	Environmentally constrained areas	No known occurrence.
6B		Generally low probability below 3 m BGL; locally high probability at < 1 m BGL near Cockle Creek in the north.
6C		No known occurrence in the west; low probability below 3 m BGL in the east.
6D		High probability at <1 m BGL in the east; low probability at 1–3 m BGL in the north, with locally shallower occurrences (<1 m BGL) near Cockle Creek and within disturbed terrain.
6E		No known occurrence.
6F		No known occurrence.

3.10 Salinity

Salt content in water and soil can accelerate the corrosion process in metal due to its nature as a good conductor for electrons. Thus, soil or water that is mildly acidic can break down concrete and steel reinforcements in buried or on ground structures. Salt is also hygroscopic; it can attract and hold water.

Based on the eSPADE portal (Department of Planning and Environment, 2024), the MCC Precinct is broadly characterised by moderate to very high salinity hazard, with lower hazard towards Cockle Creek and its tributaries and very high hazard in upland areas.

Refer to Figure 3-10 for the overall salinity hazard across the Precinct and Teralba Southgate. A higher-resolution version of this figure is provided in Figure A8 of **Appendix A – Figures**.

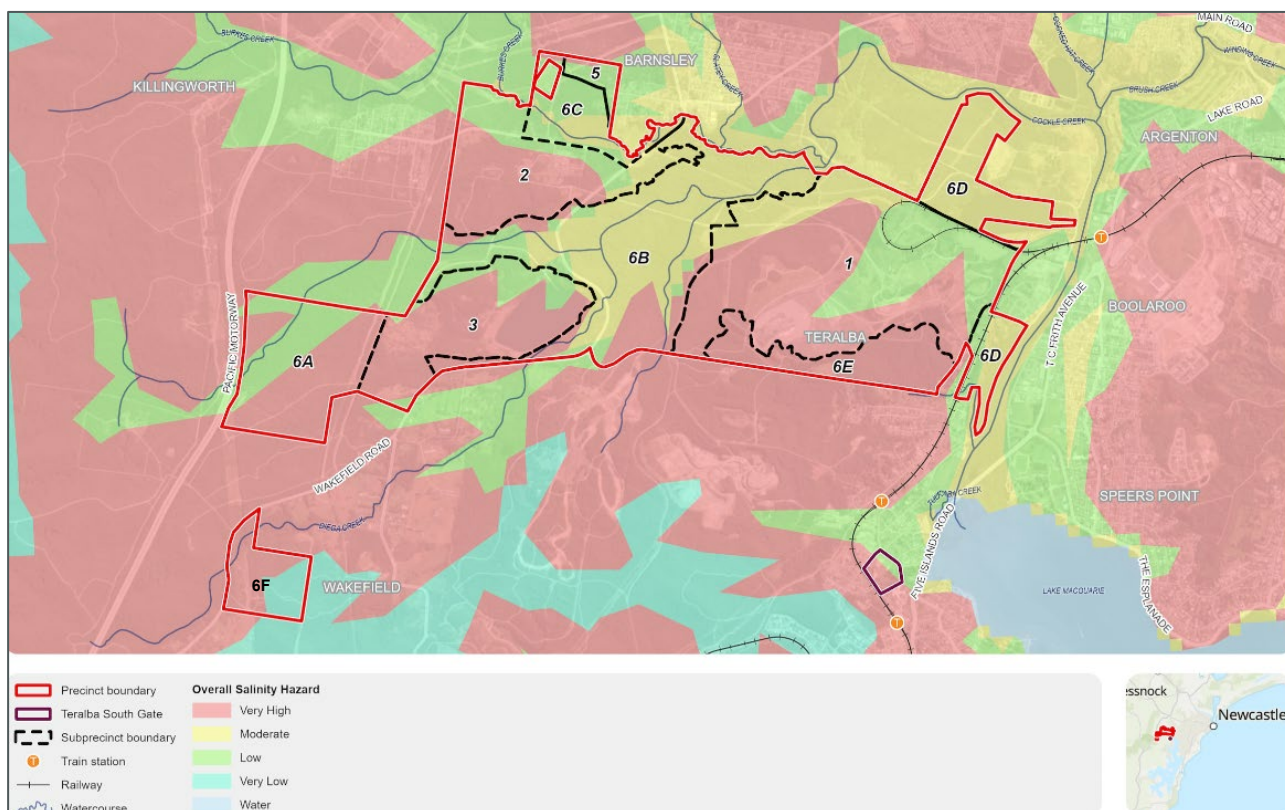


Figure 3-10 Overall salinity hazard

The overall salinity hazard for each sub-precinct is summarised in Table 3-11.

Table 3-11 Overall salinity hazard across the sub-precincts

Sub-precinct		Overall salinity hazard
No.	Name	
1	Cockle Creek	Generally very high.
2	Killingworth	Generally very high, low to moderate in the north and east.
3	West Lake	Generally very high, low in the north.
-	Teralba Southgate	Generally very high.
5	Barnsley West	Generally low.
6A	Environmentally constrained areas	Generally low near Cockle Creek and very high in upland areas.
6B		Generally moderate. Locally very high in upland areas.
6C		Generally low to moderate.
6D		Generally moderate.
6F		Generally very high in the west and very low in the east.

3.11 Groundwater

There is limited, sparse information about the hydrogeology within the Precinct and Teralba Southgate. Some publicly available data was available to the northeast of the project around the Cockle Creek estuary. There are 14 groundwater wells that have been identified in two clusters, to the south and northeast of the Precinct, based on MinView data (2024). Only wells within 100 m of the Precinct and Teralba Southgate boundary have been considered for this assessment. Groundwater levels from available logs are generally limited (refer to Section 3.13.2).

Some further data was available from around Diega Creek. However, there is some uncertainty about groundwater levels from this due to conflicting information.

It is anticipated that groundwater is expected to be shallow near Cockle Creek and tributaries in low lying areas. Perched groundwater may be encountered within granular or alluvial materials over less permeable alluvium or residual soils. High elevations are expected to be well drained, but levels could be variable and recovering where mining has influenced the groundwater regime, particularly around Diega Creek.

Groundwater recharge of the alluvium is also anticipated to be slow due to the hydrological soil group; however further testing would be required to confirm and quantify the hydraulic conductivity of the Precinct and Teralba Southgate and determine recharge rates.

3.12 Mine subsidence

According to the Mining Exploration and Geoscience Division of Regional NSW (accessed via MinView), the Precinct and Teralba Southgate are entirely underlain by the Killingworth-Wallsend and Lake Macquarie No.1 Extension Mine Subsidence Districts and bordered by the West Lake & Lake Macquarie Mine subsidence Districts, as shown in Figure 3-11. A higher-resolution version of this figure is provided in Figure A6.1 of **Appendix A – Figures**.

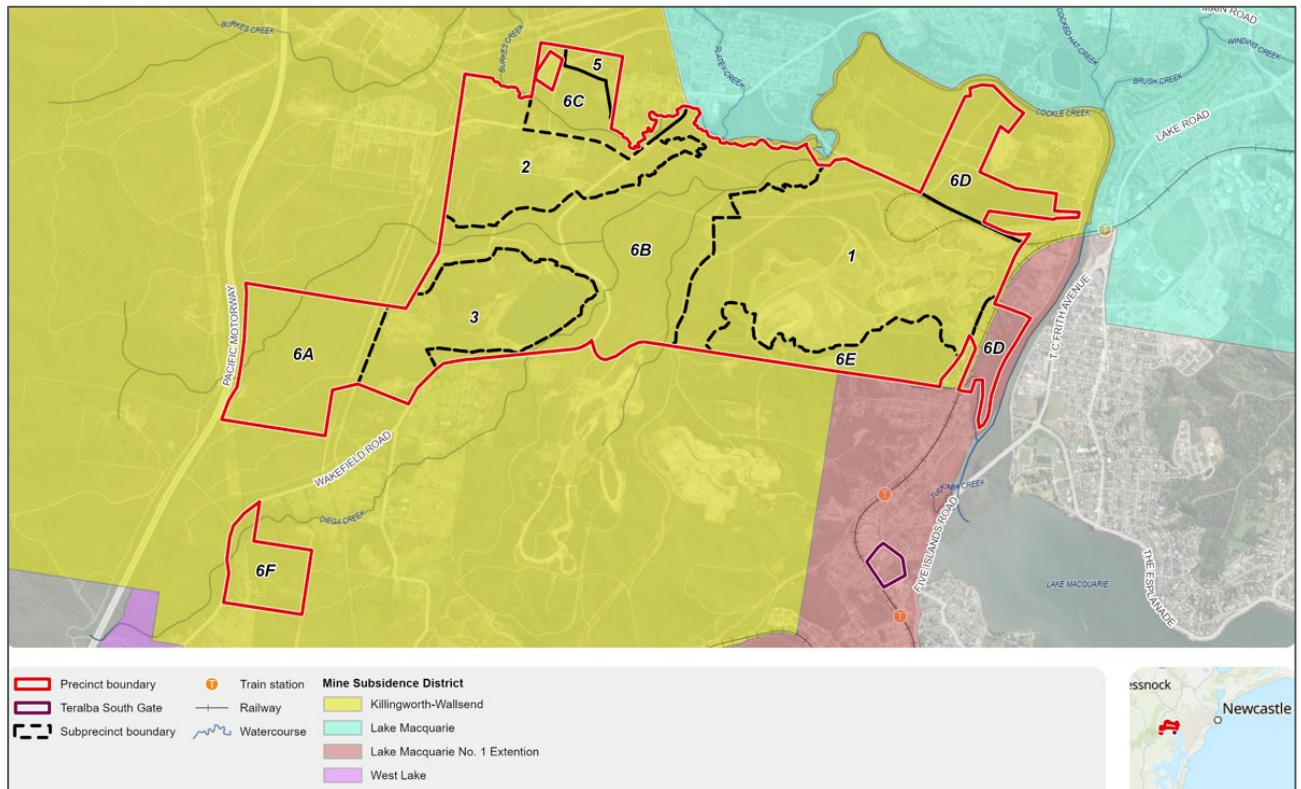


Figure 3-11 Mine Subsidence Districts

Appendix H of the “*Mining Operations Plan: Closure of the OCAL Complex*” report by SLR on behalf of Glencore (SLR, 2016) includes a plan showing the locations of underground mine entry points and extent of former underground workings. An extract of this plan is presented in Figure 3-14 and indicates that the Precinct and Teralba Southgate are extensively underlain by historical underground bord-and-pillar as well as longwall mine workings. It is noted that this report appendix also includes the underground workings, as part of Figure 1.2 of the *West Wallsend Colliery Continued Operations Project Report* (NSW Government Planning & Infrastructure, 2012).

The plan shows several aspects of note:

- Areas of mining in different coal seams are coloured
- Some areas of multiple seam workings are shown
- Areas of Longwall mining are more prevalent in the west of the Precinct
- Areas of Bord and pillar mining are more prevalent in the east of the Precinct
- Areas of underground mining do exist below the majority of the Precinct

As part of the application documents for proposed expanded mine workings at West Wallsend, the data from surface monitoring at several regions of the former mine workings is provided in the following reports:

- *Subsidence Predictions and Impact Assessment of the Proposed Western and Southern Domain Longwalls, West Wallsend Colliery, Peer Review of Report DGS WWD-012/1, Report No. MSEC433* (Mine Subsidence Engineering Consultants, 2010).
- *Subsidence Predictions and General Impact Assessment of the Proposed Western and Southern Domain Longwalls, West Wallsend Colliery, DGS Report No. WWD-012/1* (Ditton Geotechnical Services, 2010)

Existing monitoring lines were compared to predicted settlement lines to assess the modelling for the proposed new underground longwall workings. Lines WX, WN lie over the western area of the Precinct as shown in Figure 3-12. The monitoring lines above the Longwalls 11 to 18 are presented in Figure 3-13.

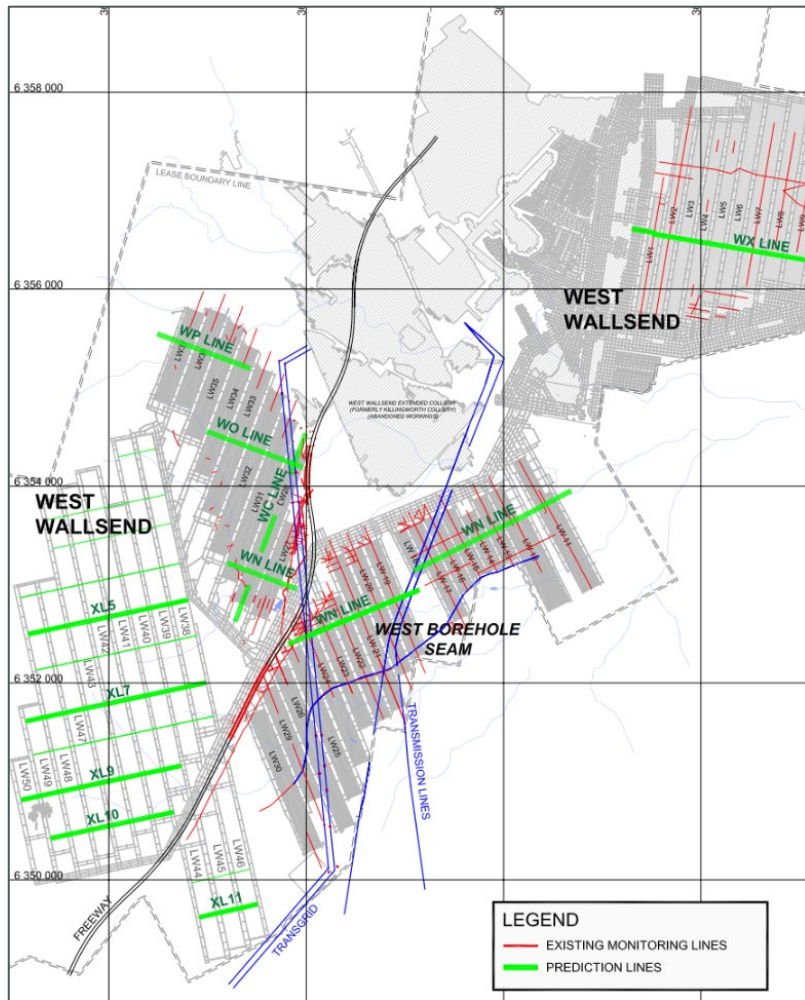


Figure 3-12 Existing and predicted settlement lines

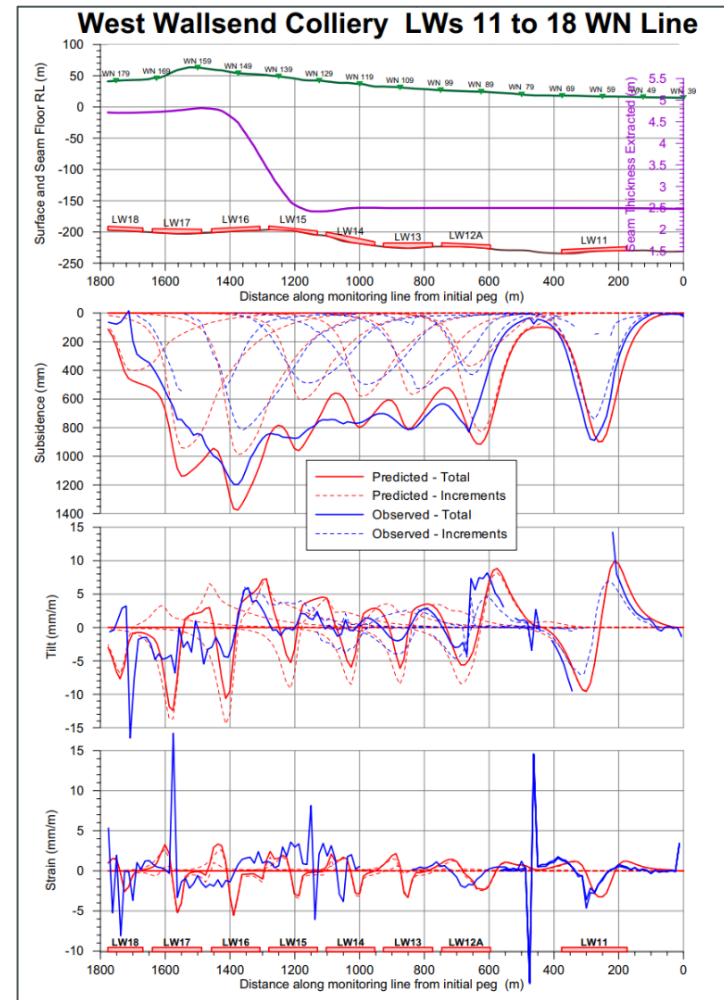


Figure 3-13 Observed vs predicted surface settlement, tilt and strain reading above Longwalls 11 to 18 of West Wallsend Colliery

The monitoring lines above the Longwalls 11 to 18 in Figure 3-13 show that the maximum record for observed surface settlement of 1.0 m was recorded over Longwall 16. The range of total settlements above Longwalls 12A to 15 ranges from 0.8 m to 0.9 m, and this variation is driven by the changes in the seam height and depth of cover. The depths to underground mine workings typically vary with local topography and bedrock geology and therefore the level of subsidence risk is also likely to vary across other areas where monitoring has not been completed. But it is noted from this data that the majority of subsidence may have already occurred above former Longwalls, which is to be expected as this is part of Longwall mining design considerations.

In addition, Section 3.2.2 of the same report provides summary information on the Historical Mining areas across the OCAL complex and refers to a study completed by Ditton Geotechnical Services (DgS) which completed a preliminary assessment of the possible long term settlement risks due to instability of the historical OCAL workings including those in the West Wallsend No. 2 and Teralba Collieries (circa 1970's to 2016) in the following seams:

Table 3-12 - Summary of worked coal seams (SLR, 2016)

Coal seam	Cover depth (m BGL)	Seam thickness
West Borehole (WBH)	Between 80 m and 320 m	Not presented in report.
Young Wallsend (YW)	Between 150 m and 380 m	Nominally 3.1 m, but typically increased to 3.3 m due to the collapse of the coal unit between the Nobby's Tuff and the YW Seam.
Borehole (BH)	Approximately 20 m to 30 m below the YW Seam. No workings below 345m	2.1 m to 2.2 m in the full width panels. 1.6 m to 1.8 m in the Longwall panels.

The variation in depth in the workings is due to the dip and faulting within the strata as well as the varying depth of cover due to hills and valleys at the surface.

The report notes that the former underground workings consist of first and second working with bord and pillar, pillar extraction and longwall panels in regions of each seam, and importantly, that *“other non-OCAL mine working also exist in higher level seams that may contribute to the long-term subsidence predictions”*

The report makes the following recommendations:

- *Based on the age and the location of the old mine workings, it is considered that the area contains several zones in which the risk of post-mining subsidence may be sufficiently ‘low’ for future developments to be designed to tolerate relatively small magnitudes of subsidence.*
- *Maximum subsidence above the OCAL mine workings due to future instability is estimated to range from < 20 mm to < 160 mm, depending on the type of workings and whether they have been flooded.*
- *Future structures and infrastructure will probably need to be designed for potential subsidence up to 150 mm, tilts < 4 mm/m, curvatures < 0.2 km⁻¹ and horizontal strains < 2 mm/m to satisfy SANSW “Safe, Serviceable and Repairable” Design Guidelines in Mine Subsidence Districts.*
- *The work by DgS was based on preliminary analysis and further detailed investigations mainly based on numerical modelling might be needed to accurately estimate the potential of further post-mining subsidence in each sub-area of the site, particularly in areas that contains multi-seam mining or First Workings adjacent to Second Workings.*
- *Areas where multi-seam mining has accorded with incomplete caving in the lower seam are assessed to have a ‘Moderate’ Risk of significant subsidence impact occurring in the future (i.e. subsidence > 150 mm).*
- *The flooding of West Wallsend No 2 Colliery may result in additional subsidence due to strata softening or goaf reconsolidation effects. The Teralba Colliery workings are already flooded and have a lower risk of additional subsidence than the dry workings do.*

- Overall, the study has identified there is 'low' risk over the majority of the study area underlain by YW and BH Seam mine workings associated with OCAL mining complex, however, further merit based assessments of a given site and proposed development may therefore be necessary in Moderate Risk Zones (i.e. Multi-seam mining locations).

Figure 3-14 is reproduced from the Mine Closure report (SLR 2016) and presents known former underground workings across the MCC Precinct and Teralba Southgate. A higher-resolution version of this figure is provided in Figure A6.2 of **Appendix A – Figures**.

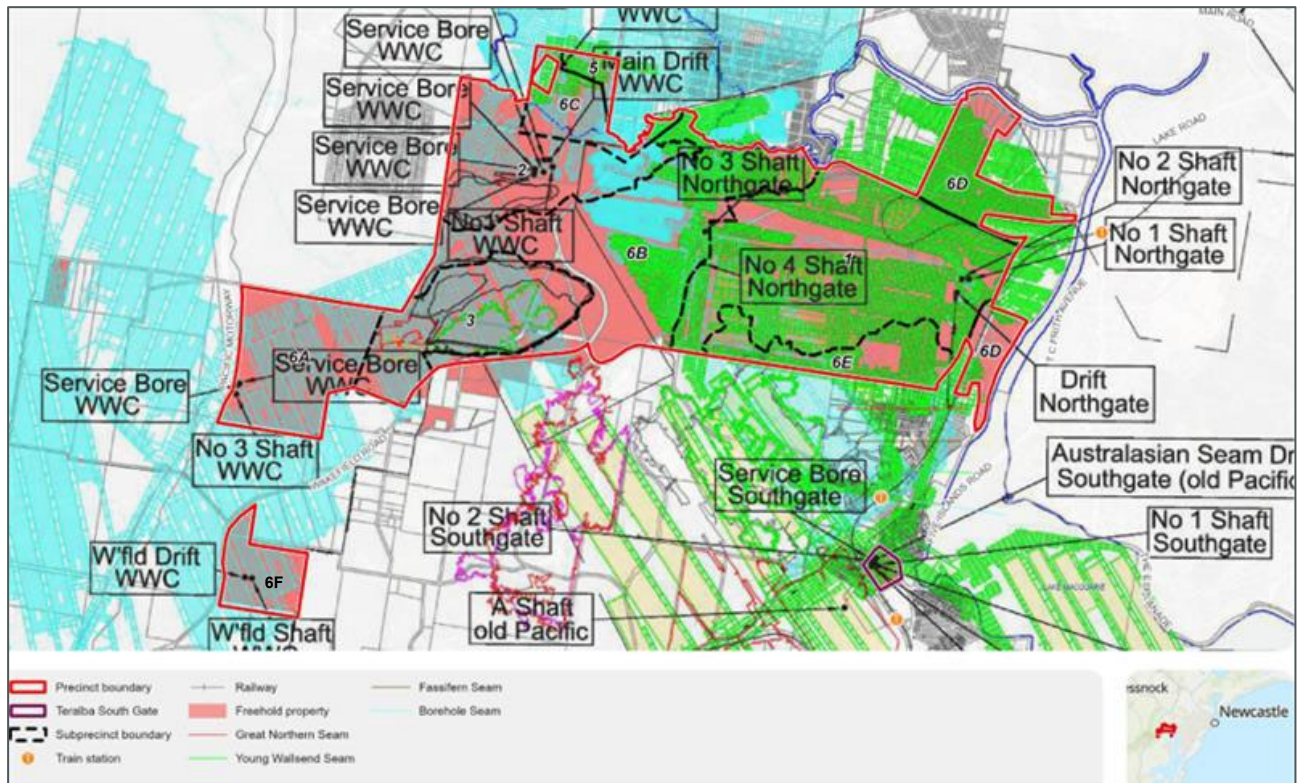


Figure 3-14 Areas of former underground mine working and surface access points (SLR 2016)

Section 2.3.8.3 of the OCAL Mining Operations Plan also provides a summary of sites that require sealing at the OCAL Complex and required sealing strategies including “installation of bulkheads and / or concrete plugs in shafts / drifts as required”. The locations of many of these are shown in Figure 3-14 above.

Given that the entire precinct is located within mapped Mine Subsidence Districts and is likely underlain by historic underground mines, at this stage mine subsidence hazard is considered to apply across the entire site, irrespective of location. The accuracy of the mapping and precise locations of the historic workings should be assumed to be approximate at this stage also.

Based on Figure 3-14, a high-level summary of former mine working types for each sub-precinct is summarised in Table 3-12, this is illustrative only at this stage and the records held by SA NSW will be required to provide the fuller picture.

Table 3-13 Mine subsidence hazard across the sub-precincts

Sub-precinct		Preliminary mine subsidence hazard
No.	Name	
1	Cockle Creek	Mixture of old workings including bord and pillar, and possibly some longwall mining. The western area, as well as near sub-precinct 6B, may have been undermined in more than one seam.

Sub-precinct		Preliminary mine subsidence hazard
No.	Name	
2	Killingworth	Mostly in the western half of the sub-precinct; not known elsewhere.
3	West Lake	Longwall workings below most of the sub-precinct.
-	Teralba Southgate	Longwall and bord and pillar workings and shafts.
5	Barnsley West	Bord and pillar workings below most of the sub-precinct.
6A	Environmentally constrained areas	Longwall mined below most of the sub-precinct.
6B		Localised to the western end of the sub-precinct; with longwall and bord and pillar workings and shafts in the central eastern region. These working may occur in more than one seam in some areas.
6C		Bord and pillar workings across most of the sub-precinct.
6D		Bord and pillar workings below entire sub-precinct.
6E		Bord and pillar workings across most of the sub-precinct and possibly limited longwall at southern edge.
6F		Longwall mined below most of the sub-precinct.

3.13 Available geotechnical information

3.13.1 New South Wales Government

The geotechnical information from the following databases have been reviewed:

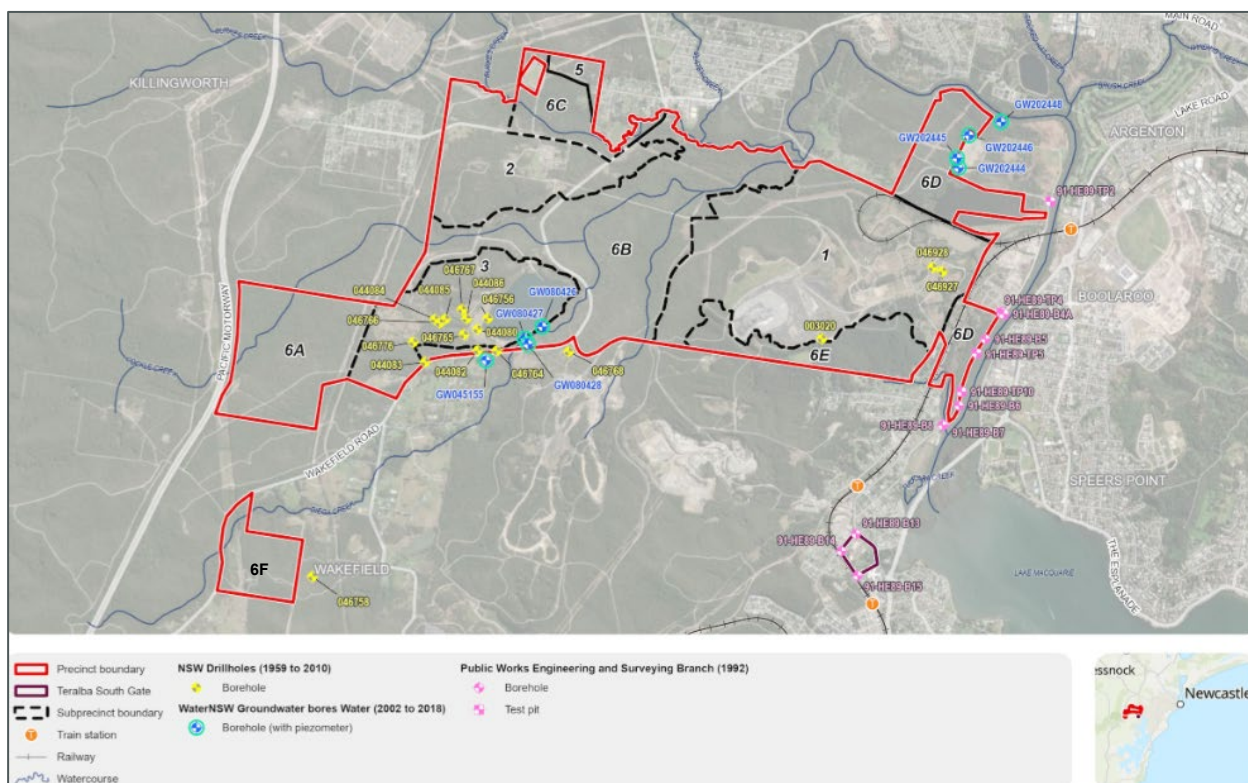
- New South Wales Government: MinView and WaterNSW portal
- NSW Public Works, State Significant Developments and Major Projects geotechnical investigations
- Aurecon: Groggi (Global Repository of Geological and Geotechnical Information)

3.13.2 Existing geotechnical investigations

The following reports have been referenced to interpret the expected ground profile and potential geotechnical constraints:

- Public Works Engineering and Surveying Branch, Lake Macquarie Sewerage Scheme, Edgeworth Effluent & Marmong Sewage, Transportation Systems, Hunter Sewerage Project Geotechnical Investigation, Ref.: 91-HE89, November 1992.
- Exploration borehole records from reports No. R00055140, R00056767, RE0011891, R00056323 for investigations undertaken between 1959 and 2010 and accessed via MinView.
- Groundwater monitoring well borehole logs accessible from the WaterNSW portal.

It is noted that only boreholes with available and exploitable information are shown on Figure 3-15. A higher-resolution version of this figure is provided in Figure A9 of **Appendix A – Figures**. Only locations within the Structure Plan sub-precinct areas or close surroundings have been included in this assessment.



A summary of the existing geotechnical information across each sub-precinct is provided in Table 3-13 and Table 3-14 for the MCC Precinct and Teralba Southgate, respectively.

Table 3-14 Summary of existing geotechnical information within or adjacent to the MCC Precinct

Sub-precinct No.	Source	Reference	Investigation locations										
			Initial investigation location ID	Adopted investigation location ID in this Report	Completion year	Relative location within/near sub-precinct	Termination depth	Fill range	Soft to firm cohesive soils range	Depth to rock (as logged)	Groundwater		
							[m BGL]	[m BGL]	[m BGL]	[m BGL]	Depth [m BGL]	Type of water level	Date
1	Minview	R00056323	SHAFT 2	046928	1959	Northeast boundary	228.4	NR	NR	5.5	NR	-	-
			1	046927	1959	Northeast boundary	225.9	NR	NR	3.1	NR	-	-
		R00055140	NORTHERN EXT 8	003020	1960	South of TSF	78.7	NR	NR	12.8	NR	-	-
2	No geotechnical information data available within or adjacent to sub-precinct.												
3	Minview	R00055140	MF14	046776	1981	Central	20.5	NR	NR	1.9	NR	-	-
			29	044084	1979	Central	26.5	NR	NR	0.9	NR	-	-
			30	044085	-	Central	19.2	NR	NR	0.9	NR	-	-
			31	044086	-	Central	28.5	NR	NR	0.9	NR	-	-
			18	044082	-	Along Wakefield Road	34.1	NR	NR	0.6	NR	NR	NR
			28	044083	-	South	10.7	NR	NR	1.2	NR	-	-
			DDH07	046756	1962	Eastern boundary	22.3	NR	NR	0.9	NR	-	-
			1	044080	-	Eastern boundary	45.7	NR	NR	0.6	NR	-	-
	WaterNSW	-	GW080428	GW080428	2002	South	24.0	NR	NR	20.8	NR	NR	NR
		-	GW080427	GW080427	2002	South	28.0	NR	NR	1.0	NR	NR	NR
		-	GW080426	GW080426	2002	South	23.0	NR	NR	5.5	NR	NR	NR
		-	GW045155	GW045155	2018	Along Wakefield Road	4.3	NR	NR	0.0	NR	NR	NR
5	No geotechnical information data available within or adjacent to sub-precinct.												
6A	No geotechnical information data available within or adjacent to sub-precinct.												
6B	MinView	R00055140	18	044082	-	Along Wakefield Road	34.1	NR	NR	0.6	NR	-	-

Sub-precinct No.	Source	Reference	Investigation locations										
			Initial investigation location ID	Adopted investigation location ID in this Report	Completion year	Relative location within/near sub-precinct	Termination depth	Fill range	Soft to firm cohesive soils range	Depth to rock (as logged)	Groundwater		
							[m BGL]	[m BGL]	[m BGL]	[m BGL]	Depth [m BGL]	Type of water level	Date
	WaterNSW	-	GW045155	GW045155	2018	Along Wakefield Road	4.3	NE	NE	0.0	NR	-	-
6C	No geotechnical information data available within or adjacent to sub-precinct.												
6D	Minview	(Public Works Engineering and Surveying Branch, 1992)	TP2	91-HE89-TP2	1991	Eastern boundary, along Racecourse Road	2.4	0.0 – 0.5	1.9 – 2.4	> 2.4	NE	-	-
			TP4	91-HE89-TP4	1991	Eastern boundary, along Racecourse Road	2.0	0.0 – 1.0	1.0 – 0.6	> 2.0	NE	-	-
			BH4A	91-HE89-B4A	1992	Eastern boundary, along Racecourse Road	5.5	0.0 – 1.0	1.0 – 4.8	> 5.5	1.7	Inferred during drilling	1992
			B5	91-HE89-B5	1992	Eastern boundary, along Racecourse Road	6.0	0.0 – 0.9	0.9 – 5.3	> 6.0	2	Inflow	1992
			TP5	91-HE89-TP5	1991	Eastern boundary, along Racecourse Road	2.0	-	1.3 – 1.6	> 2.0	1.6	Inflow	1991
			TP10	91-HE89-TP10	1991	Eastern boundary, along Racecourse Road	2.5	-	1.4 – 2.5	> 2.5	1.6	Inflow	1991
			B6	91-HE89-B6	1992	Eastern boundary, along Racecourse Road	7.0	0.0 – 1.0	1.5 – 7.0	> 7.0	1.5	Inferred during drilling	1992

Sub-precinct No.	Source	Reference	Investigation locations										
			Initial investigation location ID	Adopted investigation location ID in this Report	Completion year	Relative location within/near sub-precinct	Termination depth	Fill range	Soft to firm cohesive soils range	Depth to rock (as logged)	Groundwater		
							[m BGL]	[m BGL]	[m BGL]	[m BGL]	Depth [m BGL]	Type of water level	Date
			B7	91-HE89-B7	1992	Eastern boundary, along Racecourse Road	3.6	0.0 – 0.7	NE	1.2	2.8	Inferred during drilling	1992
			B8	91-HE89-B8	1992	Eastern boundary, along Racecourse Road	2.3	0.0 – 0.7	NE	1.4	NE	-	-
	WaterNSW	-	GW202446	GW202446	2010	Eastern boundary	5.2	0.0 - 1.9	NE	> 5.2	4.1	Standing	2010
		-	GW202445	GW202445	2010	Eastern boundary	5.2	0.0 – 1.7	NE	> 5.2	3.8	Standing	2010
		-	GW202444	GW202444	2010	Eastern boundary	5.2	0.0 – 2.2	NE	> 5.2	4.2	Standing	2010
	6E	Minview	R00055140	NORTHERN EXT 8	1960	South of TSF	78.7	NR	NR	12.8	NR	-	-
6F	Minview	R00055140	DDH09	046758	1963	100 m east	28.0	NR	NR	3.4	NR	-	-
NR – Not Recorded; NE – Not Encountered													

Table 3-15 Summary of existing geotechnical information within or adjacent to Teralba Southgate

Source	Reference	Investigation locations										
		Initial investigation location ID	Adopted investigation location ID in this Report	Completion year	Relative location within/near sub-precinct	Termination depth	Fill range	Soft to firm cohesive soils range	Depth to rock (as logged)	Groundwater		
						[m BGL]	[m BGL]	[m BGL]	[m BGL]	Depth [m BGL]	Type of water level	Date
Minview	(Public Works Engineering and	B13	91-HE89-B13	1992	Northwest boundary York St/ Pitt Street intersection	2.5	0.0 – 0.8	NE	1.8	NE	-	-

Source	Reference	Investigation locations										
		Initial investigation location ID	Adopted investigation location ID in this Report	Completion year	Relative location within/near sub-precinct	Termination depth	Fill range	Soft to firm cohesive soils range	Depth to rock (as logged)	Groundwater		
						[m BGL]	[m BGL]	[m BGL]	[m BGL]	Depth [m BGL]	Type of water level	Date
	Surveying Branch, 1992)	B14	91-HE89-B14	1992	Southwest boundary Pitt Street/Railway intersection	1.8	NE	NE	1.6	NE	-	-
		B15	91-HE89-B15	1992	South boundary, George Wright Dr/York St	1.9	NE	NE	> 1.9	NE	-	-
NR – Not Recorded; NE – Not Encountered												

4 Structure Plan preliminary ground models

The anticipated ground models across the Structure Plan sub-precincts are primarily based on the nearest existing ground investigation data to the sub-precinct and are summarised in Table 4-1. It is noted that site-specific geotechnical data is limited for much of the Precinct and Teralba Southgate, particularly within sub-precincts No.2, 5, 6A and 6C. For these areas, the anticipated ground profile has been primarily informed by a review of available geotechnical reports and regional geology mapping.

Groundwater inflows were at depth of approximately 1.5 to 2.8 m BGL in boreholes 91-HE89-B4A, 91-HE89-B5, 91-HE89-TP5, 91-HE89-TP10, 91-HE89-B6 and 91-HE89-B7, located east of sub-precinct No.6D along Racecourse Road (Public Works Engineering and Surveying Branch, 1992). These observations suggest potential shallow groundwater levels in the vicinity of Cockle Creek. Standing groundwater was also recorded at depths of 3.8 to 4.2 m BGL in monitoring wells GW202446, GW202445 and GW202444 further north of this sub-precinct (WaterNSW). Boreholes 91-HE89-B13, 91-HE89-B14 and 91-HE89-B15 within Teralba Southgate were dry upon completion. For the remaining investigation locations reviewed as part of this Report, groundwater observations were not recorded on the available logs.

Groundwater levels are expected to be relatively shallow in areas adjacent to Cockle Creek, Diega Creek and associated tributaries, particularly within low-lying areas of the MCC Precinct. Groundwater in these areas may become locally shallower following prolonged heavy rainfall.

Perched groundwater may be present in cohesionless fill or alluvial deposits where these materials overlie less permeable soils or bedrock. Groundwater conditions and hydraulic conductivity regime should be confirmed through future site investigations.

Given the limited geotechnical data available across much of the MCC Precinct and Teralba Southgate, additional investigations are recommended to verify the anticipated ground profile, improve confidence in the current interpretation, and inform future development and design following the rezoning.

Table 4-1 Expected ground models across the sub-precincts

Sub-precinct		Anticipated ground conditions		References	
No.	Name	High-level ground profile	Groundwater	Test locations No.	Other references
1	Cockle Creek	- Localised fill to the north, within TSF area and in areas identified in Section 3.2.1. - Underlain by localised alluvium in the north. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected between 3.0 and 10 m BGL within the TSF and northern areas, reducing to approximately 1.0 – 2.0 m BGL elsewhere, broadly consistent with the existing test locations within the sub-precinct.	Groundwater expected to be relatively shallow towards Cockle Creek to the north and other low-lying areas. Perched groundwater may occur within cohesionless fill and alluvial deposits.	046928, 046927, 003020	Available MinView datasets, including Seamless Geology and Depth of Regolith, historical imagery review
2	Killingworth	- Localised fill in areas identified in Section 3.2.1. - Localised alluvium in the east. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected between 3.0 and 10.0 m BGL reducing to approximately 1.0 - 2.0 m BGL in elevated southern areas.	Groundwater is expected to be relatively shallow within drainage lines and other low-lying areas.	-	Available MinView datasets, including Seamless Geology and Depth of Regolith, historical imagery review
3	West Lake	- Fill in areas identified in Section 3.2.1. - Localised alluvial deposits to the west towards Diega Creek and tributaries. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected between 1.0 and 2.0 m BGL, locally ranging from <1.0 m to 3.0 m BGL (west) and 3 m to 10 BGL (east), broadly consistent with the existing test locations within the sub-precinct.	Groundwater is expected to be relatively deep, except within local drainage lines. Groundwater expected to be shallow adjacent to Diega Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits.	46776, 44084, 44085, 44086, 44083, 46756, 44080, 44082, GW080428, GW080427, GW080426, GW045155	Available MinView datasets, including Seamless Geology and Depth of Regolith, historical imagery review
-	Teralba Southgate	- Residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected around 2.0 m BGL.	Groundwater is expected to be relatively deep. No groundwater was recorded within the boreholes of reference.	91-HE89-B13, 91-HE89-B14, 91-HE89-B15	Available MinView datasets, including Seamless Geology
5	Barnsley West	- Alluvial deposits towards Cockle Creek and tributaries. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL beneath deeper alluvial deposits.	Groundwater expected to be shallow in the vicinity of Cockle Creek and associated tributaries. Perched groundwater may occur within cohesionless alluvial deposits.	-	Available MinView datasets, including Seamless Geology and Depth of Regolith

Sub-precinct		Anticipated ground conditions		References	
No.	Name	High-level ground profile	Groundwater	Test locations No.	Other references
6A	Environmentally constrained areas	- Localised alluvial deposits near Cockle Creek and tributaries. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL in the north towards Cockle Creek.	Groundwater expected to be shallow adjacent to Cockle Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits.	-	Available MinView datasets, including Seamless Geology and Depth of Regolith
6B		- Alluvial deposits near Cockle Creek, Diega Creek and tributaries. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL in the north towards Cockle Creek. Existing test locations suggest shallow bedrock (< 1 m) towards Wakefield Road in the south.	Groundwater expected to be shallow adjacent to Cockle Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits.	044082, GW045155	Available MinView datasets, including Seamless Geology and Depth of Regolith
6C		- Alluvial deposits near Burkes Creek. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL towards Cockle Creek.	Groundwater expected to be shallow adjacent to Cockle Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits.	-	Available MinView datasets, including Seamless Geology and Depth of Regolith
6D		- Localised fill in the north. - Underlain by alluvial deposits, including estuarine deposits with highly compressive organic matter. Existing investigations recorded approximately 2.5 m of soft to firm cohesive alluvial soils on average along Racecourse Road, increasing to 5.5 m in borehole 91-HE89-B6. Alluvium thickness is to be confirmed, as bedrock was not generally encountered in the available investigations. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL towards Cockle Creek, broadly consistent with the existing test locations within the sub-precinct.	Groundwater encountered at approximately 1.5 - 2.8 m BGL along Racecourse Road, in the vicinity of Cockle Creek. Perched groundwater may occur within cohesionless alluvial deposits. Standing groundwater recorded at approximately 3.8 - 4.2 m BGL to the north.	91-HE89-TP2, 91-HE89-TP4, 91-HE89-B4A, 91-HE89-B5, GW202446, GW202445, GW202444	Available MinView datasets, including Seamless Geology and Depth of Regolith
6E		- Residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected around 2.0 m BGL.	Groundwater is expected to be relatively deep, except within local drainage lines.	003020	Available MinView datasets, including Seamless Geology and Depth of Regolith

Sub-precinct		Anticipated ground conditions		References	
No.	Name	High-level ground profile	Groundwater	Test locations No.	Other references
6F		- Localised alluvium in the north. - Underlain by residual soil/extremely weathered material associated with the Narrabeen Group and Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, broadly consistent with the existing test location near the sub-precinct.	Groundwater expected to be shallow adjacent to Cockle Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits.	046758	Available MinView datasets, including Seamless Geology and Depth of Regolith

5 Geotechnical constraints assessment

5.1 Key geotechnical criteria

The overall site setting, including slope, geology, depth of regolith, soil landscapes, ASS, salinity and mine subsidence described in the preceding sections of this Report presents geotechnical constraints that must be addressed in infrastructure planning and design, and opportunities that may be optimised from the outset.

To support the geotechnical constraints assessment, and inform the Master Plan, these constraint categories have been defined as key geotechnical criteria for the development of the overall constraint map, using their associated primary datasets listed in Table 5-1, and combined risk rankings presented in the subsequent sections and Appendix B figures.

The overall constraints map is provided in **Appendix C – Near surface geotechnical constraints map for Structure Plan**.

Table 5-1 Summary of constraint categories and associated datasets

Geotechnical criterion	Dataset(s)	Reference in Appendix B
Slope	Derived from Geoscience Australia SRTM	Figure B1
Slope stability geo zones	<i>Lake Macquarie City website, Geotechnical Slope Stability Guidelines (https://www.lakemac.com.au/Development/Planning-controls-and-development-guidelines/Development-in-geotechnical-areas)</i>	Figure B7
Geology	NSW Statewide Seamless Geology Dataset (Colquhoun et al., 2019).	Figure B2
Depth of regolith	Soil and Landscape Grid National Soil Attribute Maps – Soil Depth, (CSIRO, August 2024)	Figure B6
Soil landscapes	eSPADE 2.2 tool from the NSW Department of Planning, Industry and Environment (Department of Planning and Environment, 2024)	Figure B3
ASS		Figure B4
Overall salinity		Figure B5

5.2 Data confidence

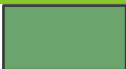
- The overall salinity mapping generally correlates with the geological mapping; however, differences in the mapped extents may result in local discrepancies in the constraint assessment. The layer is mapped at a larger scale across the state and is considered fairly low accuracy in the scale of the Precinct and Teralba Southgate but aligns broadly with the underlying geology.
- No test data available for the materials stored within the TSF. The depth profile is also not available.
- Rail loop area and anthropogenic fill: limited and incomplete GIS/data coverage. The rail loop is identified as a key data gap. At the time of this assessment, anthropogenic fill in this area is treated as moderate uncertainty/high risk, unless refined using available Glencore data to better define extent and current conditions for improved risk assessment, or through further mapping and site-specific investigation.
- Input datasets and layer for GIS have been sourced from multiple sources such as MinView, Glencore and Lake Macquarie City Council. The level of details and variability of sources / interpretations may induce variability in input dataset. The original authors limitation should be referred to if undertaking subsequent studies.
- Due to the Precinct and Teralba Southgate being above a range of historic mining sites the geology mapping is considered to be of fair to good quality with faults and local structure well understood generally compared to other regions in New South Wales.

- Coal combustion hazard (Sponcom) is not represented as part of the constraint maps as it is not mapped or available digitally at the time of writing. The risk is associated with two main forms: existing coal seams that remain in the natural geology, and coal materials that remain in mine spoils that have been stockpiled or placed as fill across the Precinct and Teralba Southgate.
- Mine subsidence: all the sub-precincts are underlain by historic underground coal workings, resulting in potential subsidence risk that must be considered in land use planning and foundation design. Potential subsidence risks are not currently quantified as further studies are required.
- Limited geotechnical investigation information for strength, consistency or thicknesses of shallow soils and rocks. Existing records are generally dated, with incomplete descriptions and limited laboratory testing, and in many locations, bedrock was not intersected, meaning soil thickness cannot be confirmed. As such, foundation conditions have not been assessed in detail. Slope stability assessment is also preliminary and based primarily on available topographic data.

5.3 Methodology

The identified geotechnical criteria were classified into three ranking levels (low, moderate, and high) based on their characteristics and relative constraints to development, including foundations, earthworks, and excavations. The risk rankings adopted for each geotechnical criterion are presented in Table 5-2.

Table 5-2 Geotechnical constraints risk ranking criteria

Risk ranking classification		Description	Examples
Ranking	Colour code		
Low		Minimal limitations to development	- Sandy soils with no organic or clay content, no disturbed terrain recorded - Low risk of ASS - Low salinity hazard
Moderate		Some limitations to development	Combination of low and high risk categories and/or inherently moderate-risk categories (e.g. mine subsidence hazard).
High		Multiple limitations to development	- Organic soils (peat), soft or clayey soils - Deep soft soils - High risk of ASS - High salinity

Following assignment of the risk rankings, a multicriteria analysis was undertaken using ArcGIS. Each criterion was assigned a numerical value (low = 1, moderate = 2, high = 3) and combined using a weighted overlay approach. Equal weighting was applied to all criteria, except slope stability associated with geo zones. The Geographic Information System (GIS) layers were then overlaid to generate a composite score, which was subsequently colour-coded in accordance with Table 5-2. Higher scores indicate increasingly challenging ground conditions for development.

The risk classification and weighting adopted for each geotechnical criterion are described in the following section. It is noted that salinity and slope stability geo zones were not reclassified for this assessment, as these datasets already include risk-based classifications. This also generally applies to ASS; however, risk classifications have been locally adjusted based on engineering judgement where site conditions in areas of disturbed terrain are expected to differ from the mapped classification.

5.3.1 Slope stability criterion

The slope stability criterion and associated geotechnical considerations and adopted weighting factor for the near surface geotechnical constraint mapping are summarised in Table 5-3, and resulting slope stability risk rankings are shown in Figure 5-1. A higher-resolution version of this figure is provided in Figure B1 of **Appendix B – Constraints maps**.

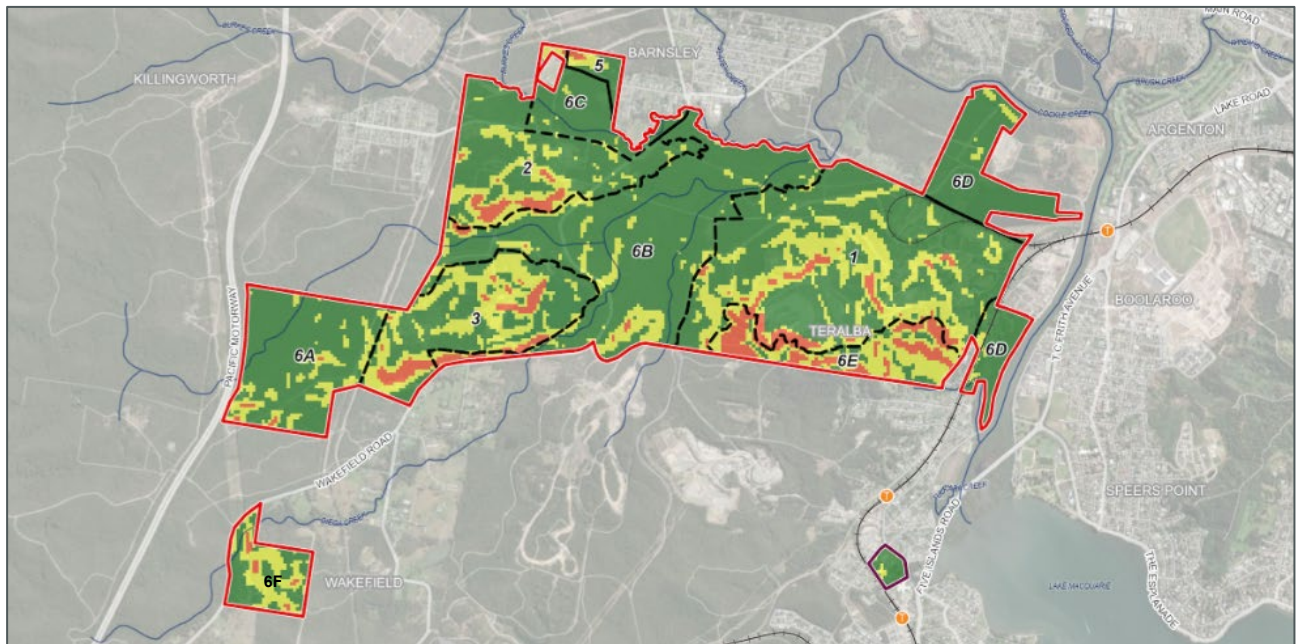


Figure 5-1 Slope stability risk mapping

Table 5-3 Slope stability criterion and weighting for near surface geotechnical constraint mapping

Slope	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
< 5 °	Minimal slope stability risk	Low	1
≤ 5 °, < 10°	Some slope stability risk	Moderate	
≥ 10 °	High slope stability risk	High	

The overall slope stability risk classification for each sub-precinct is presented in Table 5-4 with reference to the assessment presented in Section 3.1 for typical slope range and topography.

Table 5-4 Overall slope stability risk ranking for each sub-precinct

Sub-precinct No.	1	2	3	5	6A	6B	6C	6D	6E	6F	TS
Overall slope stability risk ranking ^[1]	M	M	M	M	L	L	L	L	H	M	L
Notes: L: Low; M: Moderate; H: High; TS: Teralba Southgate [1] The overall risk ranking is presented at a high level across each sub-precinct only. Localised risk may vary within individual sub-precinct areas. Refer to Figure B1 of Appendix B – Constraints maps Figure A11 Slope stability geo zones for specific risk rankings at any given location.											

5.3.2 Slope stability criterion (geo zones)

The slope stability criterion (geo zones) and associated geotechnical considerations and adopted weighting factor for the near surface geotechnical constraint mapping are summarised in Table 5-5, and resulting slope stability risk rankings against geo zones are shown in Figure 5-2 across the Precinct and Teralba Southgate. A higher-resolution version of this figure is provided in Figure B7 of **Appendix B – Constraints maps**.

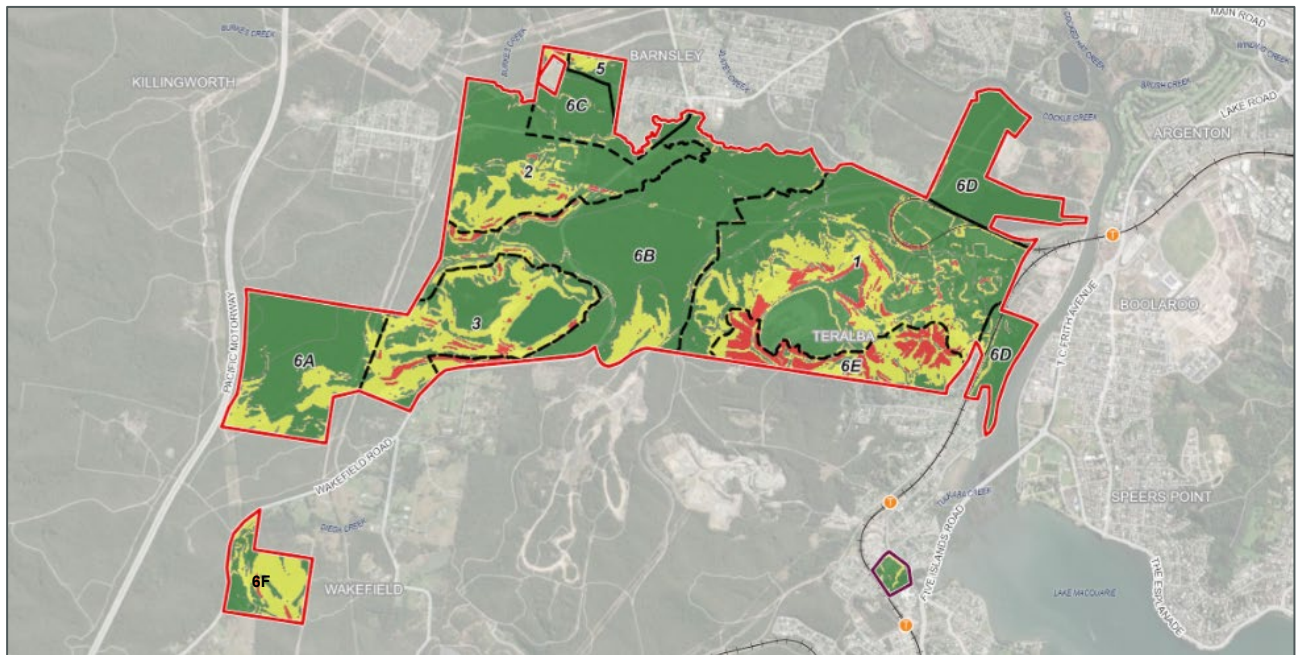


Figure 5-2 Slope stability risk mapping (geo zones)

Table 5-5 Slope stability criterion and weighting for near surface geotechnical constraint mapping (geo zones)

Geo zone ^[1]	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
Geo 5, Geo 6	Minimal slope stability risk	Low	0.5
Geo 3, Geo 4	Some slope stability risk	Moderate	
Geo 1, Geo 2	High slope stability risk	High	
Note: [1] Geo-zones relevant to the MCC Precinct and Teralba Southgate only. Conservative assumptions have been adopted where a geo zone classification overlaps two overall risk ranking categories (e.g. Geo 6), including that coal and tuff are present across the relevant rock units and that all Quaternary sediments are non-indurated.			

It is noted that the weighting factor of 0.5 applied to the near surface geotechnical constraint map reflects that the geo zones incorporate both geology and slope, which are already included as separate geotechnical criteria in this assessment based on more recent datasets for survey, as discussed in Section 3.6.

The overall slope stability risk classification for each sub-precinct is presented in Table 5-6 with reference to the assessment presented in Section 3.6 for geo zones.

Table 5-6 Overall slope stability risk ranking for each sub-precinct (geo zones)

Sub-precinct No.	1	2	3	5	6A	6B	6C	6D	6E	6F	TS
Overall slope stability risk ranking ^[1] (geo zones)	M	M	M	M	L	L	L	L	H	M	L
Notes: L: Low; M: Moderate; H: High; TS: Teralba Southgate [1] The overall risk ranking is presented at a high level across each sub-precinct only. Localised risk may vary within individual sub-precinct areas. Refer to Figure B7 of Appendix B – Constraints maps for specific risk rankings at any given location.											

5.3.3 Geology criterion

The geology criterion and associated geotechnical considerations and adopted weighting factor for the near surface geotechnical constraint mapping are summarised in Table 5-7, and resulting geology risk rankings

are shown in Figure 5-3. A higher-resolution version of this figure is provided in Figure B2 of **Appendix B – Constraints maps**.

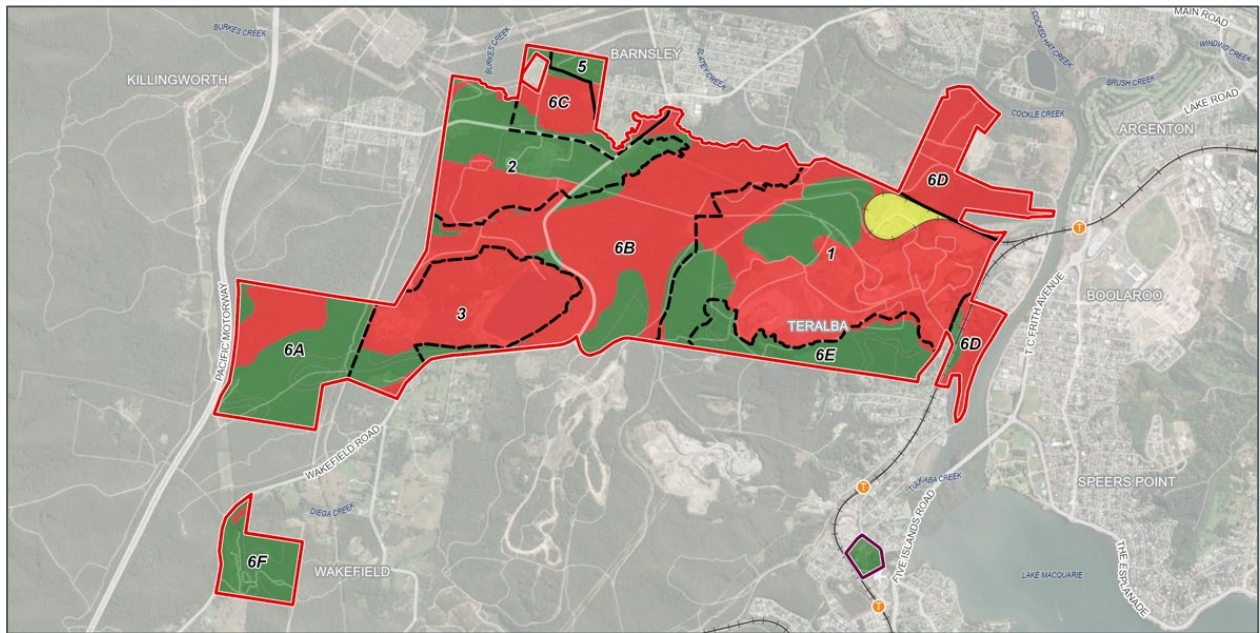


Figure 5-3 Geology risk mapping (geo zones)

Table 5-7 Geology criterion and weighting for near surface geotechnical constraint mapping

Geological category	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
Anthropogenic deposits (highly variable, may include soil, rock, building waste material)	Anthropogenic water bodies may be associated with elevated groundwater levels. Elevated groundwater levels may be associated with soft saturated soils, seepage, foundation uplift pressures, and challenges relating to excavation and slope stability requiring water management controls that may be expensive or impractical. Potential for uncontrolled fill placement to need removal and / or rework. Shallow foundations may experience excessive or differential settlement and could require preloading. Deep foundations may need increased depths, with potential excavation challenges from obstructions during excavations, uncontrolled fill, and other unpredictable ground conditions.	High	1
	Same as above, noting that the anthropogenic deposits associated with the rail loop area are not expected to be subject to excessive or differential settlement, and are therefore considered likely to present predominantly contamination constraints.	Moderate	
Alluvium (very large range of alluvial deposits that may include sands, clays and peat)	Highly compressible organic matter, specifically within estuarine deposits. These soils can cause significant total and differential settlements that can be challenging and costly to control. Highly reactive soils. Movements associated with moisture content require additional considerations in foundation design. High water table. Elevated groundwater levels may be associated with soft saturated soils, seepage, foundation uplift pressures, and challenges relating to excavation and slope stability requiring water management controls that may be expensive or impractical. Seasonal waterlogging. Changes in strength of the soils as well as volume with moisture levels can lead to defects and failures if not adequately managed in design. Highly pervious soil materials which are unstable sandy soils could present difficult hence shoring would be required.	High	

Geological category	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
Rock (conglomerate, sandstone, siltstone, claystone, coal and tuff)	Potentially highly reactive residual soils. Movements associated with moisture content require additional considerations in foundation design. Rock excavation may be required with possible constraints for earthworks, and potential limitations on material re-use due to coal. Coal outcrops are subject to spontaneous combustion; therefore, where coal seams are encountered in cut areas, excavate and backfill or blend coal for reuse in foundation treatment.	Low	

The overall geology risk classification for each sub-precinct is presented in Table 5-8, with reference to the assessment presented in Sections 3.2 and 4 for geology and preliminary ground models.

It is noted that the areas identified in Section 3.2 as having undergone historical ground disturbance within sub-precincts No. 1, 2 and 3 have been classified as high risk due to the potential presence of anthropogenic deposits in Figure B2 of **Appendix B – Constraints maps**. An exception is the area surrounding the rail loop, which has been classified as a moderate risk, as anthropogenic deposits in this area are considered unlikely to be susceptible to excessive or differential settlement.

Table 5-8 Overall geology risk ranking for each sub-precinct

Sub-precinct No.	1	2	3	5	6A	6B	6C	6D	6E	6F	TS
Overall geology risk ranking ^[1]	H	M	H	L	M	H	H	H	L	L	L

Notes:

L: Low; M: Moderate; H: High; TS: Teralba Southgate

[1] The overall risk ranking is presented at a high level across each sub-precinct only. Localised risk may vary within individual sub-precinct areas. Refer to Figure B2 of **Appendix B – Constraints maps** for specific risk rankings at any given location.

5.3.4 Depth of regolith criterion

The depth of regolith criterion and associated geotechnical considerations and adopted weighting factor for the near surface geotechnical constraint mapping are summarised in Table 5-9, and resulting depth of regolith risk rankings are shown in Figure 5-4. A higher-resolution version of this figure is provided in Figure B6 of **Appendix B – Constraints maps**.

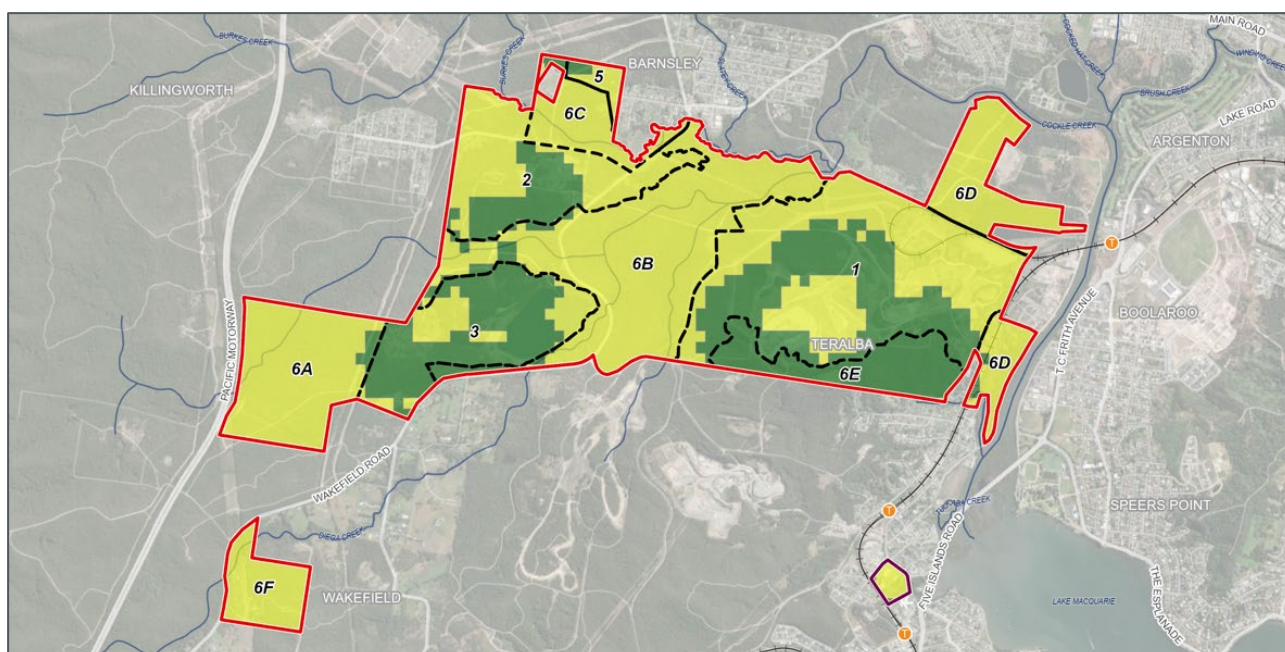


Figure 5-4 Depth of regolith risk mapping

Table 5-9 Depth of regolith criterion and weighting for near surface geotechnical constraint mapping

Depth of regolith	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
< 3 m	Settlement potential is expected to be limited. Indicative seismic sub-soil classification likely to be consistent with Class Ce.	Low	1
≤ 3 m, ≤ 20 m	Increased potential for settlement and creep, with deeper compressible soils potentially requiring ground improvement measure (e.g. preloading and wick drains). Indicative seismic sub-soil classification likely to remain Class Ce.	Moderate	

The overall depth of regolith risk classification for each sub-precinct is presented in Table 5-10, with reference to the assessment presented in Section 3.4 for depth of regolith.

Table 5-10 Overall depth of regolith risk ranking for each sub-precinct

Sub-precinct No.	1	2	3	5	6A	6B	6C	6D	6E	6F	TS
Overall depth of regolith risk ranking ^[1]	L	M	L	M	M	M	M	M	L	M	M
Notes: L: Low; M: Moderate; H: High; TS: Teralba Southgate [1] The overall risk ranking is presented at a high level across each sub-precinct only. Localised risk may vary within individual sub-precinct areas. Refer to Figure B6 of Appendix B – Constraints maps for specific risk rankings at any given location.											

5.3.5 Soil landscapes criterion

The soil landscapes criterion and associated geotechnical considerations and adopted weighting factor for the near surface geotechnical are summarised in Table 5-11, and resulting soil landscapes risk rankings are shown in Figure 5-5. A higher-resolution version of this figure is provided in Figure B3 of **Appendix B – Constraints maps**.

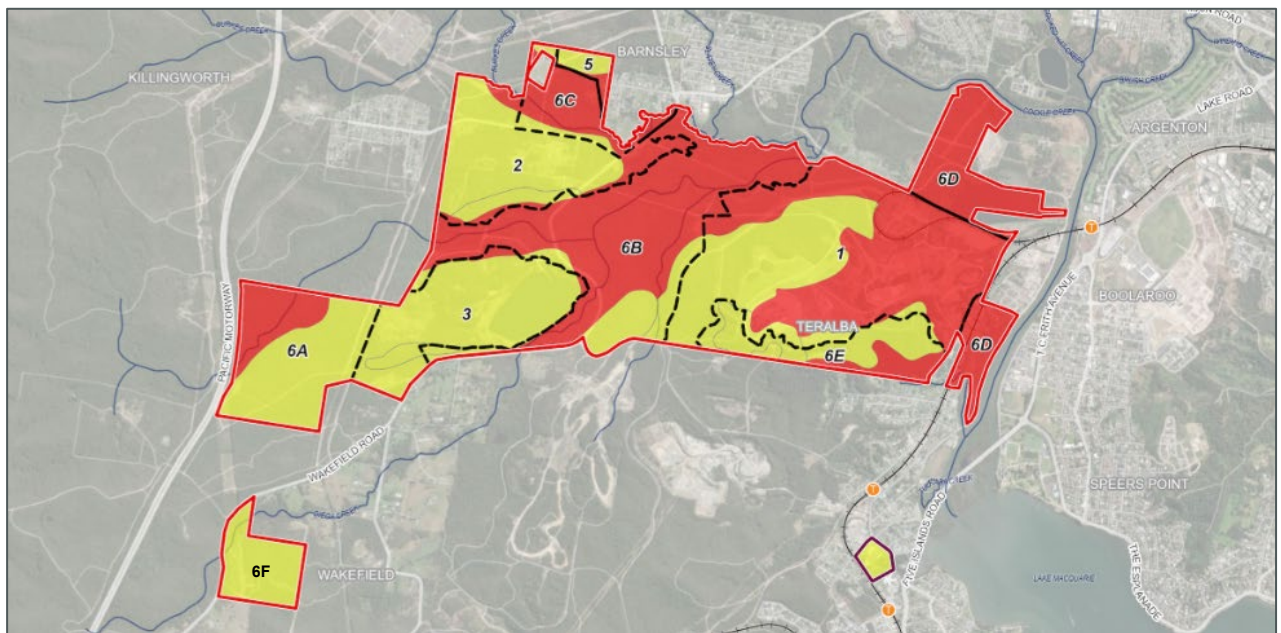


Figure 5-5 Soil landscapes risk mapping

Table 5-11 Soil landscapes criterion and weighting for near surface geotechnical constraint mapping

Soil landscape	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
Killingworth (9232ki, 9232kia)	High soil erodibility likely requiring erosion control and mitigation measures during earthworks. Seasonal waterlogging. Changes in strength of the soils as well as volume with moisture levels can lead to defects and failures if not adequately managed in design. Potentially highly reactive subsoils. Movements associated with moisture content require additional considerations in foundation design. Steep slopes inducing stability hazards, specifically within 9232kia variant. Development in areas where slopes are greater than 10 degrees is likely to require substantial earthworks, including cut-and-fill operations and localised retaining structures to achieve suitable building platforms. Mine Subsidence District, indicating a potential for underground workings and introducing a mine subsidence hazard.	Moderate	1
Cockle Creek (9232cc)	High soil erodibility likely requiring erosion control and mitigation measures during earthworks. High water table. water management controls may be more expensive or impractical. This can also lead to potential impacts on adjacent areas. Seasonal waterlogging: changes in strength of the soils as well as volume with moisture levels can lead to defects and failures if not adequately managed in design. Mine Subsidence District, indicating a potential for underground workings and introducing a mine subsidence hazard.	High	
Doyalson (9232do)	High soil erodibility likely requiring erosion control and mitigation measures during earthworks. Potentially highly reactive subsoils. Movements associated with moisture content require additional considerations in foundation design. Mine Subsidence District, indicating a potential for underground workings and introducing a mine subsidence hazard.	Moderate	
Disturbed Terrain (9232xx)	Highly variable fill composition and compaction, resulting in increased ground condition uncertainty and limited re-use potential for earthworks. Shallow foundations may experience excessive or differential settlement and could require preloading. Deep foundations may need increased depths, with potential excavation challenges from obstructions, uncontrolled fill, and other unpredictable ground conditions. Mine Subsidence District, indicating a potential for underground workings and introducing a mine subsidence hazard.	High	
Warners Bay (9232wa)	High soil erodibility likely requiring erosion control and mitigation measures during earthworks. High water table. water management controls may be more expensive or impractical. This can also lead to potential impacts on adjacent areas. Seasonal waterlogging: changes in strength of the soils as well as volume with moisture levels can lead to defects and failures if not adequately managed in design. Mine Subsidence District, indicating a potential for underground workings and introducing a mine subsidence hazard.	High	
Gateshead (9232ga)	High soil erodibility likely requiring erosion control and mitigation measures during earthworks. Mine Subsidence District, indicating a potential for underground workings and introducing a mine subsidence hazard.	Moderate	
Stockrington (9232sn)	High soil erodibility likely requiring erosion control and mitigation measures during earthworks. Steep slopes inducing stability hazards. Development in areas where slopes are greater than 10 degrees is likely to require substantial earthworks, including cut-and-fill operations and localised retaining structures to achieve suitable building platforms.	Moderate	

The overall soil landscapes risk classification for each sub-precinct is summarised in Table 5-12, with reference to the assessment presented Section 3.8 for soil landscapes.

Table 5-12 Overall soil landscapes risk ranking for each sub-precinct

Sub-precinct No.	1	2	3	5	6A	6B	6C	6D	6E	6F	TS
Overall soil landscape risk ranking ^[1]	H	M	M	M	M	H	H	H	M	M	M
Notes: L: Low; M: Moderate; H: High; TS: Teralba Southgate [1] The overall risk ranking is presented at a high level across each sub-precinct only. Localised risk may vary within individual sub-precinct areas. Refer to Figure B3 of Appendix B – Constraints maps for specific risk rankings at any given location.											

5.3.6 ASS criterion

The ASS criterion and associated geotechnical considerations and adopted weighting factor for the near surface geotechnical constraint mapping are summarised in Table 5-13, and resulting ASS risk rankings are shown in Figure 5-6. A higher-resolution version of this figure is provided in Figure B4 of **Appendix B – Constraints maps**.

It is noted that the disturbed terrains identified in Section 3.2.1 are expected to be subject to moderate ASS probability, and are therefore classified as a moderate risk.

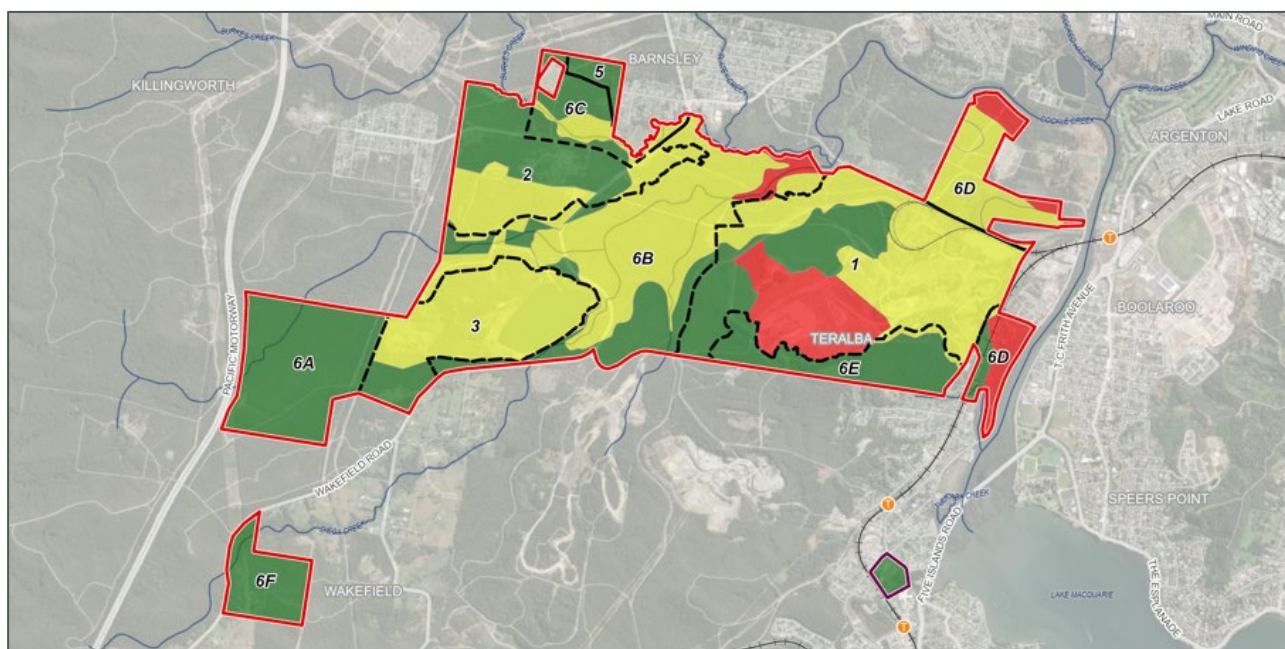


Figure 5-6 ASS risk mapping

Table 5-13 ASS criterion and weighting for near surface geotechnical constraint mapping

ASS probability of occurrence	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
No known occurrence	Minimal potential for aggressive ground conditions. Low risk during earthworks, typical spoil handling sufficient.	Low	1
Low	Potential for localised aggressive ground conditions. Excavation may intersect potential ASS requiring precautionary management and testing.	Moderate	

ASS probability of occurrence	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
High, disturbed terrain	High potential for aggressive ground conditions affecting buried concrete and steel durability. High likelihood of ASS disturbance during bulk earthworks, with acid generation and spoil management constraints requiring an ASS management plan.	High	

The overall ASS risk classification for each sub-precinct is presented in Table 5-14, with reference to the assessment presented in Section 3.9 for ASS.

Table 5-14 Overall ASS risk ranking for each sub-precinct

Sub-precinct No.	1	2	3	5	6A	6B	6C	6D	6E	6F	TS
Overall ASS risk ranking ^[1]	M	L	M	L	L	M	M	H	L	L	L

Notes:
L: Low; M: Moderate; H: High; TS: Teralba Southgate
 [1] The overall risk ranking is presented at a high level across each sub-precinct only. Localised risk may vary within individual sub-precinct areas. Refer to Figure B4 of **Appendix B – Constraints maps** for specific risk rankings at any given location.

5.3.7 Salinity criterion

The salinity criterion and associated geotechnical considerations and adopted weighting factor for the near surface geotechnical constraint mapping are summarised in Table 5-15, and resulting salinity risk rankings are shown in Figure 5-7. A higher-resolution version of this figure is provided in Figure B5 of **Appendix B – Constraints maps**.

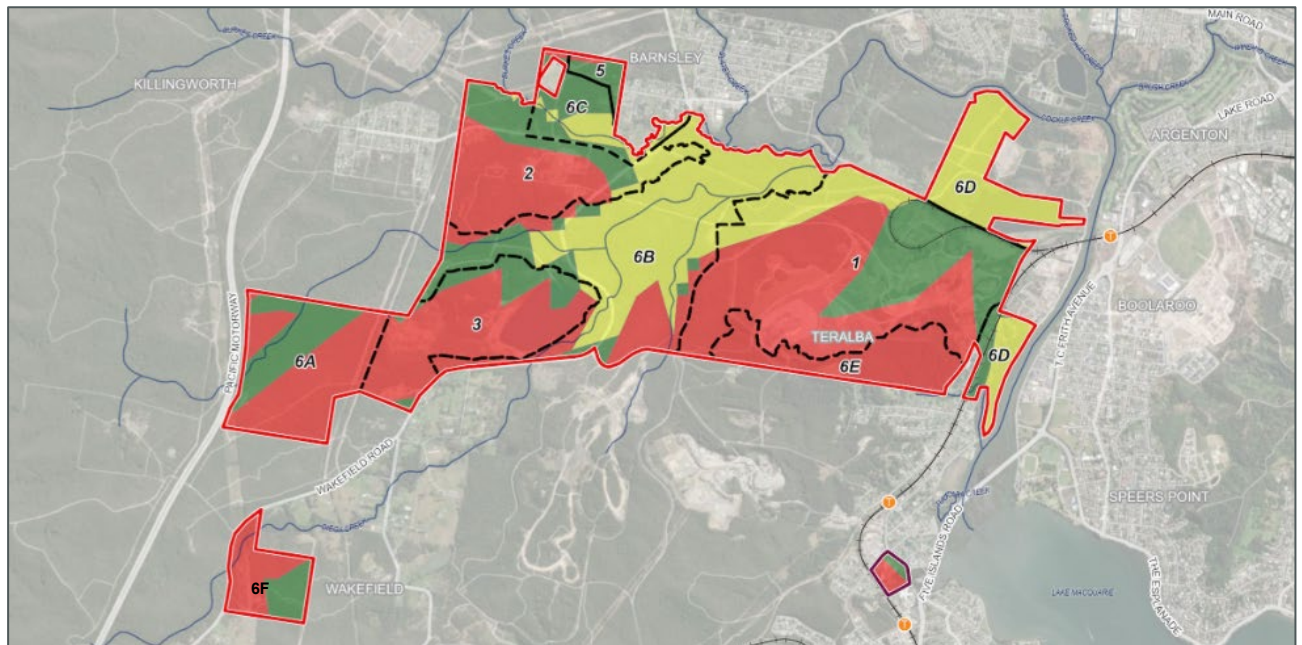


Figure 5-7 Overall salinity risk mapping

Table 5-15 Salinity criterion and weighting for near surface geotechnical constraint mapping

Overall salinity hazard	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
Low	Minimal potential for aggressive ground conditions. Low risk during earthworks, typical spoil handling sufficient.	Low	1

Overall salinity hazard	Geotechnical risk(s) for foundation, earthworks and excavations	Overall risk ranking	Adopted weighting factor
Moderate	Moderate potential for aggressive ground conditions. Earthworks and excavations may require selective spoil management.	Moderate	
Very high	Very high potential for aggressive ground conditions affecting buried concrete and steel durability. Selective spoil management, controlled dewatering discharge, and corrosion protection measures for temporary works may be required.	High	

The overall salinity risk classification for each sub-precinct is presented in Table 5-16 with reference to the assessment presented in Section 3.10 for salinity.

Table 5-16 Overall salinity risk ranking for each sub-precinct

Sub-precinct No.	1	2	3	5	6A	6B	6C	6D	6E	6F	TS
Overall salinity risk ranking ^[1]	H	H	H	L	H	M	M	M	H	H	H
Notes: <i>L: Low; M: Moderate; H: High; TS: Teralba Southgate</i> [1] The overall risk ranking is presented at a high level across each sub-precinct only. Localised risk may vary within individual sub-precinct areas. Refer to Figure B5 of Appendix B – Constraints maps for specific risk rankings at any given location.											

5.3.8 Mine subsidence criterion

Given that the entire Precinct is located within a mapped Mine Subsidence District, subsidence risk is considered to apply across the entire site. As such, Subsidence Advisory NSW will be the approving authority for any future development applications in the districts, regardless of the specific development location. It is therefore not treated as a standalone differentiating constraint between sub-precincts at this stage. Refer to Sections 7.4.8 and 8 for further details regarding the staged assessment and approval process for investigation, design and potential remediation, such as grouting works.

5.4 Summary of near surface geotechnical risk ranking per sub-precinct

Table 5-17 presents a high-level summary of the near surface geotechnical risk ranking for each sub-precinct. The near surface geotechnical risk ranking across the sub-precincts range from low to high, with sub-precincts No.1, 6B and 6Dat classified as high. As noted previously, the mine subsidence risk rating is subject to more detailed studies and is not included in the ranking process.

These rankings should be read in conjunction with the detailed constraint maps presented in Figure 5-8 and in **Appendix C – Near surface geotechnical constraints map for Structure Plan**. The maps combine the geotechnical constraints and illustrate localised variations in risk across the sub-precincts.

Table 5-17 Near surface geotechnical risk ranking per sub-precinct

Geotechnical criteria	Adopted weighting factor	MCC Sub-precinct No.										Teralba Southgate
		1	2	3	5	6A	6B	6C	6D	6E	6F	
Overall slope stability risk ranking ^[1]	1	M	M	M	M	L	L	L	L	H	M	L
Overall slope stability risk ranking (geo zones) ^[1]	0.5	M	M	M	M	L	L	L	L	H	M	L
Overall geology risk ranking ^[1]	1	H	M	H	L	M	H	H	H	L	L	L
Overall depth of regolith risk ranking ^[1]	1	L	M	L	M	M	M	M	M	L	M	M
Overall soil landscapes risk ranking ^[1]	1	H	M	M	M	M	H	H	H	M	M	M
Overall ASS risk ranking ^[1]	1	M	L	M	L	L	M	M	H	L	L	L
Overall salinity risk ranking ^[1]	1	H	H	H	L	H	M	M	M	H	H	H
Overall risk ranking ^[1]		H	M	M	L	M	H	M	H	L	L	L
Notes: L: Low; M: Moderate; H: High [1] The overall risk ranking is presented at a high level across each sub-precinct only. Localised risk may vary within individual sub-precinct areas. Refer to Figure C of Appendix C – Near surface geotechnical constraints map for Structure Plan for specific risk rankings at any given location.												

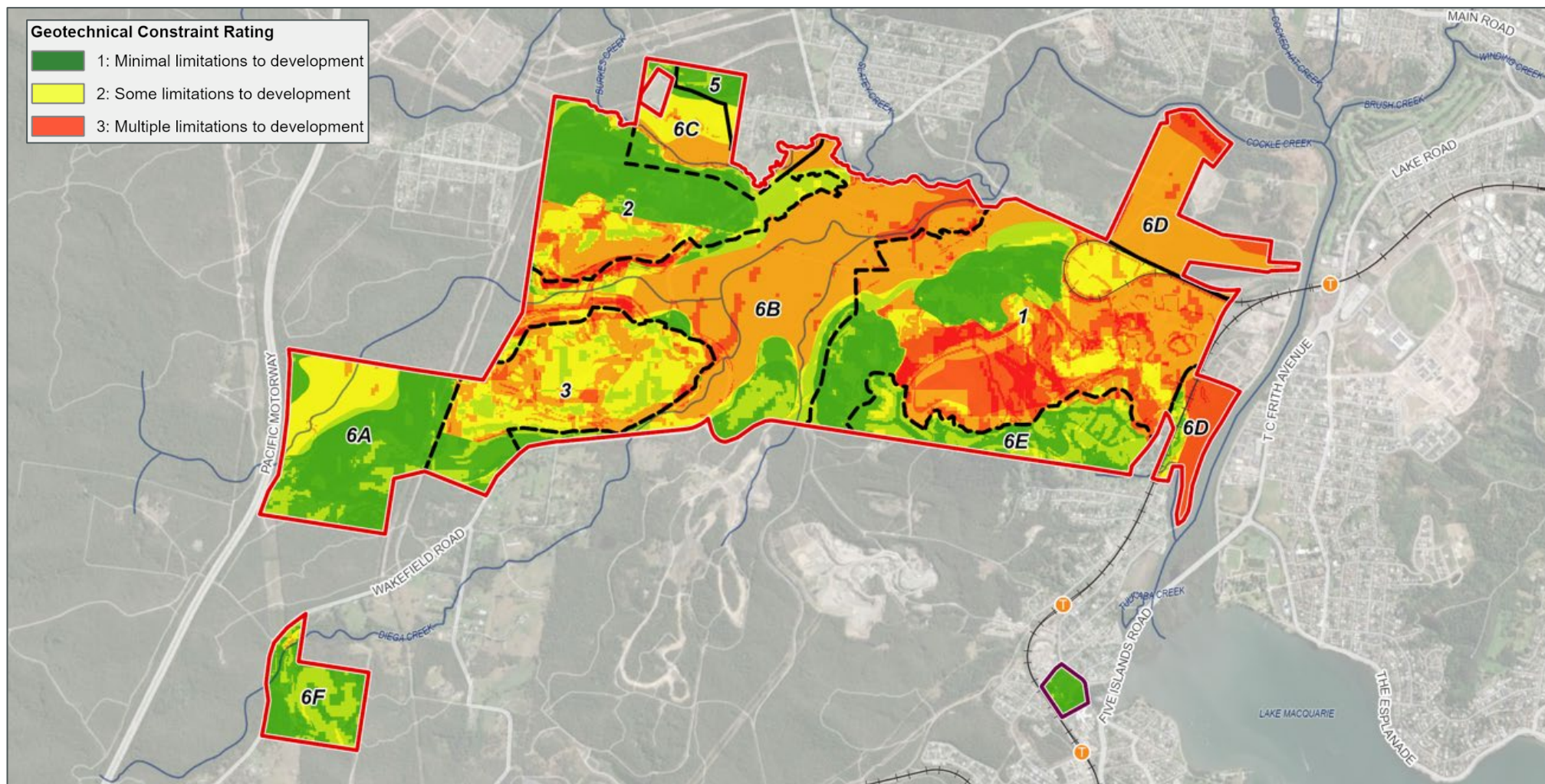


Figure 5-8 Near surface geotechnical constraints map

5.5 TSF area risk ranking

At the time of the preparation of this Report the TSF area was undergoing placement of capping to achieve the final landform. As such, final settlement has not commenced yet. Large settlements are expected and no development can occur until such time that the TSF risk rating is reduced sufficiently to allow the structure to be de-declared.

The TSF is constrained by:

- Dam Safety management requirement
- Ongoing and historical earthworks
- Hydraulically placed coal residue
- Groundwater levels with the storage
- Underlying natural ground profile that has potential regolith depths of up to 10 m.
- Steep surrounding slopes
- Variable storage depth

The ATC Williams report indicates that the depth of the storage area is variable and is expected to be formed of the original valley profile that drains to the original dam wall within the eastern side of the storage. No records are available that include estimates of long-term settlement; however, it would be expected to be in the order of metres over decades.

The storage embankments are wider on the eastern slopes and it is understood that this is where the final landform capping material is also being won from currently.

Around the TSF are natural and man-made slopes. Some of which are steep. Along the southern margin is an excavated batter.

These factors result in significant geotechnical constraints to future development, particularly due to settlement potential and variable subsurface conditions.

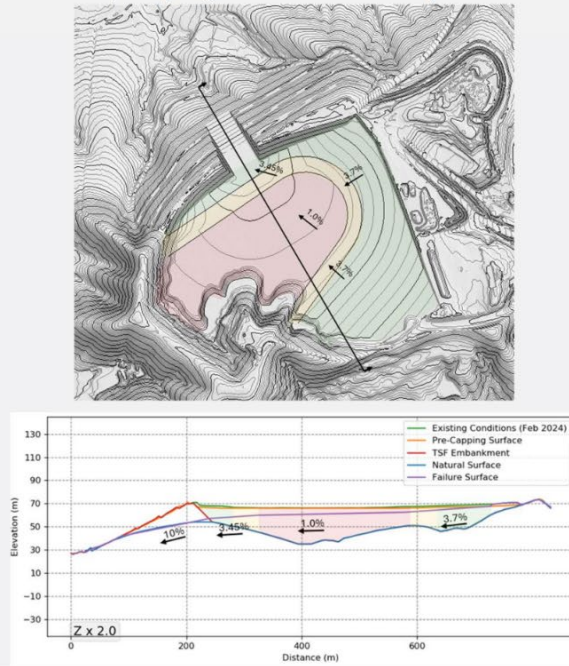
Given the current earthworks, the existing datasets are considered to provide a limited representation of geotechnical risk within the TSF area, as they ground conditions that are unlikely to correspond to the final rehabilitated landform. The risk ranking presented in Table 5-18 for the TSF footprint is therefore provided for reference only.

Table 5-18 Overall risk ranking within the TSF area (for reference only)

Geotechnical criteria	Adopted weighting factor	TSF area
Overall slope stability risk ranking	1	H
Overall slope stability risk ranking (geo zones)	0.5	H
Overall geology risk ranking	1	H
Overall depth of regolith risk ranking	1	M
Overall soil landscapes risk ranking	1	H
Overall ASS risk ranking	1	L
Overall salinity risk ranking	1	H
Overall risk ranking ^[1]		H

Simplified cross sections of the TSF from the modelled failures scenarios presented in ATC Williams report (2025) are shown in Figure 5-9 and illustrate the varying position of the neutral surface beneath the TSF. Cross sections of the northern and eastern TSF embankments are shown in Figure 5-10 and Figure 5-11, respectively. As can be seen, the eastern embankment is significantly wider and thus the ATC studies focused on failure modes through the more critical northern embankment.

Base Case (5m plug based on stability assessments)



Sensitivity (Full-height breach)

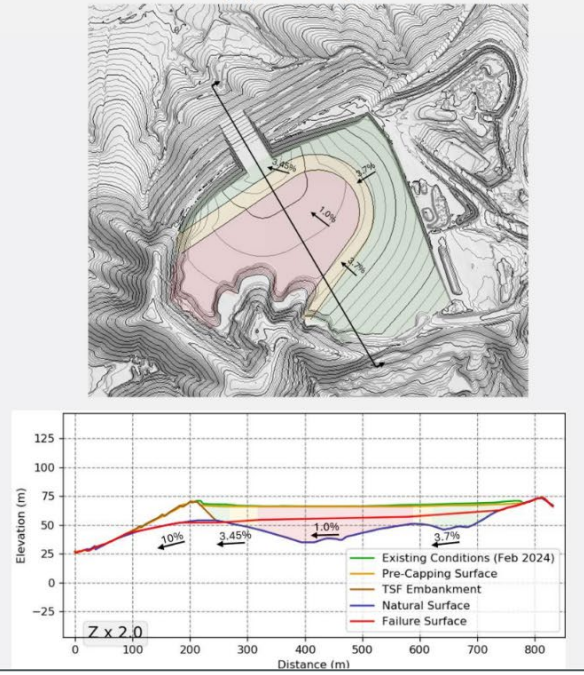


Figure 5-9 Cross sections of the TSF storage (ATC Williams, 2025)

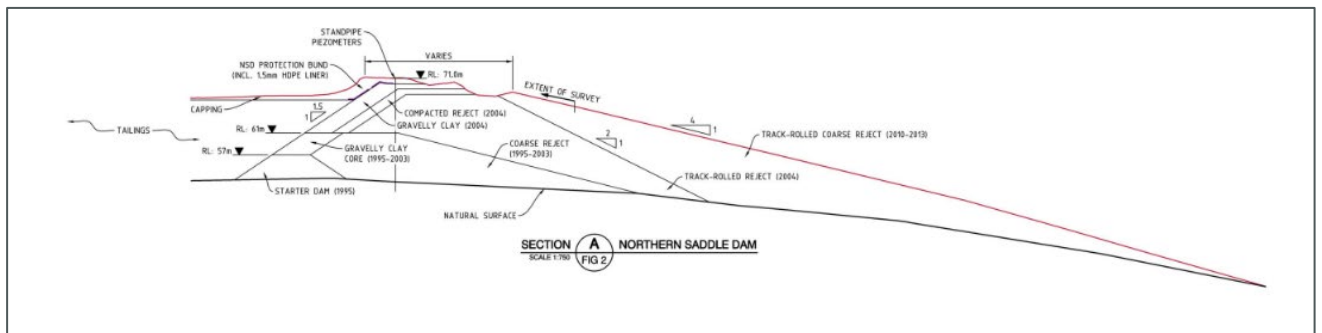


Figure 5-10 Section of eastern embankment area (ATC Williams, 2025)

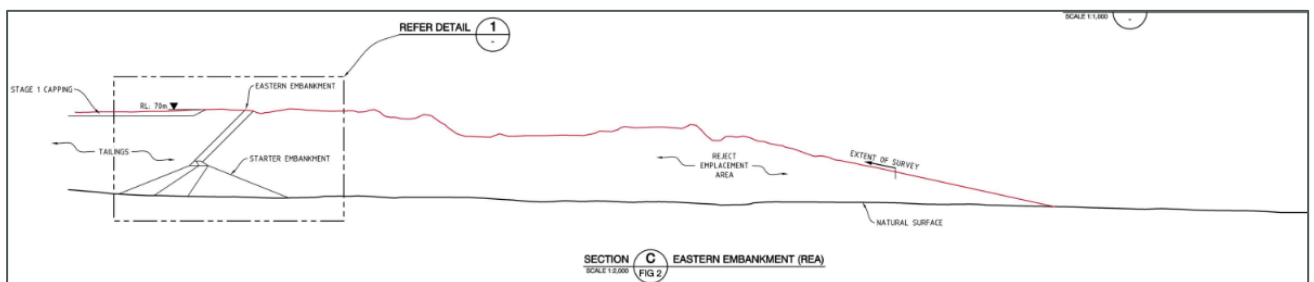


Figure 5-11 Section of eastern embankment area (ATC Williams, 2025)

6 Land use scenario

Section 7 of this Report provides a summary of the scenario development during the Enquiry by Design workshops held on 7 and 8 May 2026, then on 1 June 2026 which involved implementing visions and concepts, identifying challenges and developing innovative solutions at a precinct-wide level across all technical streams. Scenarios were developed and refined by GHD for the Precinct Structure Plan boundary. They considered land use, transport, infrastructure, PFAS, environmental, social, aboriginal heritage and economic matters in conjunction with the MCC vision. Each of the scenarios tested identifies the development limitations, constraints management and required infrastructure that would be required to support the respective structure plan's proposed development. This information was subsequently utilised at a second Enquiry by Design workshop to inform the Structure Plan.

6.1 Key scenario findings

As part of the scenario testing the methodology described in Section 5 was used to assess each option.

From a geotechnical perspective the scenarios were relatively comparable based on the testing criteria used in Section 5.1.

A key finding from the scenario testing phase was that development west of Cockle Creek and Teralba Southgate is likely to be more cost-effective, due to the greater extent of potential ASS, deep soft organic soils near the creek and tributaries. Areas of disturbed ground and anthropogenic fill within the Cockle Creek, Killingworth and West Lake sub-precincts, as well as around the coastal wetland in the north of the Precinct, may also present ongoing geotechnical constraints to development.

It is also noted that limited geotechnical data was available across the MCC structure plan area to further refine the assessment criteria (as outlined in Section 3.13). Accordingly, additional geotechnical investigations and laboratory testing will be required during subsequent stages of the MCC planning and design process to confirm ground conditions and better define the extent and distribution of the soils present within each scenario area.

7 Structure plan assessment

7.1 Approach

Reference to the Master Plan identifies the following key principles:

- Enable transition industries
- Connect infrastructure and mobility
- Preserve ecology and water
- Deliver flexible pathways and resilience

The Structure Plan has been developed by incorporating the preferred elements of the assessed scenarios, further refining areas identified for investigation, and avoiding established no-go zones. Rather than adopting a single scenario, the Structure Plan combines the key opportunities identified across all scenarios while seeking to minimise known constraints and risks. This approach aims to maximise development outcomes while maintaining flexibility for future planning, environmental and engineering requirements.

7.2 Proposed Structure Plan

The Structure Plan has been developed to support the transition of the MCC Precinct and Teralba Southgate from a former mining operation to a future employment Precinct. The Structure Plan adopts a staged development approach aligned with rehabilitation progress, infrastructure availability and risk reduction. Early development is focused on previously disturbed and serviced areas within the Cockle Creek, Killingworth and Teralba sub-precincts. Medium-term development is concentrated within Cockle Creek, Killingworth, West Lake and Barnsley West, and is dependent on further geotechnical and environmental investigations, infrastructure upgrades and utility provision. Long-term development of the TSF and dam break inundation areas is contingent on completion of rehabilitation, geotechnical stabilisation and dam safety requirements. Land release across all stages will be informed by ongoing geotechnical, environmental, flooding and servicing assessments.

The Precinct and Teralba Southgate benefit from strong regional connectivity, established transport infrastructure and its location within the Lower Hunter industrial corridor, supporting a range of future employment opportunities. The Killingworth and Barnsley sub-precincts are well suited to early employment uses linked to existing residential areas, while Teralba presents opportunities for transport-oriented mixed-use development due to its proximity to existing rail stations.

Previously disturbed and rehabilitated land across Cockle Creek, Killingworth and West Lake provide suitable areas for staged development, enabling growth to be directed away from higher-value ecological corridors. The West Lake sub-precinct also presents opportunities for recreation and nature-based tourism in association with surrounding environmentally constrained areas.

Refer to Figure 2-1 for the MCC Structure Plan Precinct and Teralba Southgate development areas and sub-precinct layout.

7.3 Geotechnical assessment of the Structure Plan

An interpretation of the geotechnical constraints associated with the Structure Plan is presented as a constraints map in **Appendix C – Near surface geotechnical constraints map for Structure Plan**. The map has been developed using the same multi-criteria analysis described in Section 5, based on the key datasets outlined in Table 5-1.

Relative land uses, anticipated structural characteristics, ground conditions and overall risk ranking for each sub-precinct are presented in Table 7-1.

Table 7-1 Sub-precinct land use intents and anticipated structural characteristics

Sub-precinct		Type	Land use intent(s)	Anticipated structural characteristics	Anticipated ground conditions	Overall risk ranking
No.	Name					
1	Cockle Creek	Developable area	General industrial, manufacturing, logistics, warehousing and freight	Shallow foundations on competent natural ground or in engineered fill, with possibly some ground improvement or deeper piles foundations for heavier load areas.	- Localised fill to the north and within TSF area and in areas identified in Section 3.2.1. - Underlain by localised alluvium in the north. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected between 3.0 and 10 m BGL within the TSF and northern areas, reducing to approximately 1.0 – 2.0 m BGL elsewhere, broadly consistent with the existing test locations within the sub-precinct. Groundwater expected to be relatively shallow towards Cockle Creek to the north and other low-lying areas. Perched groundwater may occur within cohesionless fill and alluvial deposits. Slopes typically vary between 5 and 10 degrees.	High
2	Killingworth		Light industrial, energy intensive uses, technology uses, small-medium employment uses and service commercial	Shallow foundations in engineered fill typically, with possibly some ground improvement or deeper piles foundations for heavier load areas.	- Localised fill in areas identified in areas identified in Section 3.2.1. - Localised alluvium in the east. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected between 3.0 and 10.0 m BGL reducing to approximately 1.0 - 2.0 m BGL in elevated southern areas. Groundwater is expected to be relatively shallow within drainage lines and other low-lying areas. Slopes are typically less than 5 degrees.	Moderate
-	Teralba Southgate		Medium-density residential, activated commercial and community uses	Shallow foundations on competent natural ground or in engineered fill.	- Residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected around 2.0 m BGL. Groundwater is expected to be relatively deep. No groundwater was recorded within the boreholes of reference. Slopes are typically less than 5 degrees.	Low

Sub-precinct		Type	Land use intent(s)	Anticipated structural characteristics	Anticipated ground conditions	Overall risk ranking
No.	Name					
5	Barmsley West		Light industrial / commercial	Shallow foundations on competent natural ground or in engineered fill.	- Alluvial deposits towards Cockle Creek and tributaries. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL beneath deeper alluvial deposits. Groundwater expected to be shallow in the vicinity of Cockle Creek and associated tributaries. Perched groundwater may occur within cohesionless alluvial deposits. Slopes typically vary between 5 and 10 degrees.	Low
3	West Lake	Mixed use	Conservation, passive and active recreation, nature- based tourism and complementary low-impact uses	Shallow foundations on competent natural ground or in engineered fill (minimal structural loading).	- Fill in areas identified in areas identified in Section 3.2.1. - Localised alluvial deposits to the west towards Diega Creek and tributaries. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected between 1.0 and 2.0 m BGL, locally ranging from <1.0 m to 3.0 m BGL (west) and 3 m to 10 BGL (east), broadly consistent with the existing test locations within the sub-precinct. Groundwater is expected to be relatively deep, except within local drainage lines. Groundwater expected to be shallow adjacent to Diega Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits. Slopes typically vary between 5 and 10 degrees.	Moderate
6A	Environmentally constrained areas	Conservation / recreation areas	Biodiversity corridors, low-impact recreational uses, flood, landscape amenity	No structural foundation requirement anticipated.	- Localised alluvial deposits near Cockle Creek and tributaries. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL in the north towards Cockle Creek. Groundwater expected to be shallow adjacent to Cockle Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits. Slopes are typically less than 5 degrees.	Moderate

Sub-precinct		Type	Land use intent(s)	Anticipated structural characteristics	Anticipated ground conditions	Overall risk ranking
No.	Name					
6B					- Alluvial deposits near Cockle Creek, Diega Creek and tributaries. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL in the north towards Cockle Creek. Existing test locations suggest shallow bedrock (< 1 m) towards Wakefield Road in the south. Groundwater expected to be shallow adjacent to Cockle Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits. Slopes are typically less than 5 degrees.	High
6C					- Alluvial deposits near Burkes Creek. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL towards Cockle Creek. Groundwater expected to be shallow adjacent to Cockle Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits. Slopes are typically less than 5 degrees.	Moderate
6D					- Localised fill in the north. - Underlain by alluvial deposits, including estuarine deposits with highly compressive organic matter. Existing investigations recorded approximately 2.5 m of soft to firm cohesive alluvial soils on average along Racecourse Road, increasing to 5.5 m in borehole 91-HE89-B6. Alluvium thickness is to be confirmed, as bedrock was not generally encountered in the available investigations. - Underlain by residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, locally 20 - 30 m BGL towards Cockle Creek, broadly consistent with the existing test locations within the sub-precinct. Groundwater encountered at approximately 1.5 - 2.8 m BGL along Racecourse Road, in the vicinity of Cockle Creek. Perched groundwater may occur within cohesionless alluvial deposits. Standing groundwater recorded at approximately 3.8 - 4.2 m BGL to the north. Slopes are typically less than 5 degrees.	High

Sub-precinct		Type	Land use intent(s)	Anticipated structural characteristics	Anticipated ground conditions	Overall risk ranking
No.	Name					
6E					- Residual soil/extremely weathered material associated with the Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected around 2.0 m BGL. Groundwater is expected to be relatively deep, except within local drainage lines. Slopes are typically greater than 10 degrees.	Low
6F					- Localised alluvium in the north. - Underlain by residual soil/extremely weathered material associated with the Narrabeen Group and Newcastle Coal Measures rocks. - Top of highly weathered or better rock expected at approximately 3 - 10 m BGL, broadly consistent with the existing test location near the sub-precinct. Groundwater expected to be shallow adjacent to Cockle Creek and tributaries. Perched groundwater may occur within cohesionless alluvial deposits. Slopes typically vary between 5 and 10 degrees.	Low

7.4 Constructability

7.4.1 Flood management

Riverine flooding would be expected to induce scour on any developments that border the flood risk areas or if developments are planned within these areas. Failing to mitigate flooding may cause excessive total or differential settlements in the underlying soft clays soils potentially present in the sub-precincts No.5, 3, 6A, 6B, 6C and 6D.

As such, any new development planned in or close to modelled flood regions should be designed to incorporate flood resilience features:

- Scour protection may be necessary for fill embankments or considered in foundation designs of bridges or culverts and other structures.
- New developments, such as ground floor car parking and first floor offices will reduce damage and the need for inspections and repair after flood events.
- Fill embankments should be suitably specified to use free draining materials, adequate erosion control measures to avoid fines washout, or incorporate draining base layers. The same is applicable to associated structures such as bridge abutments or retaining walls.
- Levees if required along the proposed boundaries to flood risk areas typically need to be constructed using clay rich fill or other less permeable soil material as liner material.

Reference should be made to the Flooding and Water Cycle Management Engineering Report for details related to flood management.

7.4.2 Earthworks considerations

Alluviums and residual soils are generally expected to be readily excavatable using conventional excavator equipment (20 tonne mass excavator). Where harder soil consistencies are encountered, the use of a toothed bucket / ripper tyne may be required. Suitability for re-use may be limited to general fill depending on erosion potential, reactivity, material consistency and presence of organic soils, noting that residual soils are expected to be reactive based on the constraints associated with the mapped soil landscapes. Existing fill areas may contain contaminants and are therefore considered unsuitable for reuse without further assessment.

Within the Narrabeen Group and Newcastle Coal Measures, excavatability is likely to vary depending on rock properties such as strength and degree of natural fracturing. Typically, excavation within the less weathered, higher strength and intact sandstone rock may require rock saws and/or hydraulic rock breakers. Conglomerate may require more intensive ripping due to its greater strength. Coal and tuff units may present excavation challenges and are generally unsuitable for reuse as engineered fill without further assessment.

Where slopes are greater than 10 degrees, they are generally considered unsuitable for conventional industrial and commercial developments that require large, level building pads and hardstand areas. At this stage, no preliminary assessment has been undertaken to quantify the extent of cut and fill earthworks, the likely need for retaining structures, or the associated impacts on developable land, construction costs, drainage, and overall site feasibility. A detailed earthworks assessment, incorporating terrain modelling, cut/fill balance, retaining requirements, and optimisation of development platforms, is recommended to better understand the site's development constraints and associated costs.

Where coal seams or coal bearing materials are encountered in cut areas in rock, specific management measures may be required due to the potential for spontaneous combustion and long-term geochemical instability. In these instances, coal should either be selectively excavated and removed from site or alternatively treated through controlled blending with inert materials for reuse as general fill or foundation improvement layers.

The risk of spontaneous combustion is also potentially present in reworked fill material that include coal material.

Earthworks planning should incorporate segregation of coal-impacted materials, appropriate stockpile management (including temperature and moisture monitoring where required), and controls to prevent oxidation and ignition risk.

Details regarding gas management are provided in the Contamination Engineering Report.

7.4.3 Shallow foundations

Shallow foundations are anticipated across sub-precincts No. 1, 2, 3, 4 and 5. In sub-precincts No. 1, 2, 3 and 5, where shallow groundwater conditions are expected, the use of shallow foundation solutions is likely to reduce geotechnical challenges. In these areas, founding above typical groundwater levels may minimise the need for temporary excavation support and dewatering, thereby reducing construction risk and cost. Additionally:

- The risk of differential and total settlements beneath new structures should be assessed and addressed during the design stage. If differential settlement is likely to exceed acceptable limits, foundation treatments such as removing reactive soils within the suction zone and replacing them with non-reactive engineered granular fill should be considered.
- Another solution may include surcharge preloading of proposed footprints to improve underlying soils (particularly sandy soils) to support fill pads. Preloading is also effective in reducing long-term creep deformations for soft to firm clay deposits beneath embankments. The suitability and extents of preloading will depend on the proposed infrastructure and underlying ground conditions. Typically, cohesive soils (clays) require longer consolidation periods than free draining soils such as sands. Where mixed soil profiles are encountered, ground treatment approaches will need to be adapted accordingly.
- The surcharge period required to achieve acceptable settlement performance in accordance with serviceability requirements depends on the thickness of the soft/compressible clay deposits and the magnitude of applied preload. While increasing surcharge loads can reduce consolidation timeframes, care must be taken to avoid instability in weak ground conditions. Typical preload durations may range from 3 to 18 months; however, project specific analysis is required to confirm suitable durations. Vertical drains may be installed to accelerate consolidation. The use of lightweight fill materials in combination with surcharge loading is also an accepted construction approach. Where final levels permit, non-engineered fill may be used as surcharge material above required grade levels.
- For light industrial developments such as warehousing and offices buildings, shallow foundations are generally expected to be sufficient. However, heavily loaded point structures within warehouses, such as gantry cranes, will likely require deep foundation solutions.

7.4.4 Deep foundations

It is recommended that heavy structures requiring deep foundations, such as basement car parks (which may be associated with land uses in sub-precincts No. 1 and 2), adopt driven piles or Continuous Flight Auger (CFA) pile solutions where practicable. These methods generally reduce the volume of excavated material and the need for groundwater management compared to bored piles. Additionally:

- Pile foundations will need to be designed to account for uplift forces associated with flooding, as well as potential negative skin friction in consolidating soils.
- Where founding conditions extend into rock within a mine impacted strata, rock socketing may be suitable; however, localised mine grouting or void treatment may be required. Structural systems should also accommodate potential ground variability through appropriate flexibility.
- Stakeholder requirements, including rail and road authorities, may require additional design constraints on foundation selection, construction methodology and allowable movements, which will need to be confirmed during design.

7.4.5 Utilities

Utilities associated with light industrial, commercial, residential and general industrial development, primarily within Sub-precincts No. 1, 2, 4 and 5, should consider the following:

- Underground services are typically installed within trench excavations. Shallow groundwater conditions, together with the potential presence of ASS, may increase construction complexity and cost.
- Most service trenches are expected to require shoring and dewatering, particularly in areas of shallow groundwater.
- Surficial soils are expected to be easy to excavate. Where shallow rock may be encountered in sub-precincts No.2, No. 4, and in localised areas of sub-precinct No.1, the excavatability of rock will depend on its strength, weathering and degree of fracturing. Excavation of less weathered, higher-strength rock may require specialised equipment such as rock saws or hydraulic rock breakers.
- Above ground utilities such as overhead power lines may require deep foundations to prevent differential settlement issues.

7.4.6 Pavements and roads

The surficial soils across much of the Precinct, particularly soft alluvial deposits, are expected to exhibit low California Bearing Ratio values and therefore limited subgrade strength.

Organic soils may be present near Cockle Creek and its tributaries, while highly reactive clays within alluvial and residual soils are expected across much of the Precinct. These conditions may require ground treatment measures such as removal and replacement or ground improvement. Where road embankments or fill platforms are proposed over compressible soils, preloading and/or basal reinforcement may be necessary to manage settlement associated with long-term consolidation.

7.4.7 Acid Sulfate Soils

Areas identified as having a higher potential for ASS, particularly within sub-precincts No.2, 6B, 6C and 6D, will require ASS testing as part of future site investigations. Where ASS are confirmed, preparation and implementation of an ASS management plan is likely to be required during construction.

7.4.8 Mine subsidence

Given the historical mining activities across the entire MCC Precinct and Teralba Southgate and Mine Subsidence District zoning, future development applications would trigger a referral to the SA pursuant to the Coal Mine Subsidence Compensation Act 2027 (integrated development).

Initial engagement with Subsidence Advisory NSW has occurred at the time of writing and the advice received from the agency will be captured in the finalisation report.

Optioneering of layouts may potentially reduce subsidence risks and thus approval requirements. Development will need to follow Subsidence Advisory NSW procedures and approval conditions, including staged review and engagement.

The proposed development type will govern the approval pathway and may take the form of the guidelines or the Merit Assessment Policy (SA NSW version 2 5 April 2023).

The accuracy of the mapping and precise locations of the historic workings should be assumed to be approximate at this stage. To illustrate a typical application process for a mine subsidence affected development area the following steps are summarised below:

1. Identify affected properties on SixMaps – Lot and DP etc. of affected portions or adjacent portions where applicable
2. Search publicly available databases for current mining / exploration leases

3. Meet with SA NSW to get informal advice on classification of site with respect to Mine Subsidence conditions
4. If applicable, lodge RFI with SA NSW to get formal notification of Mine Subsidence requirements (if any). This may include any, or all, of the followings:
 1. Confirmation that the site is in a proclaimed Mine Subsidence district
 2. Prescriptive advice on specific types of development that are allowed / not allowed
 3. Prescriptive advice on the Mine Subsidence Surface Movement Parameters that must be applied to development of the site (e.g. Total maximum subsidence, ground tilt, ground strain, radius of curvature etc.)
 4. Requirements for additional site investigation (e.g. Drilling to identify existing mine voids in the case of existing bord and pillar workings, drilling to identify existing site geology and geotechnical conditions etc.)
 5. Requirements for additional analysis and report on predicted mine subsidence induced surface movement parameters
5. Request mining records from the NSW resource regulator (a small payment is often required)
6. Preparation of a desktop review of historical mine layouts and assessment of the quality of the records and associated risk assessment
7. If applicable, preparation of a pillar stability assessment numerical modelling report
8. If applicable, geotechnical investigation to confirm extent, depth and condition of workings
9. If applicable, preparation of a Mine Subsidence Impact Statement (MSIS) for submission with SA NSW
10. Follow up meeting with SA NSW to discuss the recommendations of the MSIS and agree on final design requirements
11. SA NSW issue final conditions for the development

The initial step of the desktop based assessment will provide a better understanding of the overall mining history and associated subsidence risks and as such is a key first step to understand development pathways and clarity on require further works for development approval and costs.

Typically, the following subsidence parameters would be specified in the conditions of approval:

- Vertical Subsidence (x mm)
- Compressive Strains (y mm/m)
- Tensile Strains (z mm/m)
- Tilts (w mm/m)
- Radius of Curvature (v km)

Subsidence Advisory NSW will review the subsidence risk specifically for a proposed development and select the requirements based on the policy associated with that type of development. As such, the guidelines and polices are tailored to development types, and developments with tighter or lower tolerances to settlement types will have more requirements to achieve approval in general. So early engagement can help inform options to adjust layouts before progressing to a more developed design stage (e.g. Concept Design)

As a general guide for mitigation measures, elements of any developments should be designed as far as practicable and economical, to maintain safe, serviceable operation in the event of subsidence and reduce the cost of repair through:

- Selection of flexible and lightweight materials rather than brittle materials

- Provision for articulation, adjustment and ease of replacement or repair
- Reducing transfer of ground strains to structure footings
- Consideration of post-subsidence safety and functionality of drains and surface water storages elements such as stormwater detention basins, swales and gravity pipes

Mine subsidence and design to manage the associated risks is common in the Hunter region and specialist firms are available to support development applications and construction work such as grouting of underground voids. As the Precinct is located within the Lake Macquarie LGA it is potentially eligible for funding assistance through the Lake Macquarie Mines Grouting Fund, with application processes outlined in Council guidelines (Program Guideline V1, March 2025) though this would need to be confirmed with LMCC.

Should geotechnical investigations be deemed necessary to improve the accuracy of mapping of the underground layouts or to confirm if workings have already collapsed then these would typically comprise boreholes drilled through the seams in question. Whilst the depths of the seams are deep, typical methods only retrieve samples or rock core from the zone of interest and as such the upper zone of each borehole is typically not sampled and can be drilled quickly. Standard geotechnical methods and testing within the boreholes is then complimented by Mine subsidence specific methods which can include:

- Geophysical downhole logging (Density, Neutron and Gamma scanning)
- Downhole camera inspection
- Sonar mapping
- Boreholes installed with high density PVC pipe and capped for future grouting works.

If mine grouting works are deemed to be required for new developments this is typically completed from the surface via grout injection boreholes. Modern techniques include low viscosity grout to create or increase pillar sizes within the voids and thus reducing the volumes of grout required and also grouting in flooded voids.

The development designer is required to certify on the drawings that improvements covered by the approval will remain safe, serviceable and repairable in the event of the structures being subjected to the specified subsidence.

Across the Precinct, areas above longwall panels would be expected to have already undergone settlement associated with mining and, as such, more likely endorsed by SA (subject to conditions). Bord and Pillar workings are considered therefore to have a higher likelihood of remaining and potentially more variable subsidence risk associated with this older method of mining.

Some regions have multiple seams already mined and varying methods and are therefore expected to have more complicated assessments and development approvals.

Where significant thick and strong rock units exist above the former working (e.g. Munmorah Conglomerate) these may provide better ability to span the ground subsidence and decrease the surface settlement amounts.

Opportunities include:

- Undertaking a Precinct-wide subsidence risk assessment to support developers' early investment assessments
- Coordinating with Lake Macquarie Council to simplify or pre-approve selected application stages and identify potential funding opportunities.

7.4.9 TSF area

No development can occur on or downstream of the TSF until the TSF is 'de-declared'. Once the structure is de-declared, ground investigations and design considerations will be required.

Geotechnical investigations are expected to potentially include the following test types:

- Cone Penetrometer Tests (CPT)

- Oedometer tests
- Monitoring of phreatic surface

Reporting may include analysis of total and differential settlements and potential development of 3D modelling to understand critical regions and zoning for treatment.

Design is likely to include:

- Pre-loading with general fill material (to be designed to suit the underlying fill thicknesses and material types)
- Inclusion of wick drains to allow pore water pressures to dissipate more rapidly
- Monitoring

Opportunities may exist to support or accelerate de-declaration of the TSF, including:

- Dewatering studies, to explore the options to reduce the groundwater levels in the storage and thus the risk of dam failure in modelled scenarios.
- Modifying the embankments to reduce the risk of failure (and increase factors of safety) – this could include earthworks such as addition of toe buttressing or general reprofiling of the embankment slope to achieve lower Factor of Safety.
- Adjustments to the agreed final landform (currently being developed by Glencore) to introduce terracing or similar developable areas on the eastern embankment, above those regions where fill thickness is minimal over the former natural surface and off the dam structure, as shown indicatively in Figure 7-1.



Figure 7-1 Potential initial developable areas around eastern edge of TSF (blue)

Any proposed development in the downstream dam break impact area is subject to different constraints and would be also subject to revision of the assessed area being updated.

Alluvial sediments area present in the majority of this area and geotechnical investigations to inform design and planning would likely include most of the standard methods such as boreholes, test pits or CPTs. The potential for shallow or perched groundwater would mean that groundwater monitoring would also be essential.

Reference should be made to the Dam Break Engineering Report for further details related to the dam risk area.

8 Recommendations

The majority of the Structure Plan sub-precincts have limited existing geotechnical information available. It is therefore strongly recommended that these areas be subject to targeted geotechnical investigations to support future development post rezoning.

The scope of investigations should be proportionate to the proposed land use intensity, sub-precinct characteristics and constraint profile, and structured to inform both precinct-scale planning and site-specific development outcomes.

These should be integrated and applied in conjunction with the Dam, Flooding, Water Cycle Management and Contamination Report recommendations.

The proposed investigation scope and alignment with the Precinct application areas, and to be coordinated with other disciplines such as Contamination, is summarised in Table 8-1.

Table 8-1 Summary of recommendations

Investigation	Purpose	Primary application area
Cone Penetrometer Testing (CPT)	Characterise alluvial soils, fill and soft ground conditions, particularly within the TSF area and alluvial area where soft to firm or organic soils may be present. CPTs provide continuous profiling, dissipation testing, and indicative permeability and groundwater data. A nominal spacing of approximately 250m is suggested. CPT methods are generally preferred where appropriate as they generate minimal spoil, reducing handling and disposal requirements, including potential costs associated with contaminated drilling arisings.	TSF area, alluvial zones (Cockle Creek, drainage corridors), soft/rehabilitated landforms.
Borehole drilling and sampling (including gas assessment where coal seams present)	Validate CPT results, obtain soil/rock samples, confirm depth to bedrock and assess subsidence risks, enable installation of monitoring wells. Where coal seams are intersected, in-situ gas sampling and monitoring (e.g. canister or equivalent methods) should be undertaken to assess potential gas presence.	Subsidence-prone areas proposed for development such as West Lake, Killingworth and Barnsley West and areas with variable fill.
Mine subsidence assessment	Development applications within Mine Subsidence Districts require approval from Subsidence Advisory NSW before starting work. As such, consultation with Subsidence Advisory NSW is recommended during early planning to confirm site-specific requirements and approvals and thus confirming feasibility. Development will need to follow Subsidence Advisory NSW procedures and conditions, including staged review and engagement. The assessment of historic mine workings is typically desk based initially then intrusive drilling investigations and testing to verify the mine conditions followed by settlement analysis and grouting design (plan).	Areas prone to subsidence such as sub-precincts planned for development, including Barnsley West, Killingworth and West Lake.

Investigation	Purpose	Primary application area
Groundwater monitoring wells	Establish groundwater regime and support stability, settlement and dewatering assessments. Installed within selected boreholes to record groundwater levels and support settlement and dewatering assessments, with in-situ permeability testing where required.	Alluvial areas, TSF surrounds, low-lying land around such as that around Cockle Creek and Killingworth.
Test pitting	Assess shallow subsurface conditions and near-surface material properties to support pavement design and assess potential material reuse, including bulk sampling of near-surface soils.	Early activation areas including Teralba Southgate, Killingworth, Cockle Creek.
Geophysical surveys (such as seismic refraction)	Characterise ground profiles and estimate depth to bedrock in proximity to creek crossings.	Creek crossings, infrastructure corridors.
Acid Sulfate Soil (ASS) Testing	Identify and assess ASS risks in susceptible areas	Floodplains, wetlands, low-lying areas which occur across the Precinct, including Cockle Creek.

All investigations should be undertaken in accordance with AS 1726:2017 *Geotechnical Site Investigations*.

For investigations within subsidence prone areas, such as sub-precinct planned for development (including Barnsley West, Killingworth and Weste Lake), consultation with Subsidence Advisory NSW during early planning is recommended to confirm site-specific requirements and approvals. Development will need to follow Subsidence Advisory NSW procedures, including staged review and engagement.

9 Conclusion

The Geotechnical Assessment considers that the Precinct can support future urban and employment uses for commercial and industrial uses; however, rezoning must be explicitly aligned with geotechnical constraints, risk levels, and rehabilitation. The site is highly heterogeneous, characterised by legacy mining impacts, variable fill, alluvial soils, and subsidence risk, meaning that development suitability is location dependent.

Lower-constrained areas such as Teralba Southgate and Killingworth present clear opportunities for urban land use such as residential and mixed use, while moderate constraint areas can be progressively released subject to detailed investigations and engineered solutions. High constraint areas, including the TSF, floodplains, and deep alluvial zones should be reserved for conservation, open space, or deferred long-term uses, with development informed by geotechnical investigations.

The desktop analysis highlights that key geotechnical risks including mine subsidence, settlement, flooding, slopes, areas of historic filling, ASS and salinity occur within the Precinct and inform the required site-specific detailed investigations infrastructure design following rezoning at the DA stage to support future development. Investigations should be tailored to land use intent and constraint profile of the sub-precinct outlined in this report. It is considered possible that further subsidence assessment work may indicate that development types with low tolerance for ground movement or structural deformation could prove uneconomical.

The Tailings Storage Facility represents a long-term redevelopment constraint, requiring completion of rehabilitation, settlement stabilisation and regulatory de-declaration before any development can be contemplated in this area. Similarly, floodplain and estuarine environments which occur within the Precinct will remain constrained and are better suited to conservation and recreational land uses that don't impact flood conveyance and increase risk.

In consideration of the above, future development must be informed by detailed geotechnical assessment addressing:

- Mine subsidence
- Settlement and fill behaviour
- Slope stability
- Ground conditions and rehabilitation status
- Earthworks

Overall, the appropriate planning response is to implement a flexible, adaptive rezoning framework that:

- Prioritises development in low-risk, serviced, and previously disturbed areas
- Enables conditional development in moderate-risk areas with engineered mitigation
- Restricts or defers development in high-risk environments
- Integrates geotechnical constraints into statutory planning controls and staging triggers

This approach ensures that rezoning balances development potential with physical site limitations, supports safe and cost-effective delivery, and aligns with the long-term transformation of the Precinct into a sustainable post-mining landscape.

It is noted that, while additional investigation, design and construction measures may be required beyond those typically associated with a greenfield development, these are not expected to present significant constraints to development.

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Appendices



Appendix A – Figures

Figure A1 Site plan

Figure A2 Topography

Figure A3 Slope

Figure A4 Geology

Figure A5 Soil landscapes

Figure A6.1 Mine Subsidence Districts

Figure A6.2 Mine subsidence (extents of all underground workings)

Figure A7 Acid Sulfate Soils

Figure A8 Salinity

Figure A9 Available geotechnical information

Figure A10 Depth of soil and regolith

Figure A11 Slope stability geo zones

Figure A1 Site plan

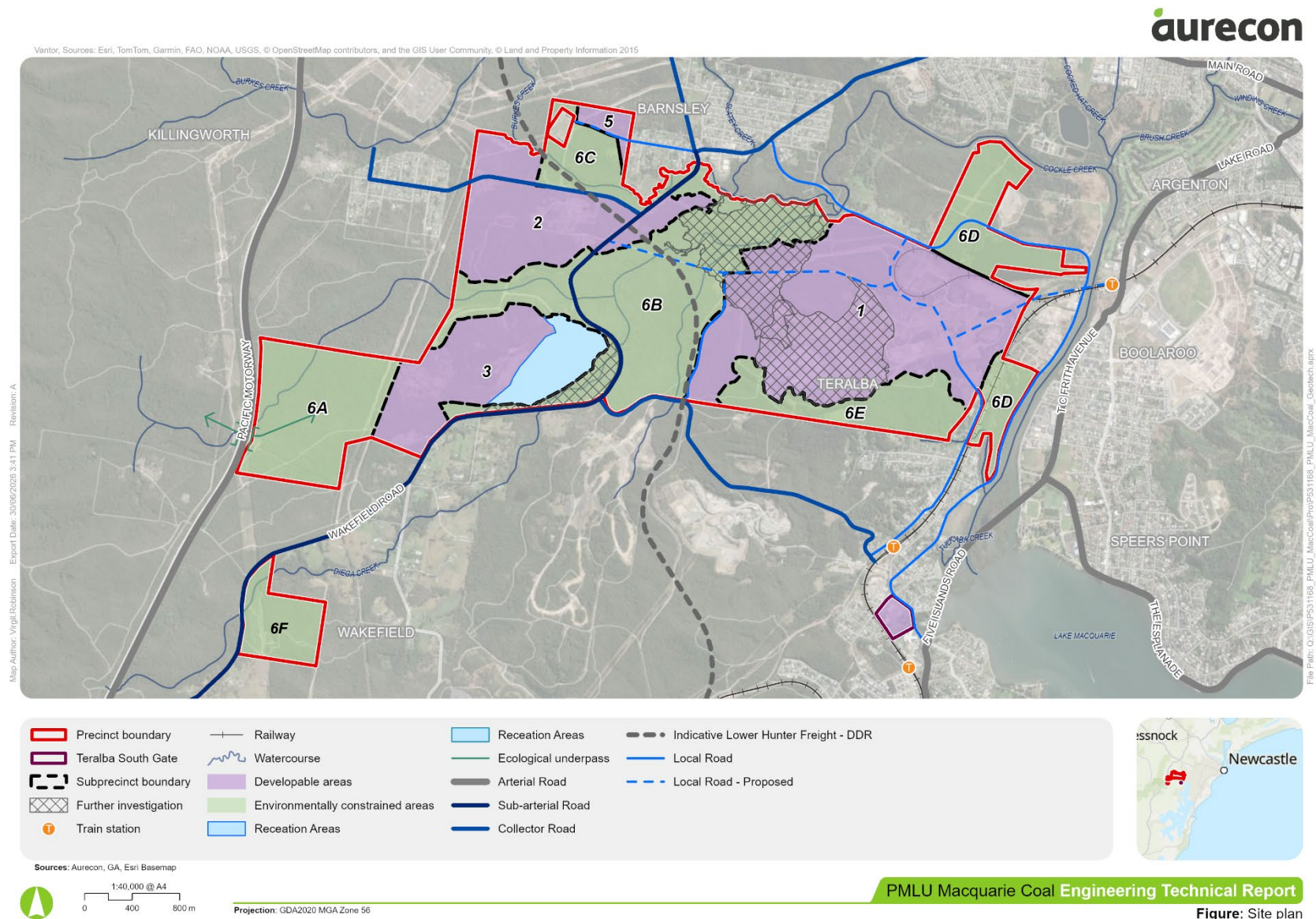


Figure A2 Topography

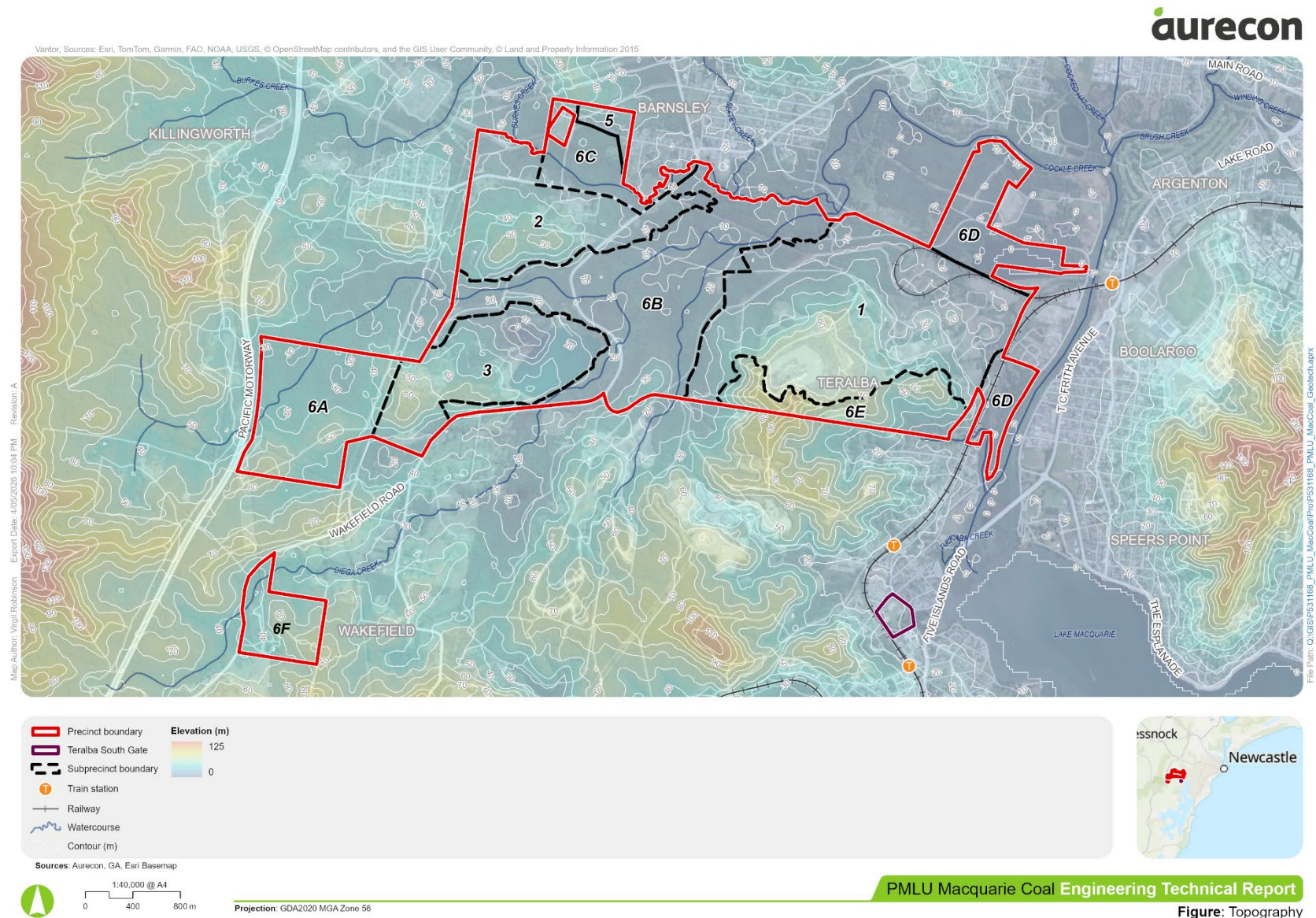


Figure A3 Slope

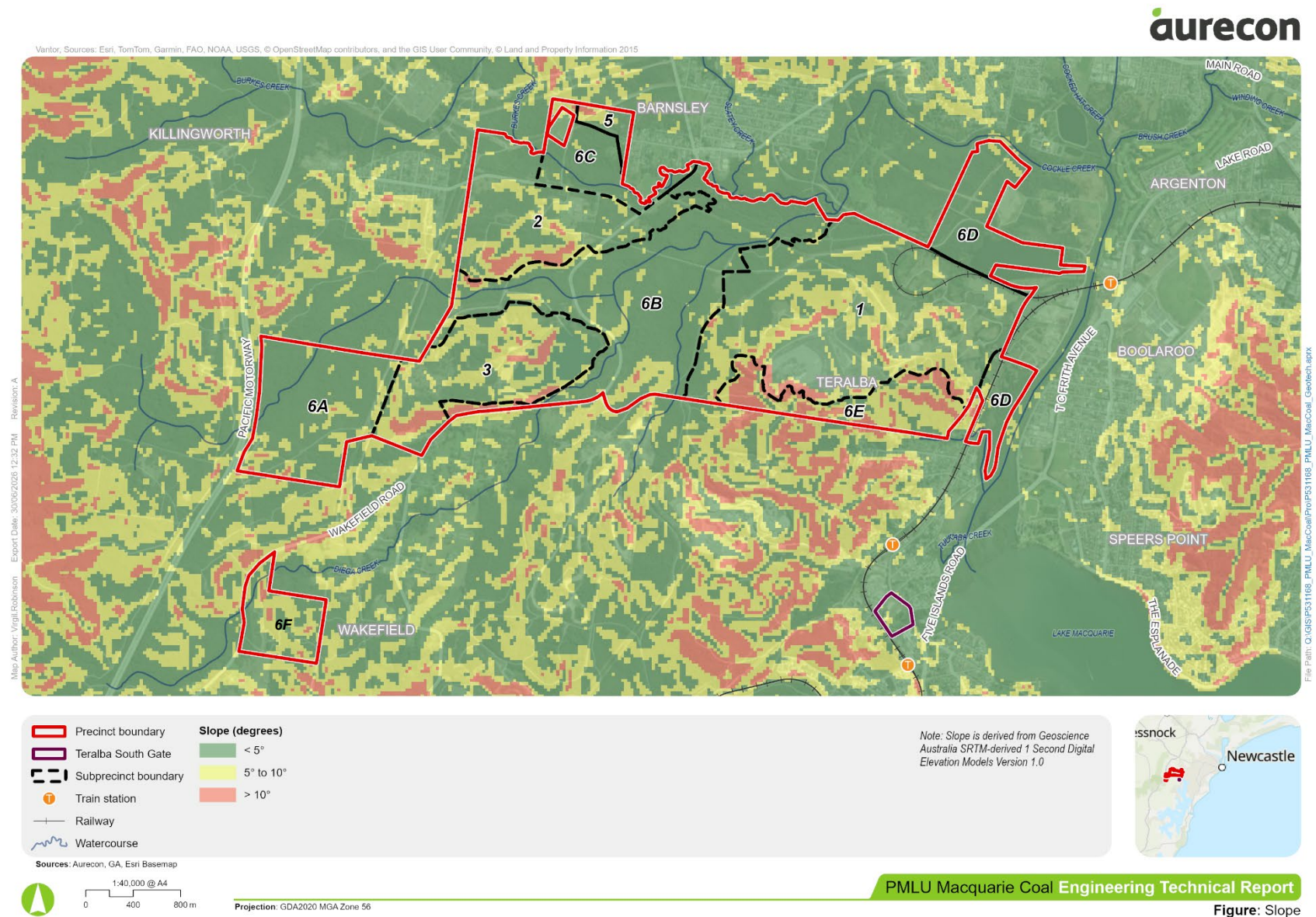


Figure A5 Soil landscapes

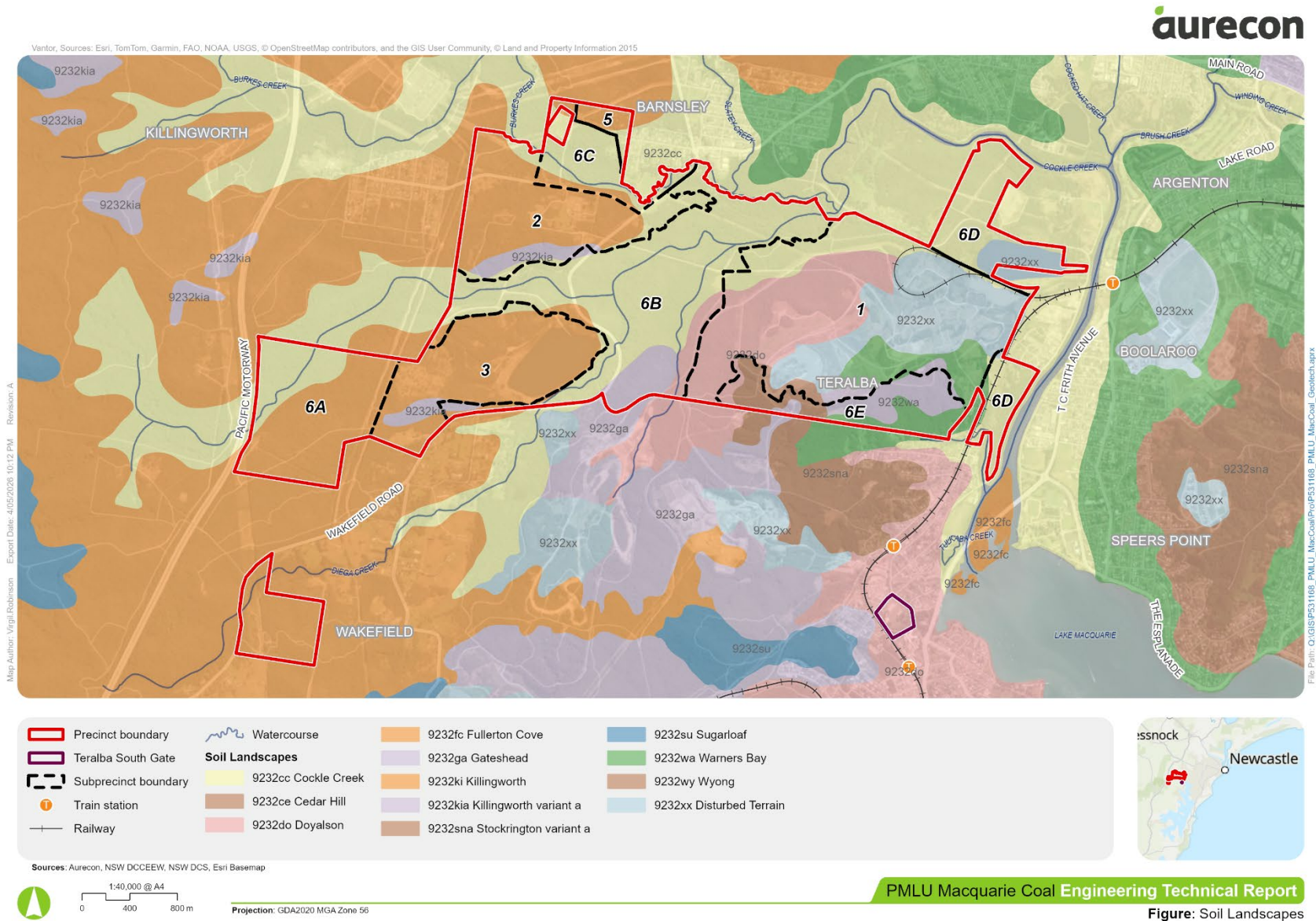


Figure A6.1 Mine Subsidence Districts

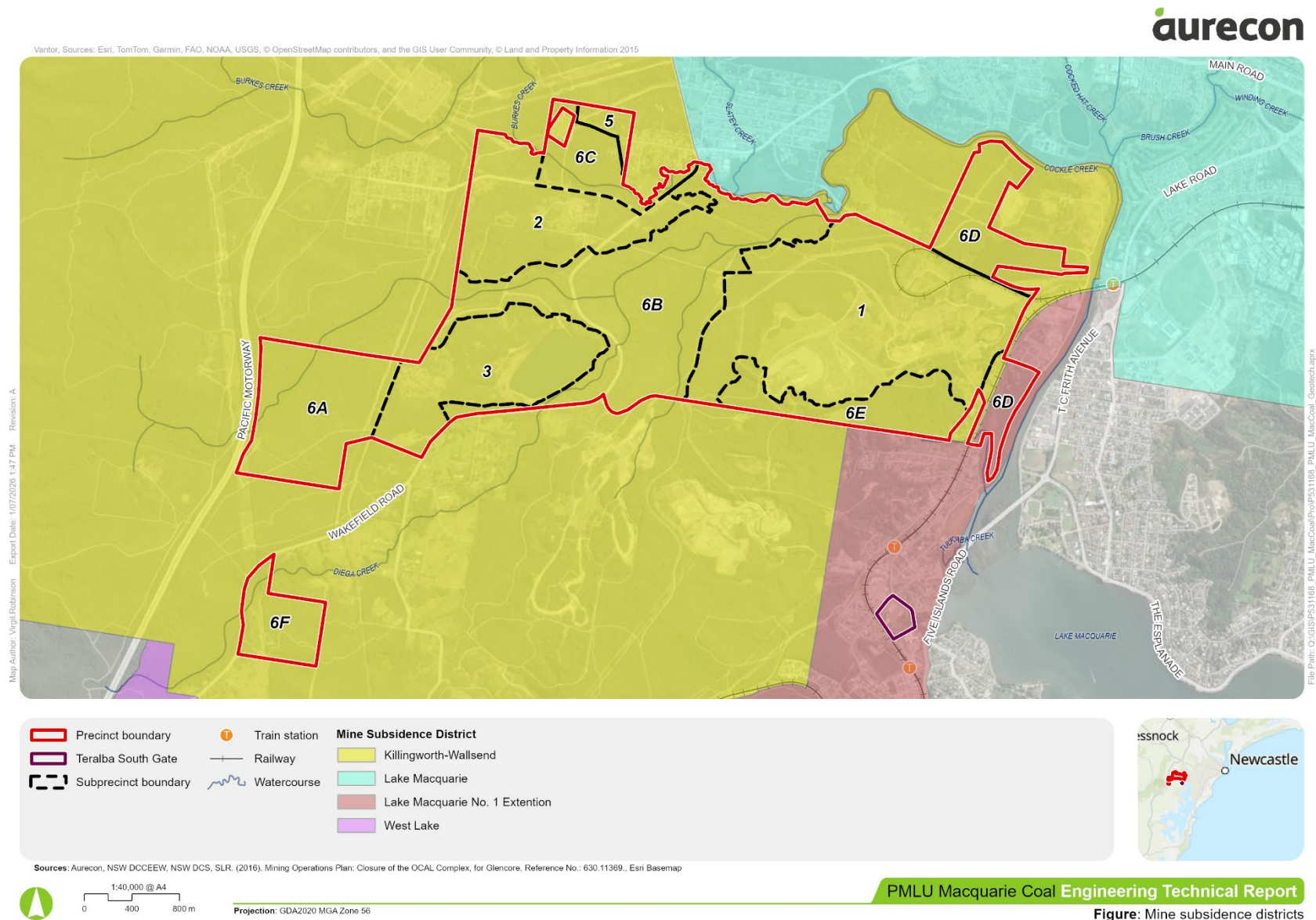


Figure A6.2 Mine subsidence (extents of all underground workings)

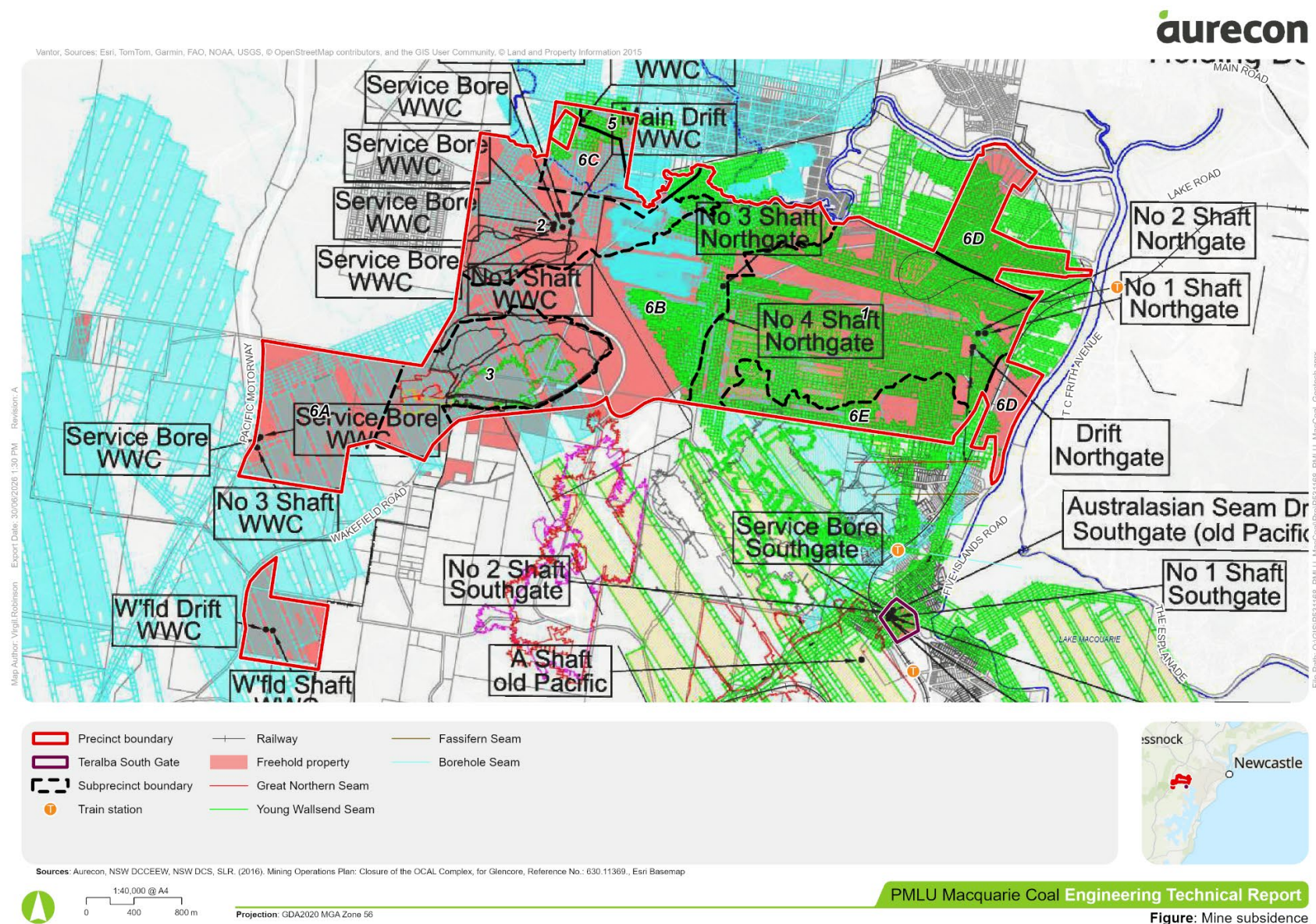


Figure A7 Acid Sulfate Soils

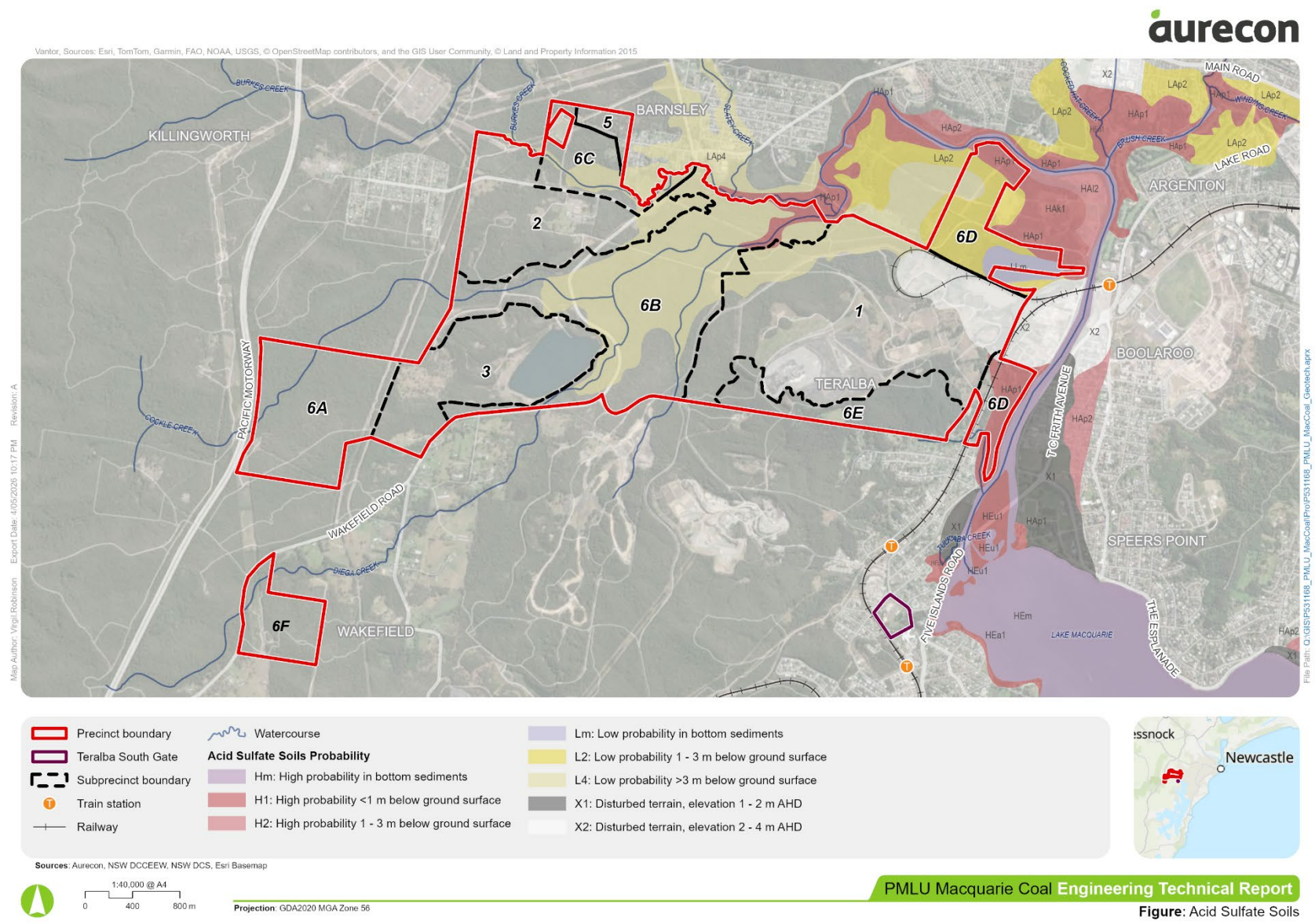


Figure A8 Salinity

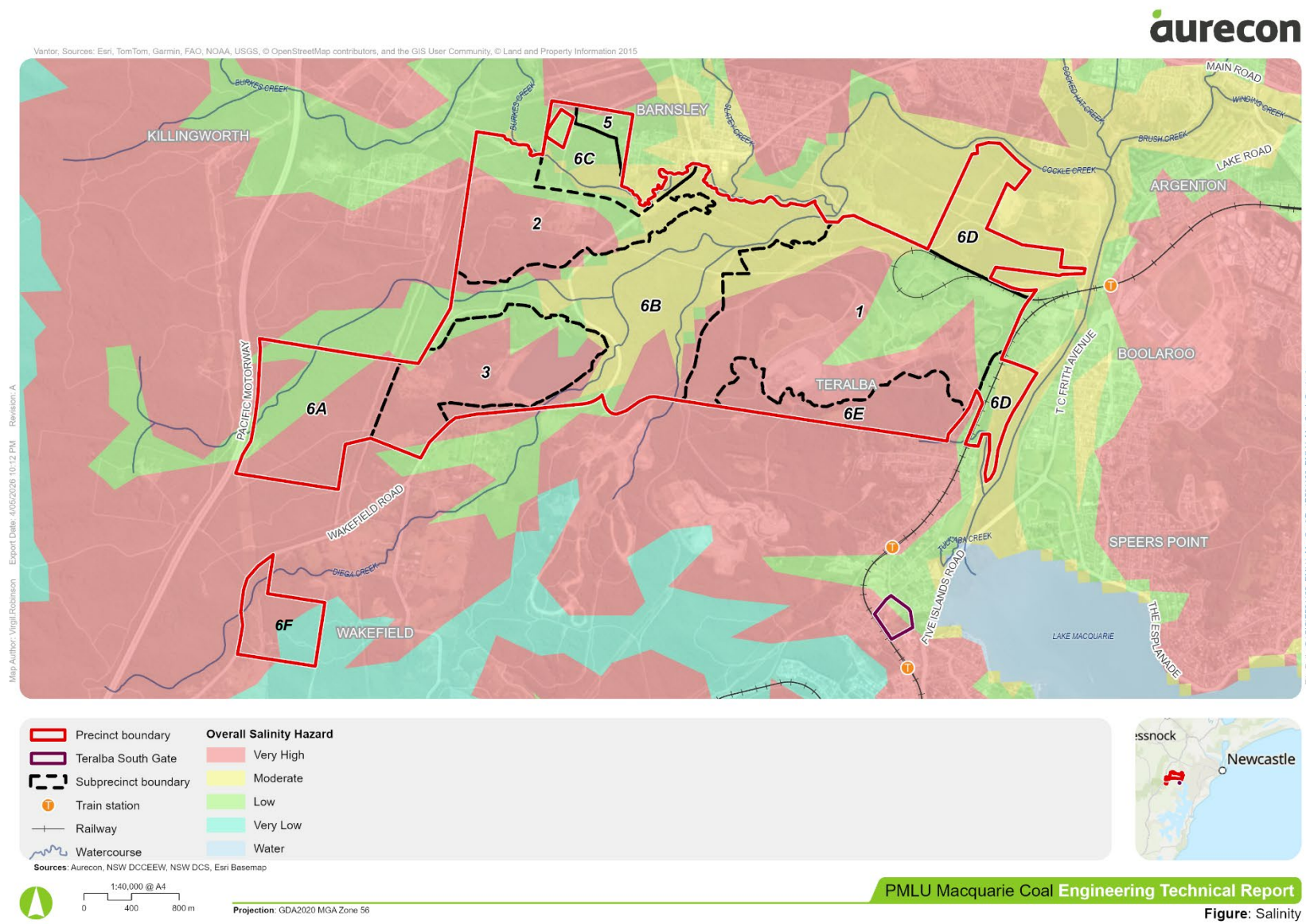


Figure A9 Available geotechnical information

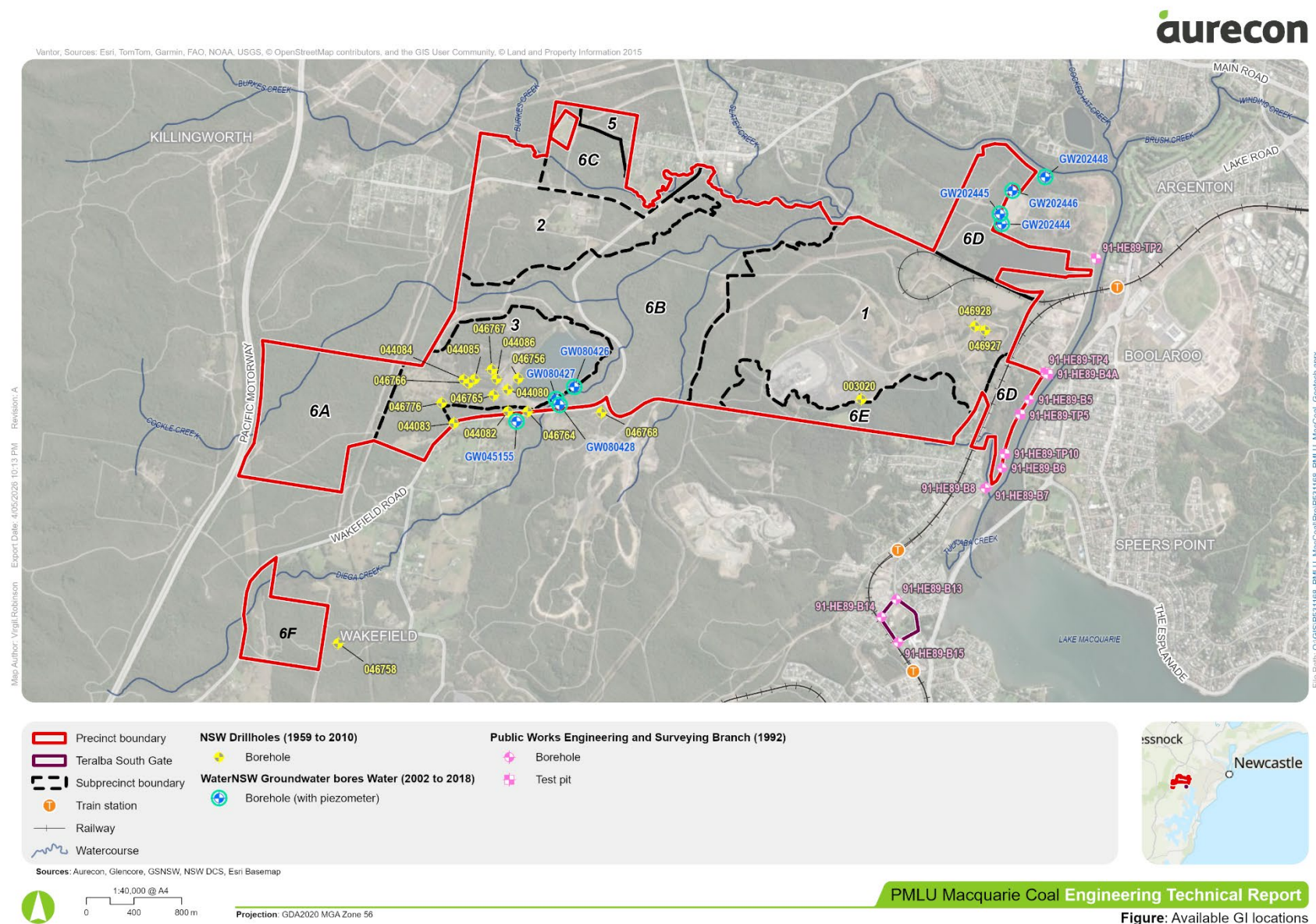


Figure A10 Depth of soil and regolith

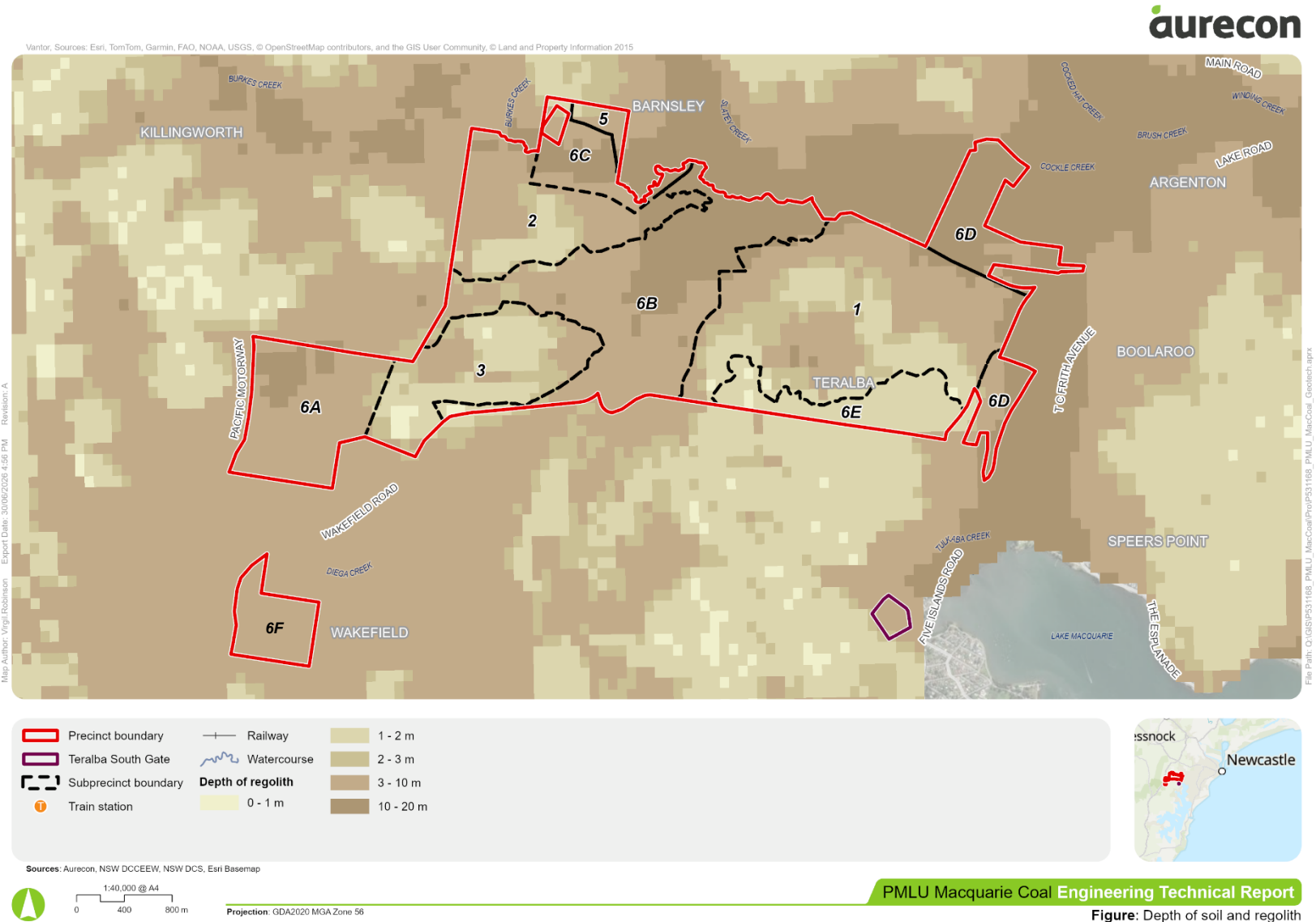
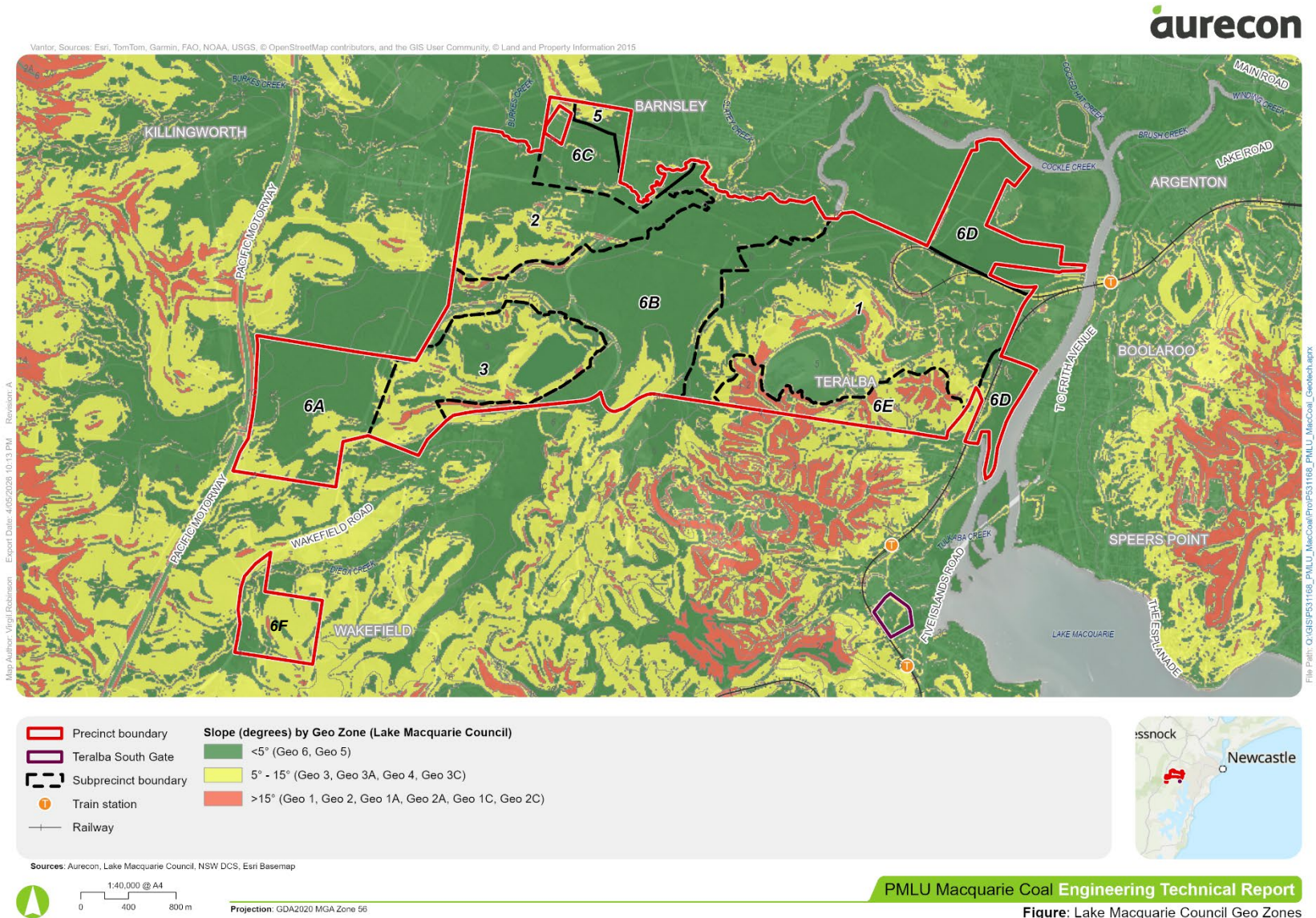


Figure A11 Slope stability geo zones



Appendix B – Constraints maps

Figure B1 Slope

Figure B2 Geology

Figure B3 Soil landscapes

Figure B4 Acid Sulfate Soils

Figure B5 Salinity

Figure B6 Depth of soil and regolith

Figure B7 Slope stability geo zones

Figure B1 Slope

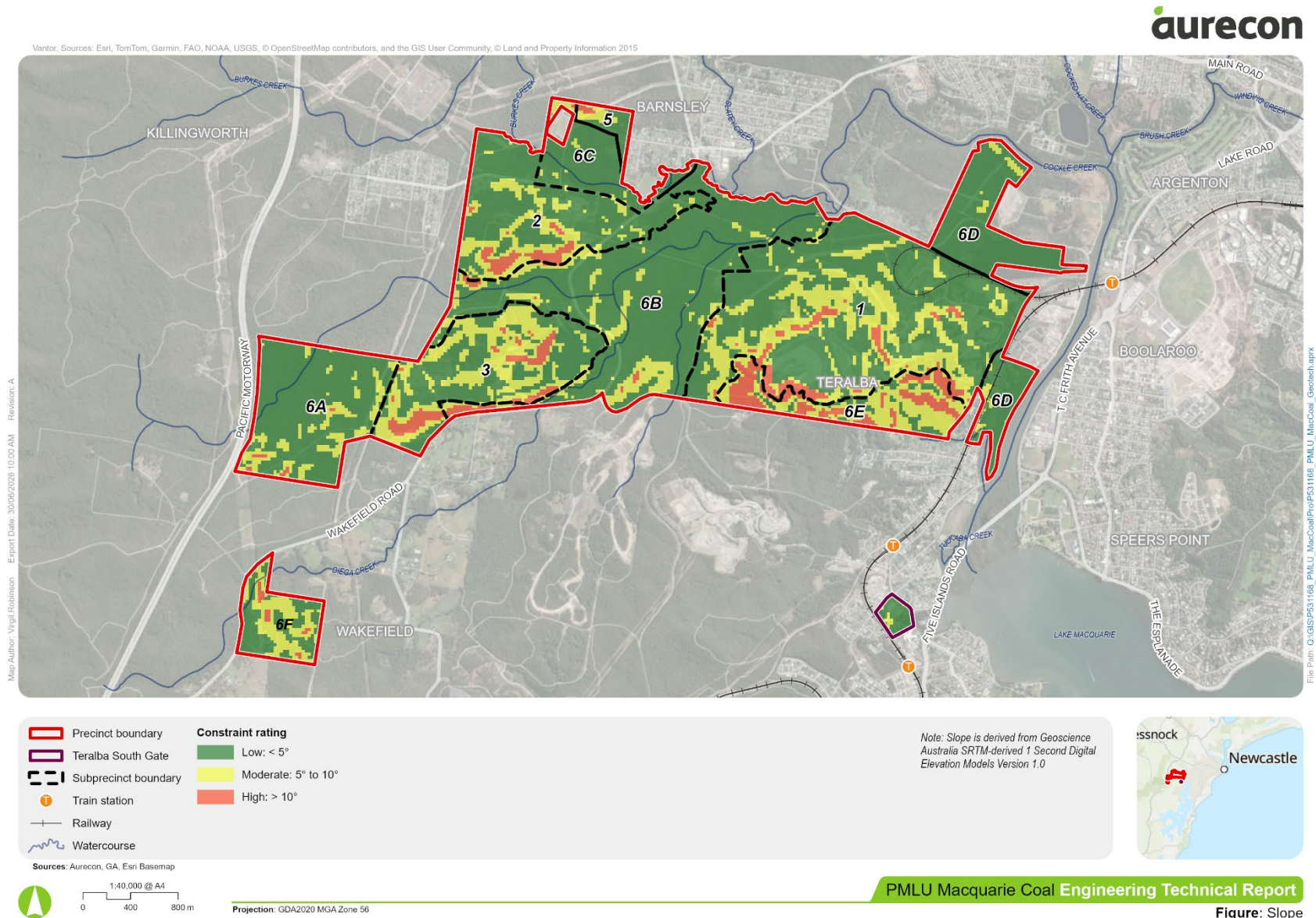


Figure B2 Geology

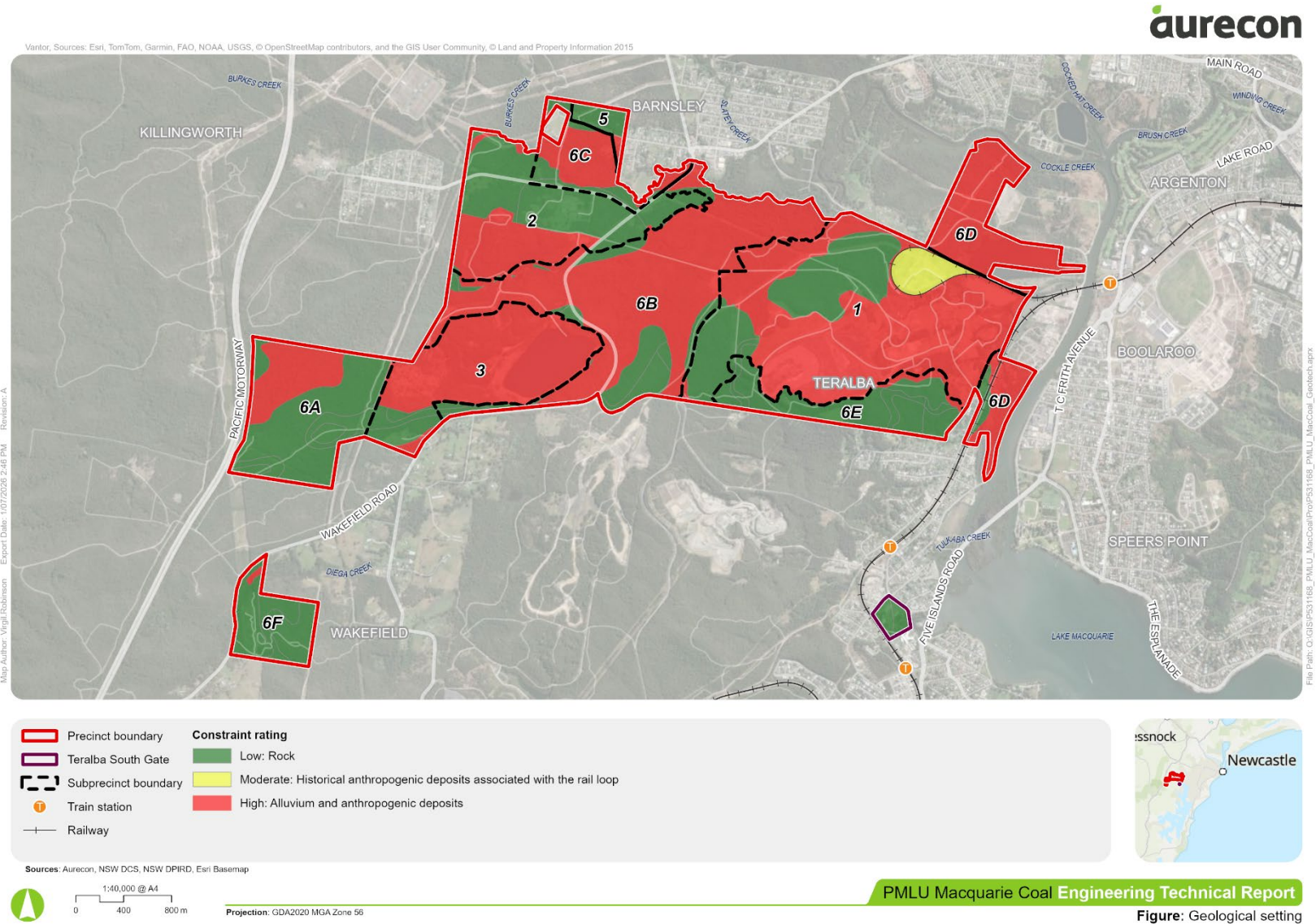


Figure B3 Soil landscapes

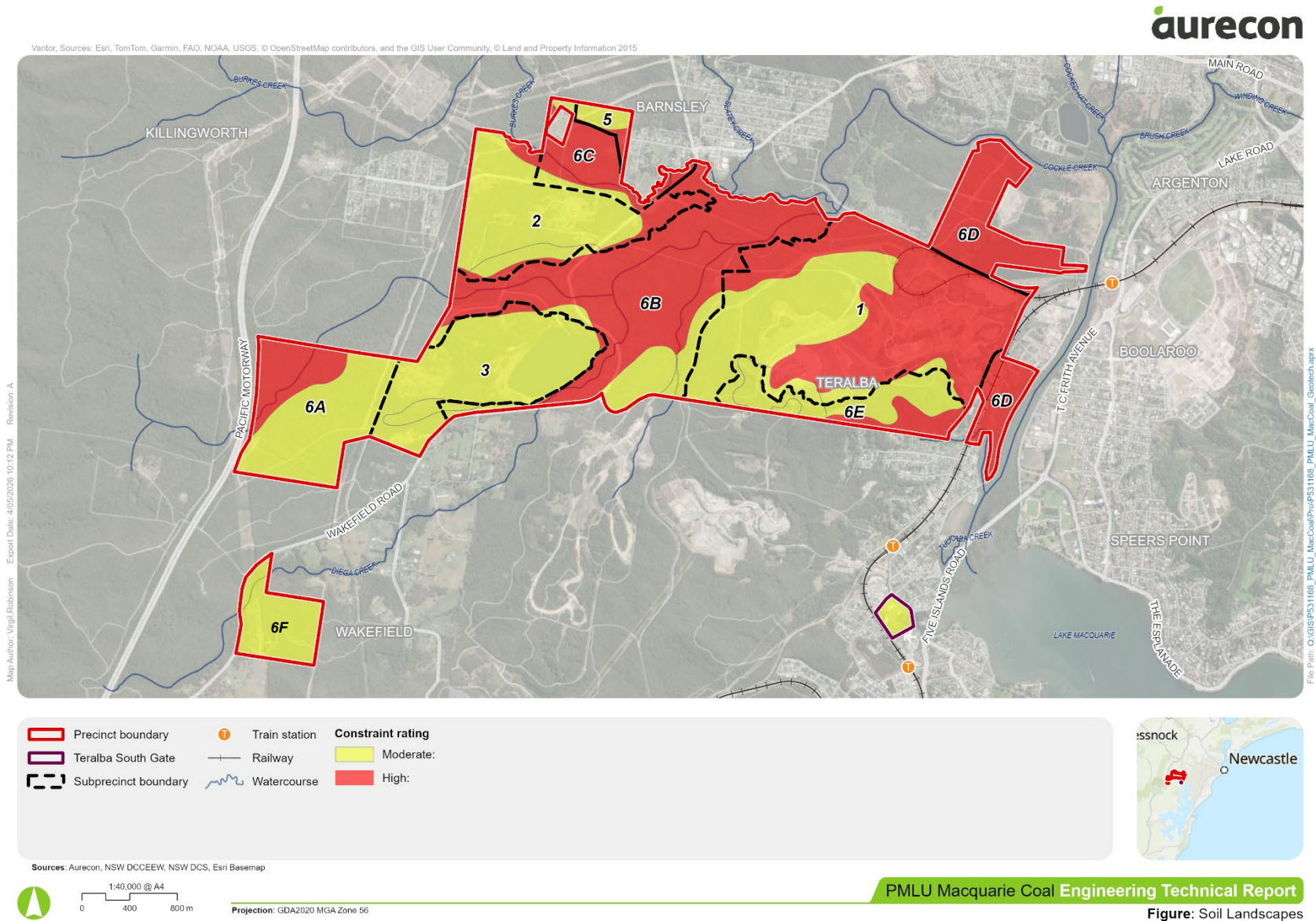


Figure B4 Acid Sulfate Soils

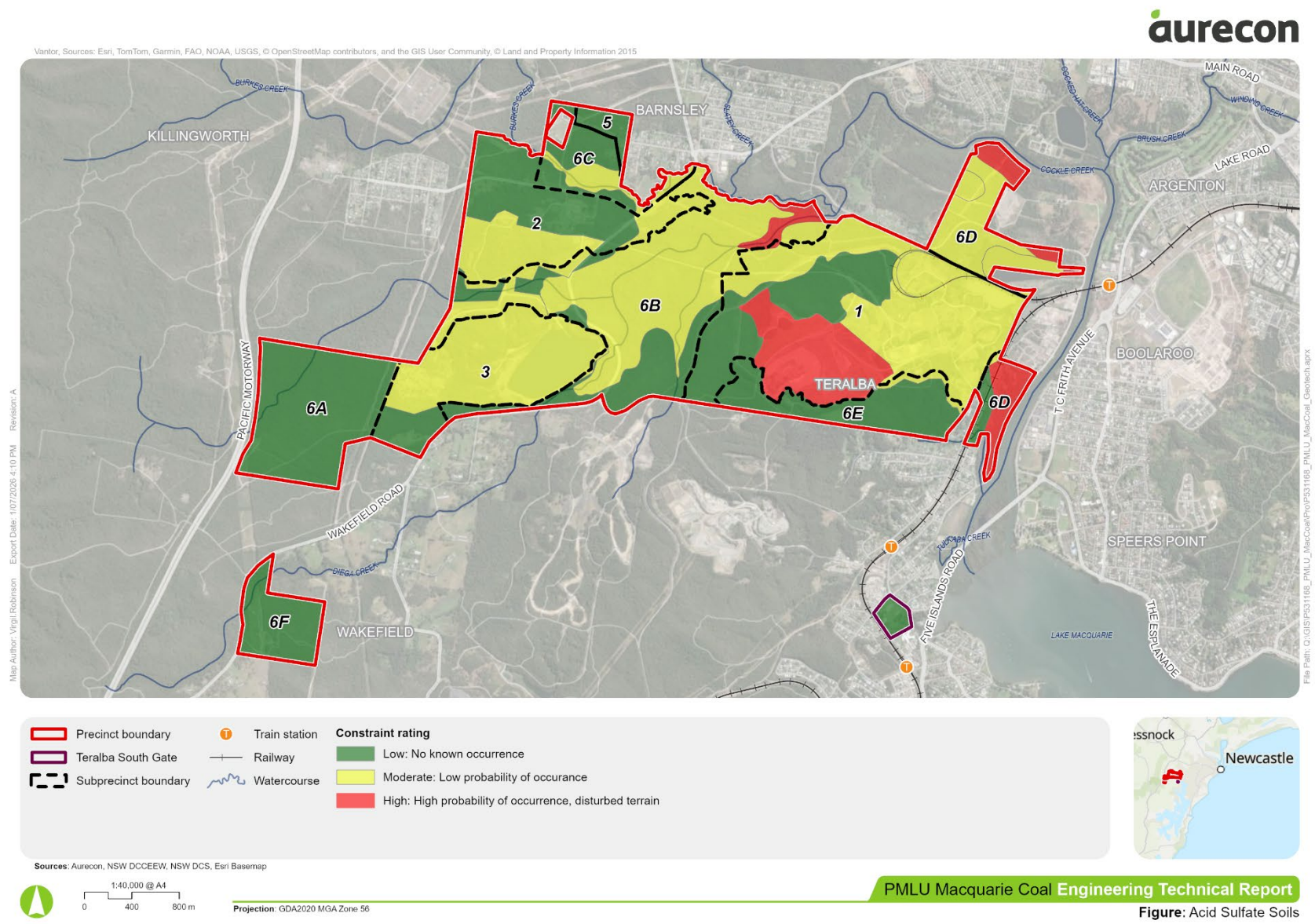


Figure B5 Salinity

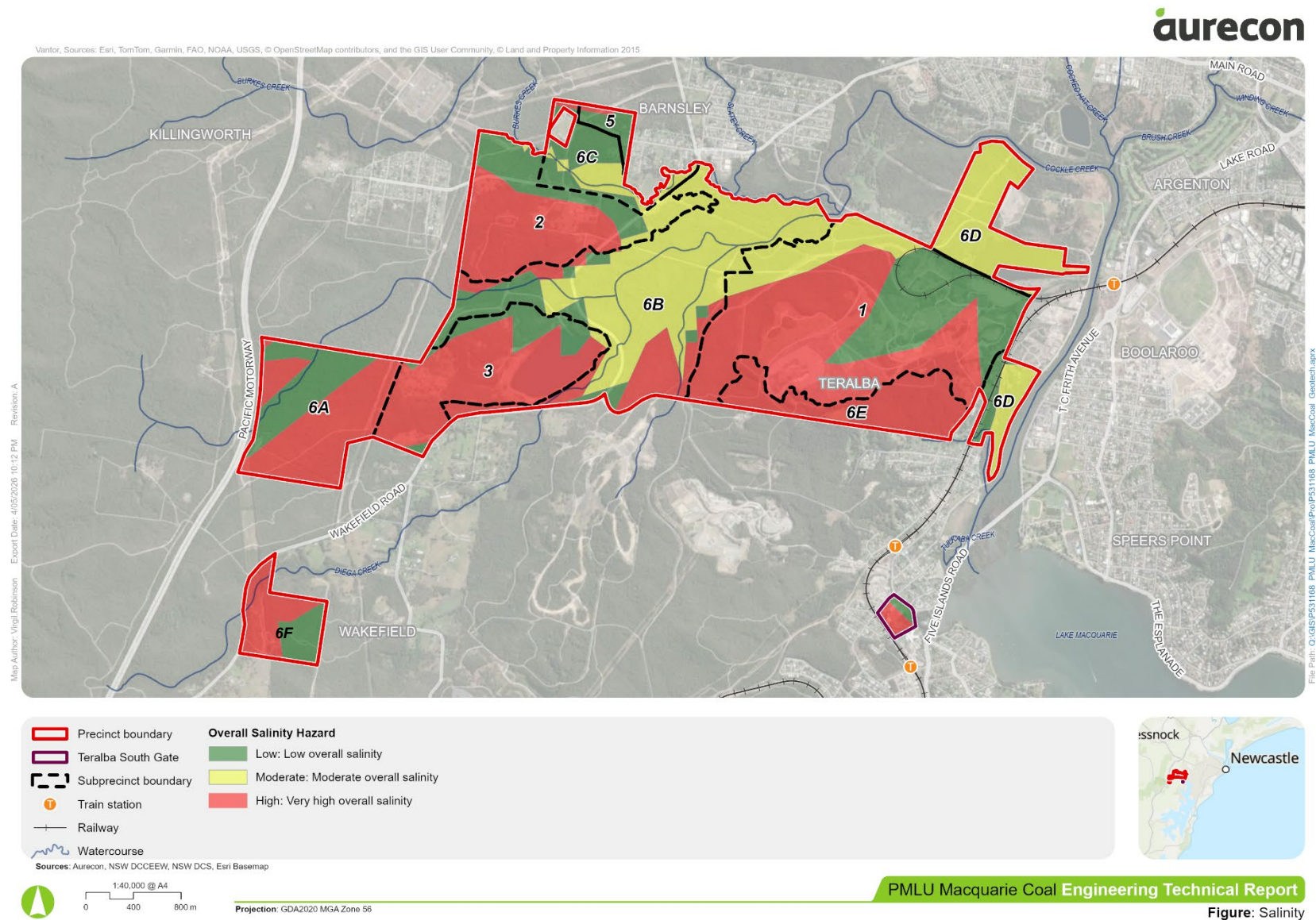


Figure B6 Depth of soil and regolith

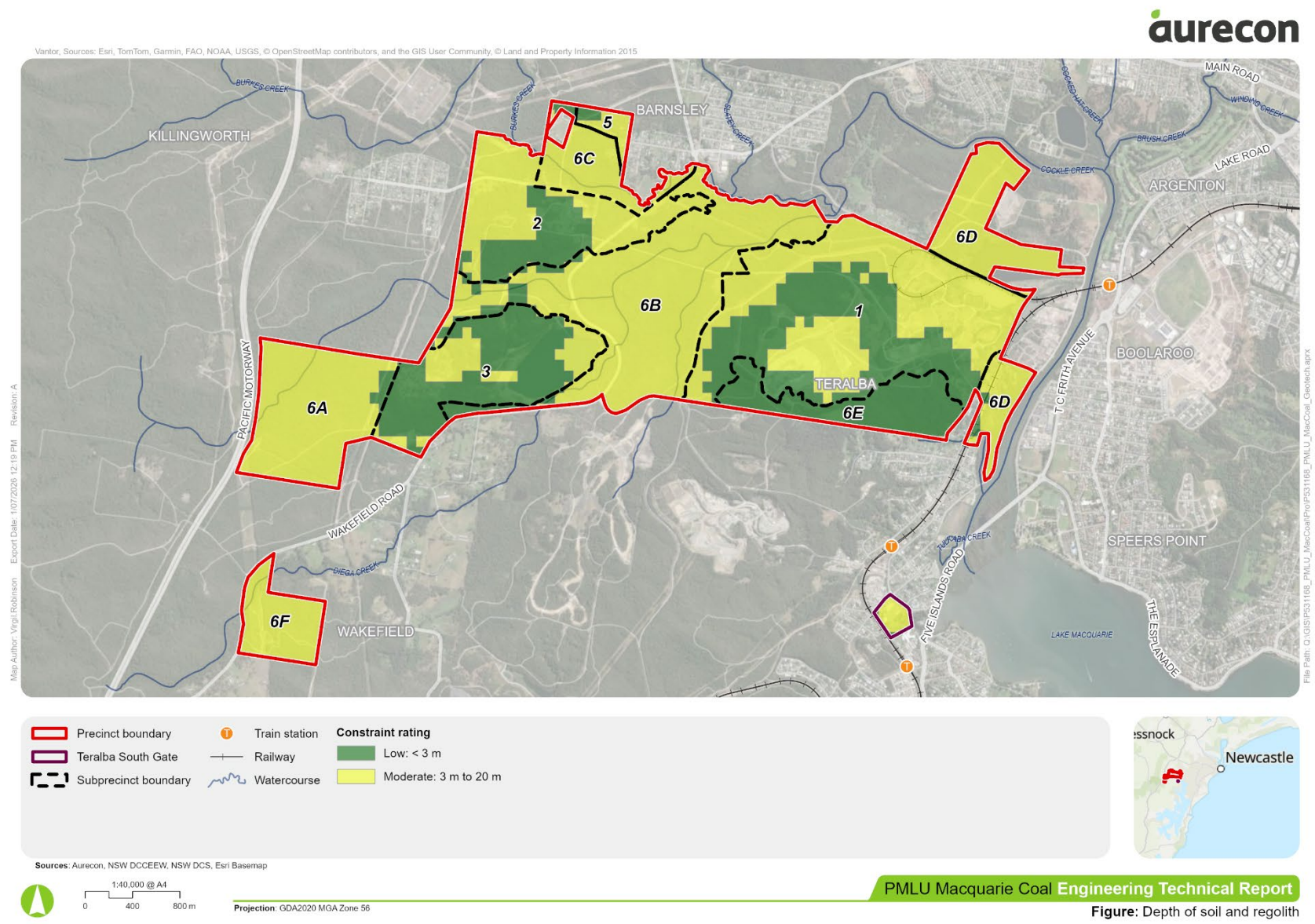
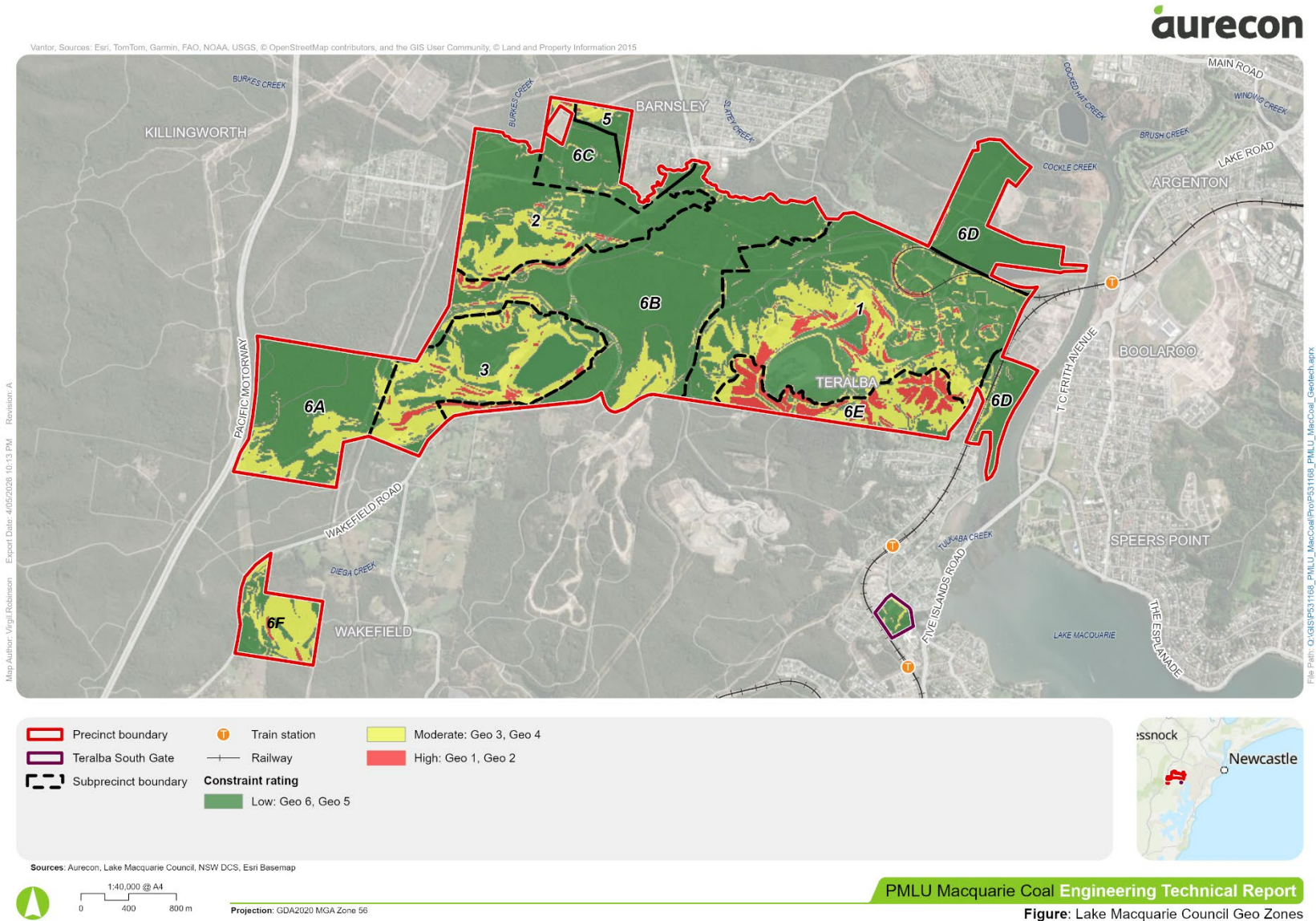
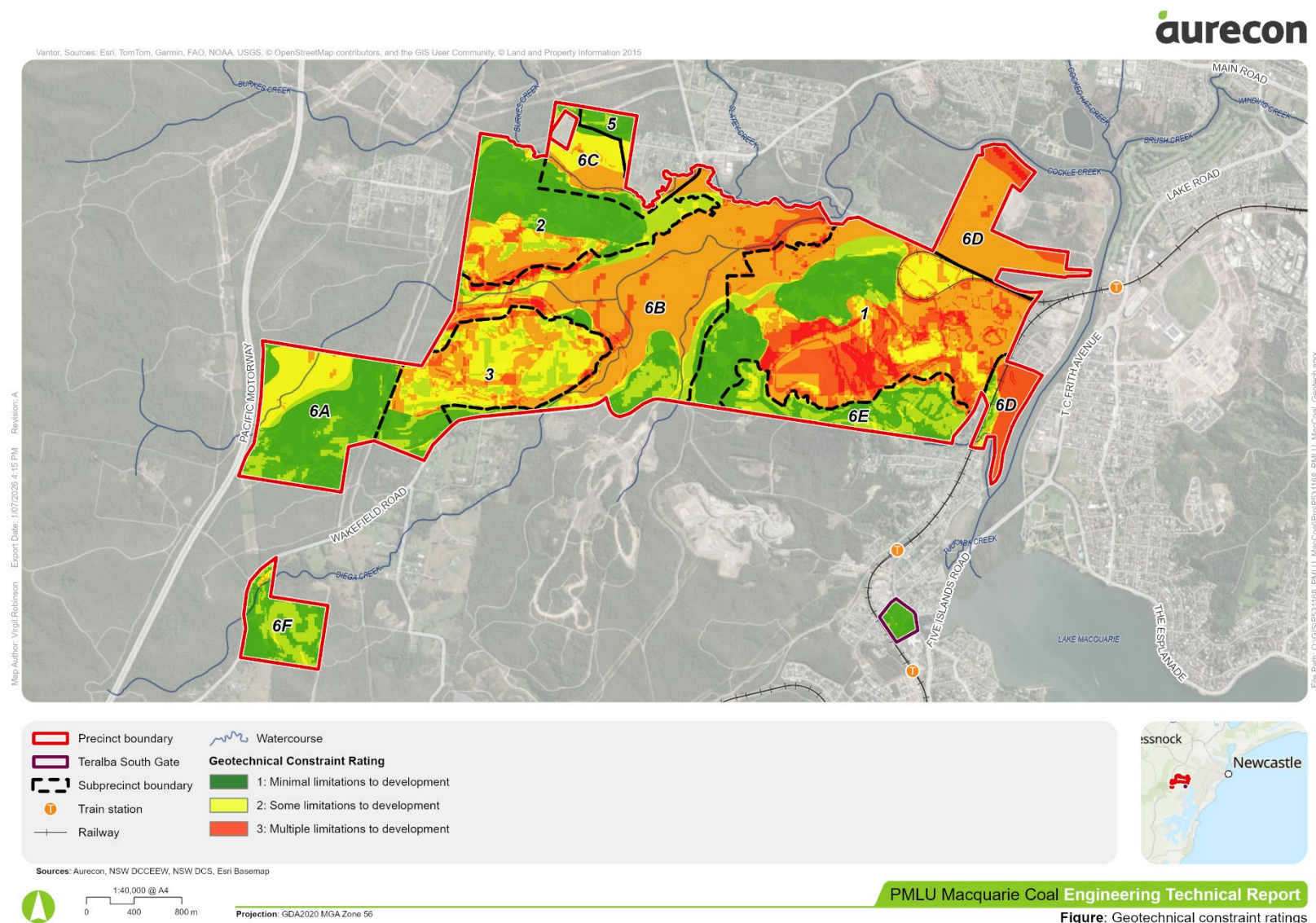


Figure B7 Slope stability geo zones



Appendix C – Near surface geotechnical constraints map for Structure Plan

Figure C Near surface geotechnical constraint map for Structure Plan



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